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Repairing Williams System 11 Pinball

1986 to 1990, Part One

Also this guide can be used for Williams System 9 games.

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Scope.

This document is a repair guide for Williams System 11 (and System 9) pinball games made from 1986 (High Speed) to 1990 (Dr.Dude). Some Bally games from 1988 to 1990 are also included as Williams bought the Bally pinball name in 1988. Updates of this document are available for no cost at <http://pinrepair.com> if you have Internet access. **This document is part one of three** (part two is [here](#), and part three is [here](#)).

IMPORTANT: Before you Start!

IF YOU HAVE NO EXPERIENCE IN CIRCUIT BOARD REPAIR, YOU SHOULD NOT TRY AND FIX YOUR OWN PINBALL GAME! Before you start any pinball circuit board repair, review the document at <http://pinrepair.com/begin>, which goes over the basics of circuit board repair. Since these pinball repair documents have been available, repair facilities are reporting a dramatic increase in the number of ruined ("hacked") circuit boards sent in for repair. **Most repair facilities will NOT repair your circuit board after it has been unsuccessfully repaired ("hacked").**

If you aren't up to repairing pinball circuit boards yourself or need pinball parts or just want to buy a restored game, I recommend seeing the [suggested parts & repair sources web page](#).

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Bibliography and Credit Where Credit is Due.

Many of the ideas in this repair guide are not original. Lots of people contributed to this document, and I just want to say, "thanks!" Below are a list of the resources used in the development of this guide. Some resources/people may have been innocently left out. If this is the case, and an idea is here that was originally yours, please notify me and I will make sure to give you credit!

- Tom Cahill at Williams, who provided me with lots of telephone support!
- [Jerry Clause](#), who provided tons of tips and tricks.
- Jonathan Deitch.
- Tom Callahan and his web site at www.repairconnection.com.
- Mr. Johnson and his web site at www.aros.net/~rayj/action/tech. Ray's postings and tips were most helpful.
- Duncan Brown. Duncan provided lots of tips and tricks.
- Tuukka Kalliokoski's web page at www.flipperit.net/tkalliok/flipperi/index_en.html.
- Rob Hayes, who's advice and proof reading were very appreciated.
- David Gersic, who also did proof reading and provided some tips.
- John Robertson and his posts & tips helped mucho grande.
- Pinball Liz Tech Reprints #1 to #6, August 1995 to August 1996, for their tips and tricks.

Some people question whether I wrote all this material myself. I did, but of course like everyone, my repair techniques and ideas are gathered not only from my own experience, but from work that others in this hobby do and share at shows, on the internet, etc. So if you're the originator of some cool trick or tip in this document, and I'm not giving due credit, just let me know and I'll add you to the list of contributors above.

1a. Getting Started: Experience, Schematics

What Repair Experience Is Expected?

Little experience in fixing pinballs is assumed. Basic electrical knowledge is helpful, but not necessary. I do assume you can solder and use the basic features of a Digital Multi-Meter (DMM) such as measuring voltage and resistance. Please see <http://pinrepair.com/begin> for details on the basic electronics skills and tools you will need. This document should help if you just bought your first (or second, or third) pinball "as-is", and hope to fix it.

Got Schematics?

Having a schematic for your game would be ideal, but sometimes you can fix your game without it. If you don't have a schematic, order one from one of the sources listed in the [suggested parts & repair sources web page](#).

Online schematics are available too:

- [System11b schematics](#) (Banzai Run)
- [System9 schematics](#)
- [System7/8 Troubleshooting](#) (Jun 1984)

1b. Getting Started: Necessary Tools

Fixing electronic pinball games will require a few tools. Luckily, most are not that specialized and are easy to get.

Non-Specialized Tools Required:

- Work Light: clamp style lamp
- Screwdrivers: small and medium size, phillips and flat head
- Nut Drivers: 1/4", 5/16", and 11/32"
- Wrenches: 3/8", 9/16", 5/8" required, other sizes suggested
- Allen Wrenches: get an assortment of American sizes
- Needle Nose Pliers
- Hemostat. Handy for holding parts and springs. Best to have both the curved and straight versions if possible.
- Right Angled Screwdriver: both phillips and flat head.

Specialized Tools Required:

These specialized electronics tools are needed. Please see <http://pinrepair.com/begin> for details on the basic electronics tools you will need.

- Alligator clips and wire. You can buy these at Radio Shack, part number 278-001, \$3.69.
- Soldering Iron.
- Rosin Core 60/40 Solder.
- De-soldering tool.
- Digital Multi-Meter (DMM).
- Logic probe.
- Hand Crimping Tool: Molex WHT-1921 (part# 11-01-0015), Molex part# 63811-1000, Amp 725, or Radio Shack #64-410.

Cleaning "Tools" Required:

- Novus #2 or MillWax (for cleaning playfields and rubber)
- Novus #3 (for polishing metal parts)
- A paste wax (Treewax) or hard Carnauba wax (for waxing playfields and cleaning rubber)

You can get Novus at many places (my local grocery store sells it), or from any good pinball vendor. I don't recommend MillWax, but others like it (mostly because they have been around for a LONG time and are used to it). Do not use any Wildcat products or CP-100! They react with plastic and can yellow ramps and lift mylar. Treewax paste wax or Meguires Carnauba Wax can be gotten at Kmart or the local hardware.

1c. Getting Started: Parts to Have On-Hand

When fixing electronic pinballs, I would highly recommend having some parts on-hand to make things easier and cheaper. All these parts are available from a pinball retailer.

Parts to have:

- #47 light bulbs: have 20 or so around. Fifty is plenty to do most games. I suggest using #47 bulbs instead of #44 bulbs, as they consume less power and produce less heat.
- #555 light bulbs: have 20 or so around. Fifty is plenty to do most games.
- #906 or 912 flash bulbs: have 10 or so around.
- #89 flash bulbs: have 10 or so around.
- #1251 flash bulbs: (used on some system 11 games).
- Fuses: I would have five of any needed value on hand at all times.
Get 250 volt fuses, not 32 volt (regardless of what the manual may say). Radio Shack sells fuses for a decent price. Slow-blo fuses are known as MDL fuses. Fast-blo fuses are known as AGC fuses. At minimum you'll need:
 - 1/10 amp slo-blo (not all games use this)
 - 1/8 amp fast-blo (newer games with 14 digit displays)
 - 1/4 amp slo-blo
 - 1/2 amp slo-blo
 - 2 amp slo-blo
 - 4 amp slo-blo
 - 5 amp slo-blo
 - 7 amp slo-blo
 - 8 amp slo-blo
- Nylon Coil Sleeves: the longer 2 3/16" length (part number 03-7066-5) are used when rebuilding flippers. The 1.75" length (part number 03-7066) are used for pop bumpers, etc. Sleeves with a lip (part number 03-7067-5) and tubing on each side (known as an "inline" sleeve) are used on the knocker, ball popper, etc. Also 03-7067-7 sleeves are used on drop target reset coils.
- Ball shooter sleeve, part number 03-7357.
- Flipper assembly, part number A-15848.
- Flipper Plunger/Link: used when rebuilding flippers (part number A-15847 or A-10656).
- Flipper Link Spacer Bushings: these small bushings go inside the flipper links (part number 02-4676).
- Flipper Coil Stops: used when rebuilding flippers (part number A-12390).
- Flipper EOS Switch: part number 03-7811.
- Flipper Cone Return spring: part number 10-376. I prefer not to use this style of return spring (I update all the system 11 games I work on to the newer WPC fliptronics style of return spring).
- 1/4" Heat Shrink Tubing: this is used on the flipper pawl when rebuilding flippers.
- Shooter Spring: the short chrome spring on the outside of the shooter mechanism (part number 10-149). These rust and look like crap in short order.
- 1 1/16" Pinballs: a new pinball will make your playfield last longer.

- Leg Levelers: replace those old crummy looking leg levelers with brand new ones. 3" are used on solid state games.
- Rubber Rings: you can order game-specific ring kits with exactly the rings you need. Don't forget to get flipper rubbers and a shooter tip.
- Remote battery holder.

Electronic Parts:

- Diode: 1N4004 (switches and coils)
- Diode: 1N4148 used on CPU board for battery circuit.
- Diode: 1N4764 and 1N4763 for HV power supply 100 or 91v 1 watt
- Diode: 1N4730 for power supply 3.9v 1 watt.
- Diode: 1N4735 for CPU board 6.2 volts 1 watt.
- Capacitors: 2.2 mfd 250 volt non-polarized (Williams part number 5045-12098-00). These are used on the EOS switches.
- Bridge Rectifiers: 35 amp, 200 volt (or higher) bridge rectifiers around, with lug leads. The industry part number is MB3502. Power supply circuits use these.
- 15,000 mfd or 18,000 mfd 20 volt cap. Used for the +5 volt filter cap.
- 39k Ohm 2 watt "flameproof" resistors: HV power supply circuit.
- 5 ohm 10 watt resistors: flasher resistor.
- 330 ohms 7 watt resistors: if your game is Fire! or before, have some of these around as they often break from the under the playfield flash boards.
- 27 ohm 5 watt sand resistors: CPU board lamp matrix.
- .156" Trifurcon connector pins, header pins, and housings. Used to repair burnt connectors.

Transistors:

- TIP102 for solenoids and lamp matrix. Replaces original TIP122.
- 2N4401 pre-driver for TIP102/122.
- TIP36c for high power solenoids.
- TIP42 for lamp matrix columns.
- 2N6427 for lamp matrix columns.
- 2N5060 for lamp matrix rows.
- 2N3904 for switch matrix.
- 2N5401 for power supply.
- 2N6059 for the power supply (replaces the original 2N6057).

Chips:

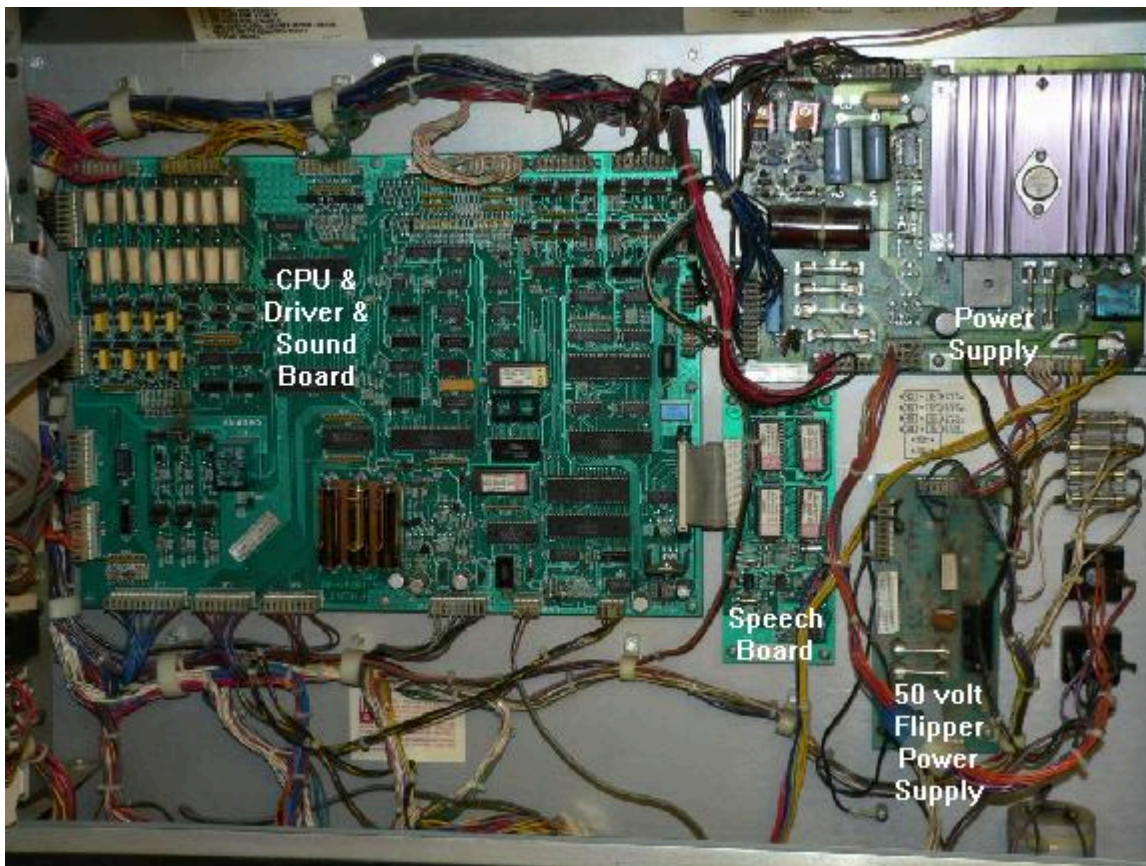
- 6808/6802 CPU chip. The 6808 can replace the harder to find 6802.
- 6821 PIA chip. Have several around as the CPU board uses six (or seven) of these PIA chips.
- 74LS244 chip: used on the CPU board for the switch matrix.
- 7402 chip used on the CPU board for special solenoids circuit.
- 7406 chip for lamp matrix rows.
- 7407 chip used on the CPU board for special solenoids circuit.
- 7408 chip used on the CPU board for the lamp matrix columns and solenoids.
- 4049 CMOS chip for the Display Master Controller board.
- 4050 CMOS chip for the Display Master Controller board.
- 4011 CMOS chip for switch matrix.
- 6116 or 2016 or NTE2128 2k by 8 CMOS static 24 pin RAM chip. This RAM chip holds settings and bookkeeping totals.

Order the transistors and diodes from many sources. I would suggest All parts and schematics should be ordered from someone on the [suggested parts & repair sources web page](#).

1d. Getting Started: Different System Generations

There are essentially four different generations of the Williams system 11 CPU board (five if System 9 is included). It is important to know which generation boardset is in a game, because components and circuit boards change with each generation. The differences in the four generations has mostly to do with the sound. With each new generation of system 11, the sound section was modified to use fewer and less expensive components. Eventually Williams moved all sound components to a separate sound board.

A System9 CPU backbox from a Comet. Picture by Scott.



- System 9: this was really the earliest release of System 11. System9 was only used on three games (Comet, Sorcerer, Space Shuttle). The difference between a System9 CPU board and a System 11 CPU board is minimal:
 - CPU board: the system9 CPU board has less ROM memory (up to one 27128 CPU EPROM and a 2732 EPROM opposed to System 11's two 27256 or 27512 CPU EPROMs). If the appropriate sized System9 ROMs are used, a System 11 or 11A CPU board can be used in a System9 game (and the system9 speech board will not be needed). Note a System 11B or 11C CPU board will not work in a System9 game because of the missing sound circuits. It is best to use the original System11 CPU board though, because only the original Sys11 CPU board has the segmented power-on LED (system11A only has individual LEDs, which are useless for any System9 diagnostics).
 - Sound & Speed: System9 used a separate speech board (which is the older System 7 speech board sometimes jumpered for larger EPROMs). The sound portion of the sound card is built-in to the system9 CPU board (and the CPU 2764 or 27128 at U49 position is used for the sound ROM). The speech card was changed during the production run of Space Shuttle and jumpers for 2732 Eproms were added (though some System9 speech boards do not have jumpers).
 - Score Displays: System9 uses the System7 master score display panel (system9 uses four 7-digit numeric only displays). Comet used the old system7 master display panel C-8863 with ribbon cables, but this was changed during Comet's production run to the D-10749 board. This new board replaced the ribbon cables with .156" headers connectors (in the Comet manual you will find only the schematics for the new D-10749 master Display card but I know that many still have the older cards installed). The new D-10749 board was only used in Comet as Williams went back to a ribbon cable design instead of the .156" Molex connectors.
 - Power supply: the system9 power supply is the same as a late system7 power supply. It is also the same as an early system11 power supply, except for the G.I. plugs which are different on system11.
 - 50 volt power: System9 uses a 50 volt power supply board for the flippers just like late system7 and early system11 games.

System9 games did suffer from an ailment inherited from system3 to system7 games: the dreaded white and black playfield connectors. These can be accidentally mixed, causing severe damage to the CPU board (the game and sound ROMs will die, as will at least one PIA and some sort TTL chips like the 74154 at U9 display control - nearly any chip that is power sensitive). When assembling the two large rectangle black and white connectors, make sure the wire colors match (the plug colors should also match, but double check the male/female plug wire colors). These connectors are less of a problem on system9 games largely because the backbox is now hinged, and does not get removed often. If these two

plugs are reversed, a quick way to tell is the right flipper will energize as soon as the game is turned on (but hopefully you have not turned the game on with these two plugs reversed!)

System9 also had some jumpers too. For example, W10 (above U50) would only be installed if the CPU board was in a Strike Zone game. Also Sorcerer used different ROM jumpers than Space Shuttle or Comet (but the two Sorcerer ROMs at U19 2732 and U20 2764 can be reformatted into a single 27128 at U20 so it uses the same jumpers as Space Shuttle or Comet).

Installing a System11 CPU board in a System9 Game.

Yes this work if the ROMs are programmed correctly. The system 11 CPU board is installed "upside down" in a system9 game. The location of the battery holder makes a lot more sense this way (and the CPU board silkscreen writing is positioned so you can read it!) Any leaked acid will not flow over the whole board (bad idea from Williams to change the orientation in System 11). When a System11 CPU board is installed in a System9 game, no ribbon cables go to the master display board, and there is no sys11 background sound board, and the system9 speech board is not needed. All the other connectors are identical. The seven segment display on the system11 CPU board does need some slight rewiring to decode the system9 ROMs. This information and ROMs thanks to Frank G.

The stock system9 ROMs need some reconfiguration to work in a system11 CPU board. Here are the system9 ROMs formatted for a system 11 CPU board:

- [Comet ROMs for System11 CPU board](#)
- [Sorcerer ROMs for System11 CPU board](#)
- [Space Shuttle ROMs for System11 CPU board](#)

In addition, a jumper will need to be set on the System11 CPU board to work in most system9 games: remove jumper W8 and insert jumper W9. Without this modification the game will work, but there won't be any sound. This happens because the sound reset in System11 is controlled by the main CPU chip, and it toogles a PIA output. System9 sound does not work this way.

System11 jumpers and how they can be set to run a system9 game.

JUMP S11 CPU	FUNCTION	GAMENAME.....		
		HIGH SPEED S11	ALLEY S11 CATS, SORCERER, STRIKE S9 ZONE	COMET, SPACE SHUTTLE S9
W1	oscillator to system	in	in	in
W2	memory configuration	in	out	in
W3	"	out	in	out
W4	"	in	out	out
W5	+5VDC to pullup resistor on opto input	in	in	in
W6	+12VDC to pullup resistor on opto input	out	out	out
W7	jumper for German settings (USA games:)	in	in	in
W8	sound proc RESET from proc-control RESET	in	out	out
W9	sound processor RESET from system RESET	out	in	in
W10	+18VDC to sound amplifier	out	out	out
W11	+12VDC to sound amplifier	in	in	in
W12	address-buffer enable	in	in	in
W13	data-buffer enable	in	in	in
W14	1 mS <u>IRQ</u> signal	in	in	in
W15	2 mS <u>IRQ</u> signal	out	out	out
W16	BLANKING to input 'D' of LED decoder	in	out	out

A System11 CPU backbox from High Speed. On the insert panel is the score display Master Display board. The Large board is the CPU board. The board at the upper left is the background sound board. The top right board is the power supply. And the lower right board is the 50 volt power supply board.



A System11 score displays from High Speed. Two 7-digit alphanumeric displays, two 7-digit numeric displays, and one 4-digit ball/credit display.



System11 Board System Generations.

Short method to ID different system11 MPU boards:

- "System 11" - connector at 1J15 (upper left corner).
- "System 11A" - U1 present (audio amp), no connector at 1J15.
- "System 11B" - U1 not populated, no connector at 1J15.
- "System 11C" - entire sound section in upper left not populated, no connector at 1J15.

Better explanation of all the different System11 MPU boards:

- System 11: first generation of system 11, part number D-10881. Seven segment diagnostic display. This board has a complete, but unused opto switch circuit at connector 1J9 (which paralleled the switch connector at 1J8). High Speed and Grand Lizard used a background sound board D-11297. Road Kings used sound board D-11298. CPU speaker connector 1J15 is present and used, as the CPU sound amplifier U1 is utilized. The CPU board's sound section handled the main speech & sound, but also used an additional background sound board. All original system11 games used five score displays (top two were 7-digit

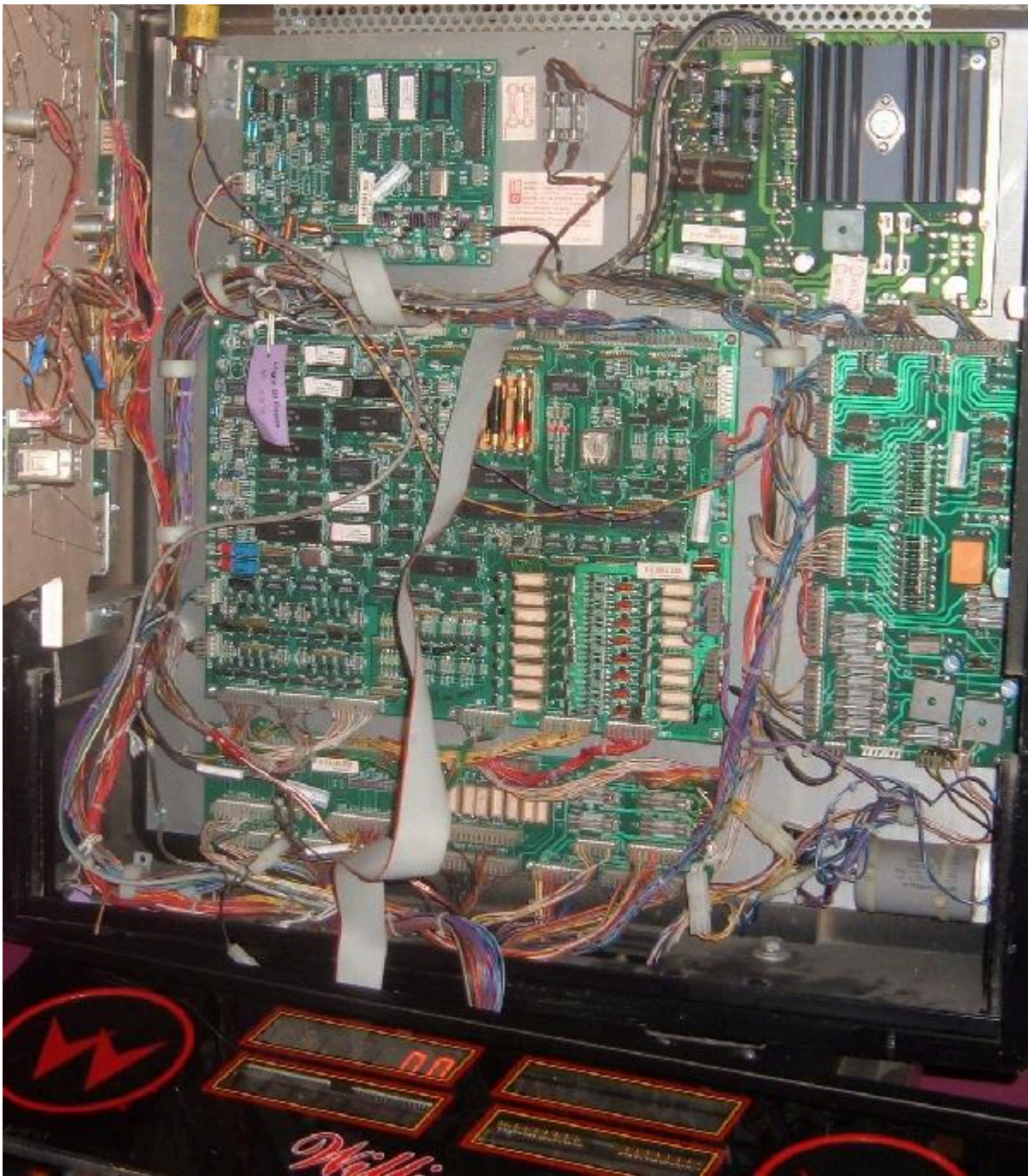
alpha-numerics, bottom two were 7-digit numeric, and one 4-digit numeric credit/ball display). The master display panel (D-10877) was separate from the discrete displays. Game EPROMs used a 27256 at U27 and a 2764 at U26. With Road Kings the U26 EPROM increased to a 27128. But the System11 board can use a pair of 27256 EPROMs without any jumper changes. The System11 MPU board can be used in any system11-11c game, with a couple of small modifications (see below for more details).

A System11A CPU backbox from F-14. The Large board is the CPU board (the missing sounds circuitry and be seen at the upper left corner of the CPU board). The board at the upper left is the sound board. The top right board is the power supply. And the lower right board is the 50 volt power supply board. Note the score display panel does not have a 4-digit ball/credit display. The relay on the CPU board is for the flippers. The relay on the power supply is for the GI. And the relay mounted above the CPU board (center) is for the triple police lights on top of the backbox.

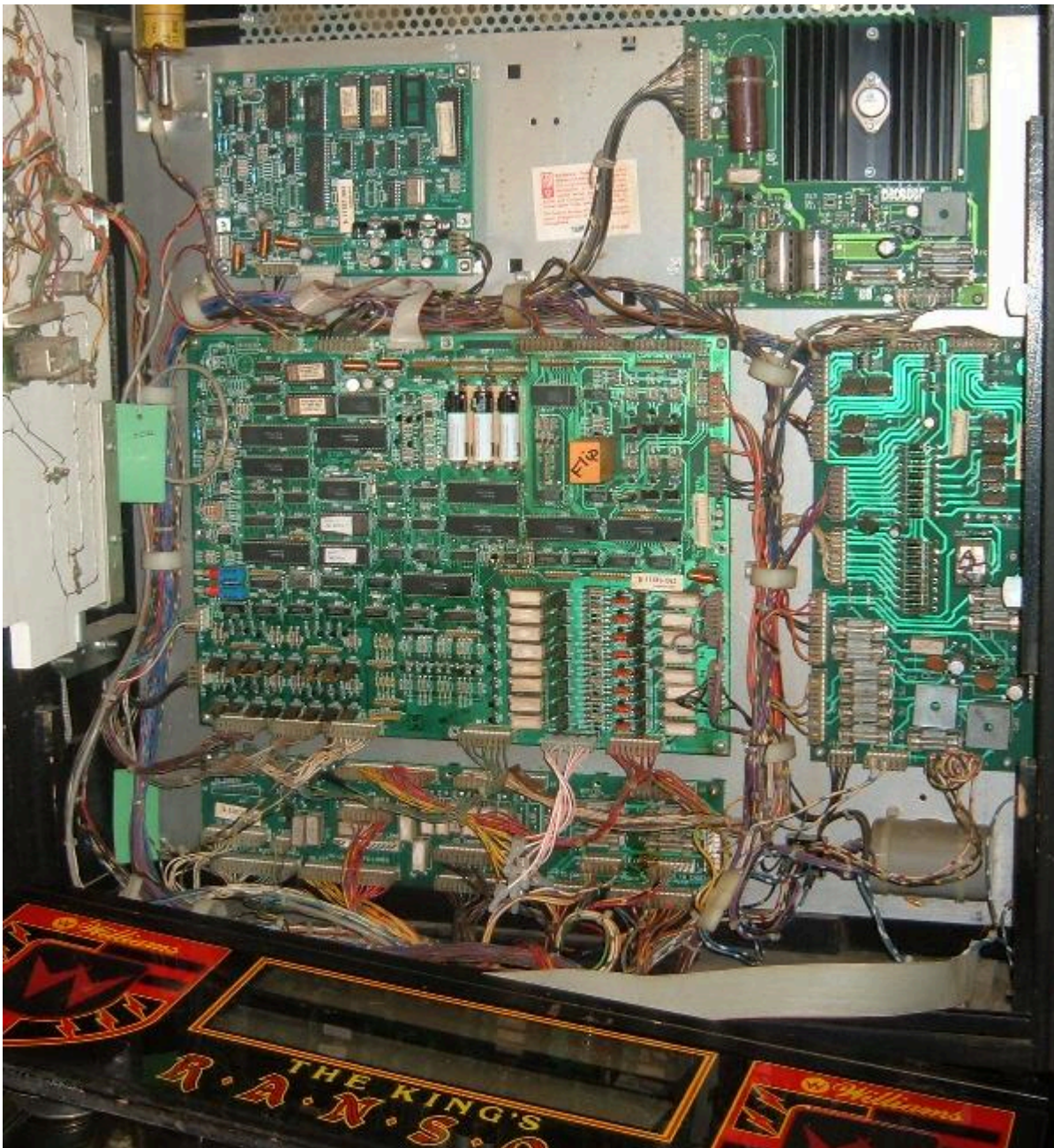


- System 11A: second generation of system 11, part number D-11392. Six zener diodes ZR3-ZR8 (1N5234 or 1N4735, 6.2 volts) added to the CPU board for the special solenoid switch input connector. A ground connection was added to connector 1J18 pins 6,7. The seven segment diagnostic display was changed to three LEDs. These games used audio board D-11581 which handles the speech and sound and has it's own sound amplifier. (Exception: Pinbot used sound board D-11298.) The initial system11 background sound board was no longer used. Because the new system11a sound board has its own ampplier, the System11A CPU boards have their speaker connector 1J15 was removed and the CPU sound amplifier U1 is not used (1J15 must be added back to the CPU board for it to work in earlier system11 games). Strangely Williams left the unused amplifier in place for system11a (perhaps to make it backward compatible to system11). The opto switch connector at 1J9 was removed. Jumpers W16, W17 were added to ground pin 38 of the 6802/6808 microprocessor U24. If a Motorola microprocessor is used, W16, W17 should be connected. If a Hitachi HD680x mpu is installed, jumpers W16 and/or W17 should be removed. Other brands are not affected by this and work fine with pin 38 tied to ground. Note that W17 is for the speech MPU chip and W16 for the main MPU chip. Pinbot and Millionaire system11A games used five score displays (top two were 7-digit alpha-numerics, bottom two were 7-digit numeric, and one 4-digit numeric credit/ball display) all combined onto a single circuit board (D-15414 I believe). F-14 and later system11A games only had four displays (two alpha, two numeric) because the credit/ball display was dropped. All four displays were mounted on a single circuit board (D-11609 or D-11610). Game EPROMs used a 27256 at U27 and a 27128 at U26. The System11a MPU is the most versatile of all system11 boards. Can be used in all games sys11-sys11c (except for a small modification needed to use it in the original sys11 games, see below for more details).

A System11B CPU backbox from Swords of Fury. The old larger power supply (pre-Taxi) is used in this game, with the whole bottom section of the power supply unpopulated (no 18 volt output for lamp matrix, no 25 volt output for low-power coils, not GI relay or GI connectors). There is no board mounted GI relays, but a GI relay can be seen on the backbox insert panel (there's one under the playfield for playfield GI). On the interconnect board, there are no 4N25 chips mounted. Instead a switch is used on the flipper button to detect a lane change. Later games with interconnect boards will have these 4N25 chips so a flipper lane change can be detected without a mechanical switch.



A System11B CPU backbox from Black Knight 2000. The new smaller power supply (Taxi and later) is used in this game, along with two 16 character alpha-numeric displays. Note there is no board mounted GI relays, but a GI relay can be seen on the backbox insert panel (there's one under the playfield for playfield GI).



- System 11B: third generation of system 11, part number D-11883-xxx (where xxx is the game number). The sound *amplifier* section (U1) was removed (along with its associated capacitors and resistors) from the upper left corner of the CPU board. But the rest of the sound circuit is still present. These games also used audio board D-11581 (except for Space Station which used sound board D-11298, and Jokerz which used a stereo sound board), and often a Sound Overlay board, so the unused U1 sound amplifier and associated parts were removed (these must be added back to use this CPU board in a system11 game). System 11B had the opto switch circuit and SRC6 removed, as these were really never used. Also changed was the values of some resistors and capacitors in the sound section. Most polystyrene capacitors were replaced with less expensive ceramic caps. Special solenoids now CPU controlled and games have Auxiliary power driver board (actually these both started with "Big Guns, the last system 11a game). Also transistors Q42-Q49 were changed from two rows of four transistors to a single row of eight, and replaced with 2N5550 (2N5551) transistors. Resistors R71-R78 connected to them were changed to 470 ohm. This made the board a bit more robust. System11b games up to Swords of Fury used four displays (top two were 7-digit alpha-numerics, bottom two were 7-digit numeric) mounted on a single circuit board D-11610. System11b games Taxi and later used two 16-digit alpha-numeric displays mounted on a single circuit board D-12232. Some later games had the displays mounted separate (D-12308) from the master display board. Game EPROMs used a 27256 at U27 and a 27128 at U26. Starting with Swords of Fury U26 changed to a 27256 (though the later Taxi still used a 27128 at U26). Sys11b is a good MPU board, but can't be used in earlier games without adding some sound amp components back to the board. Can be used in sys11c games though without any mods.

System 11b CPU boards also had an option to replace the two original 6116 (2k) RAMs at U23 (sound) and U25 (game) with a 6264 RAM (8k) at either location. The original 6116 RAM comes in a 24 pin dip package, and the 6264 RAM comes in a 28 pin package (the four additional solder pads for the 6164 can be seen on every 11B and 11C CPU board). To install 6264 RAM chips, add two 28 pin sockets at U23/U25:

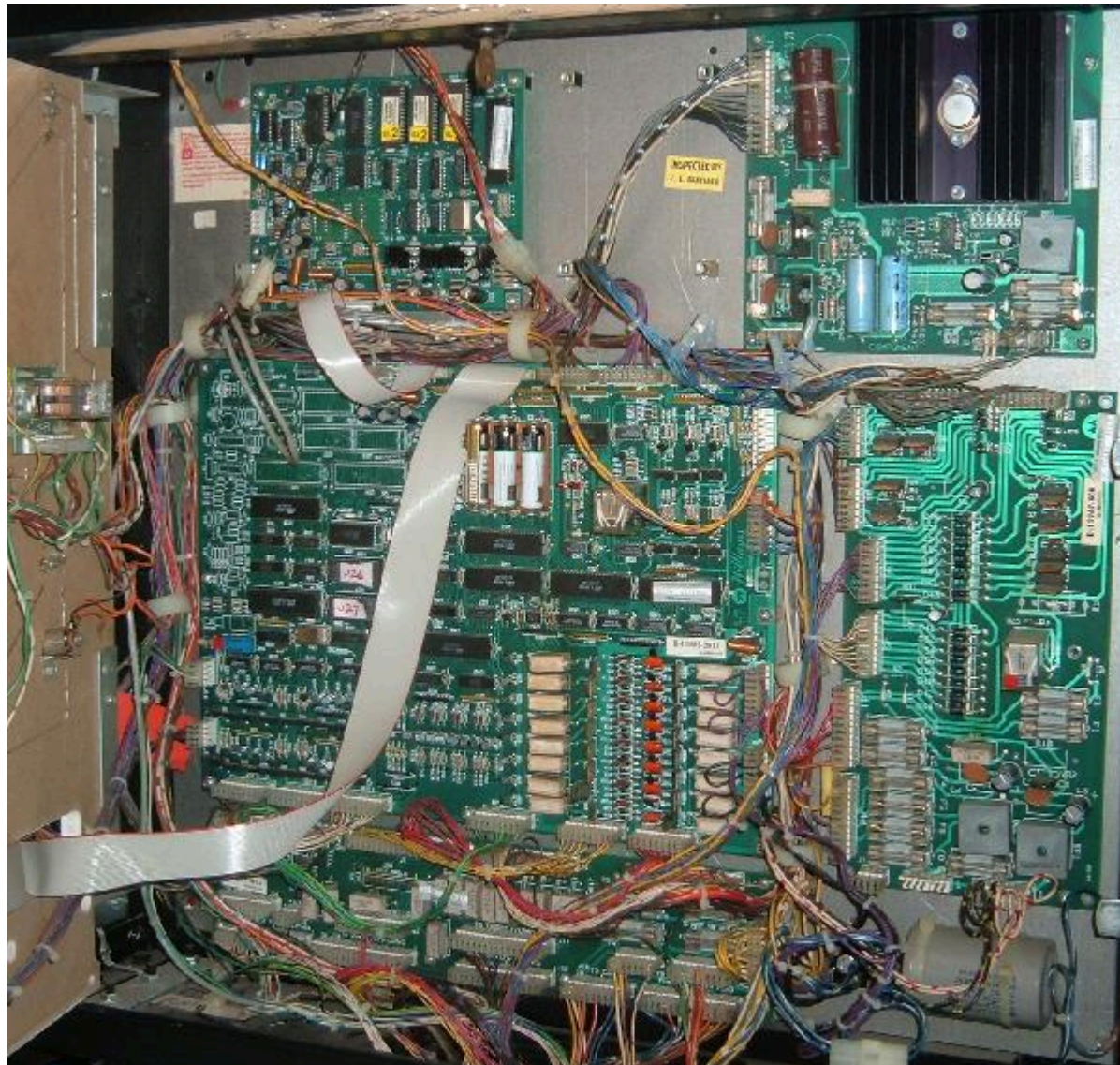
6264 at U23 (sound): W18=in, W19=out

6264 at U25 (game): W6=in, W5=out

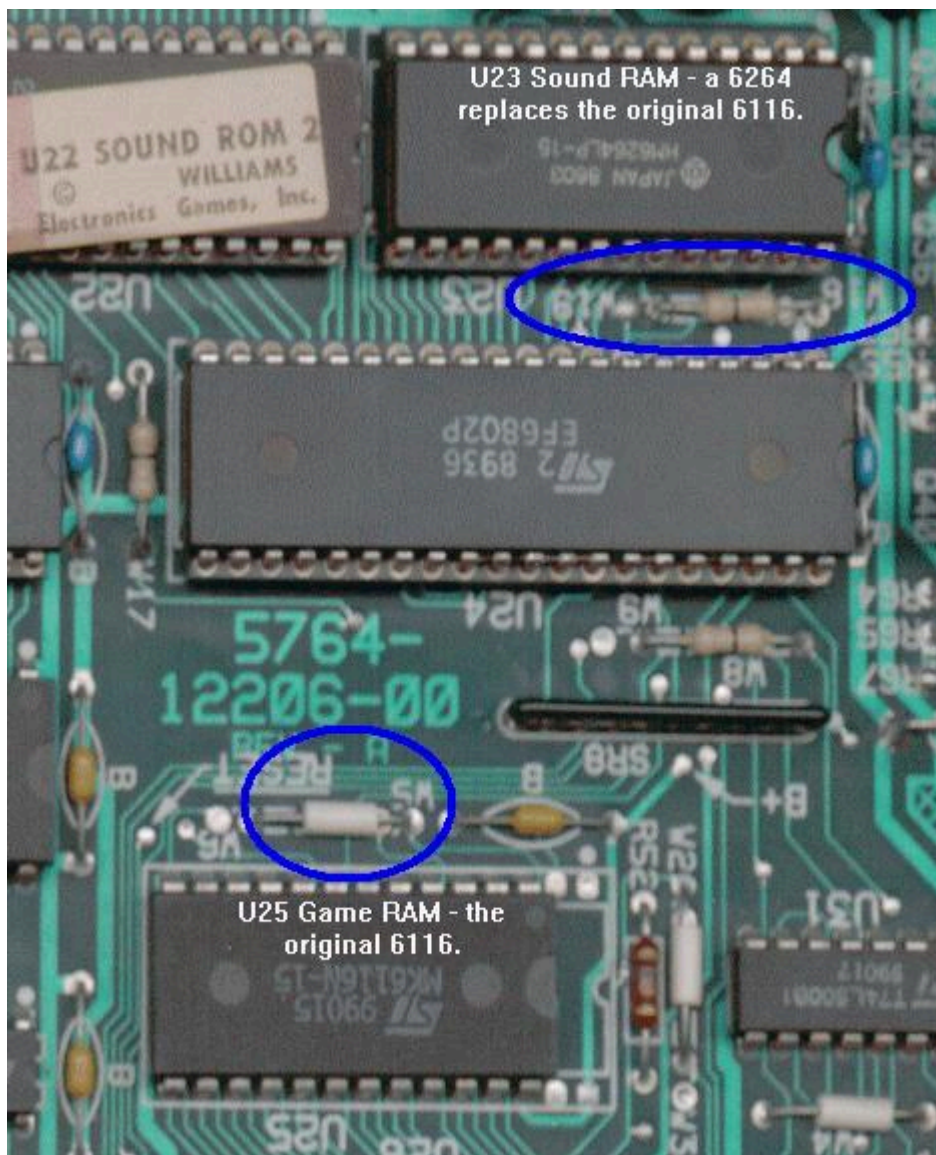
6116 at U23 (sound): W19=in, W18=out

6116 at U25 (game): W5=in, W6=out

A System11C CPU backbox from Pool Sharks. Note the lack of any sound circuitry or EPROMs in the upper left corner of the CPU board.



Here's a system11b CPU board that is using a 6264 for the U23 sound circuit, and a 6116 RAM for the CPU. To accomodate the 6264 at U23, jumper W18 was installed and W19 removed. Since the original 6116 game RAM at U25 is still used, jumper W5 is installed and jumper W6 is removed.
Picture by Frank.



- System 11C: fourth generation of system 11, part number D-11883-xxx (where xxx is the game number). Note this is the same CPU part number as system 11B! And also most system11c game manuals have system11b schematics, but with a note that the entire sound circuit was eliminated (all audio and sound now processed on the separate sound/audio board{s}). System 11c boards can *only* be used in system 11c games because all sound circuits have been removed from the CPU board. There are no sound ROMs at U21 and U22 and no 6802/6808 CPU at U24 and no 6821 at U9, and all of the other circuitry from the top left corner of the board has been moved to the D-11581 sound board. Because of this, the easiest way to tell a system 11C CPU board is the lack of sound EPROMs at U21 and U22. System11c games used two 16-digit alpha-numeric displays driven by the master display board D-12232. The displays were sometimes mounted separately on a different board D-12308. Game EPROMs used a 27256 at U27 and a 27256 at U26.

The least flexible CPU boards is system11c, since the sound circuit is completely removed, and a separate sound board is used. This board can only really be used in system11C games because of this. Populating the missing sound section on a system11C board would be difficult, as some of the sound chips are hard to find (and expensive). But the system11c CPU board can be used in many sys11/11a/11b games if the accompanying D-11581 sound is used too, though I haven't quite figured out how to do that (there are two sound ROMs on the System11b CPU board, and only one empty ROM socket on the D-11581 sound board, so some creative ROM combining may be needed).

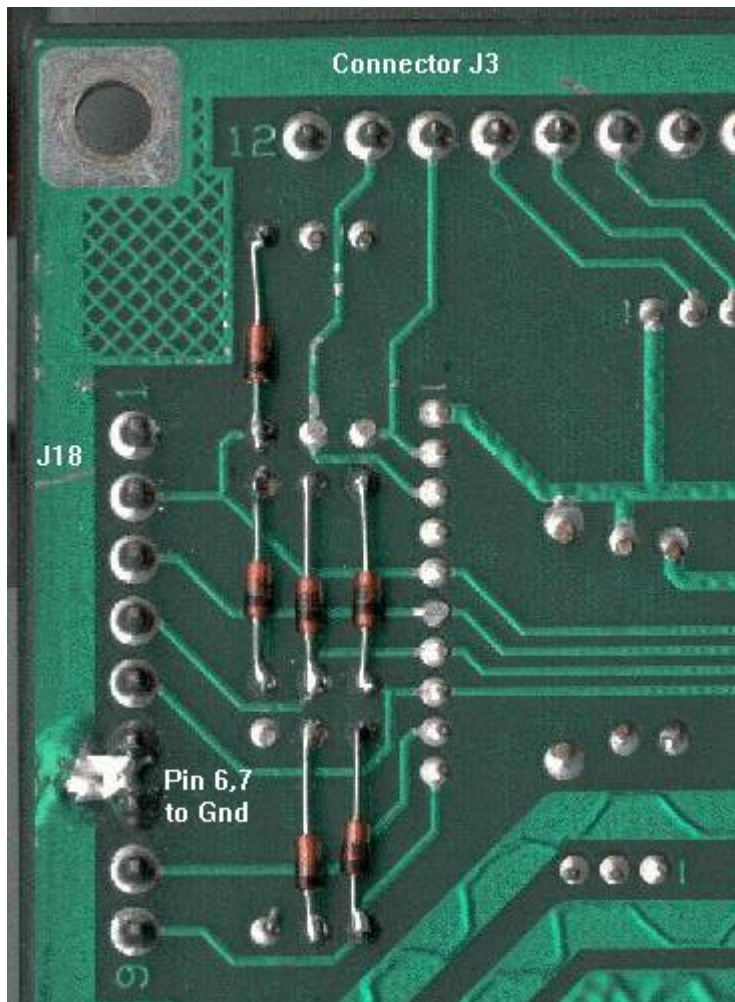
The most flexible CPU board is system11A. The system11A CPU board can be used in any 1986-1990 sys11, sys11b or sys11c game. The only exception is if the sys11A CPU board is going into an original sys11 game (High Speed, Road Kings, or Grand Lizard). In this case, connector 1J15 (four pins, .156" molex male header) must be added to the sys11a CPU board (that's all that needs to be done). To use a system 11a board in a system 11c game, don't use connectors 1J16 (volume control) and 1J15 (speakers). Also the sound ROMs at U21/U22 and CPU at U24 and PIA at U9 are not used, since they are moved to the sound board.

The original system11 CPU (as used in High Speed, Grand Lizard, Road Kings) is also very flexible, after the CPU board is modified slightly. It can work in any sys11, sys11a, sys11b or sys11c game with these modifications. CPU connector 1J18 pins 6,7 must be run to ground, otherwise the special solenoids will not work in sys11b and later games. Also there are six 1N5234 or 1N4735 6.2 volt zener diodes that should be added to the back of the CPU board for the special solenoid switch inputs. If using a system11 board in a system11c game, just don't use CPU connectors 1J16 (volume control) and 1J15 (speakers), and the sound ROMs U21/U22 and 6802/6808 U24 and PIA U9 are not used.

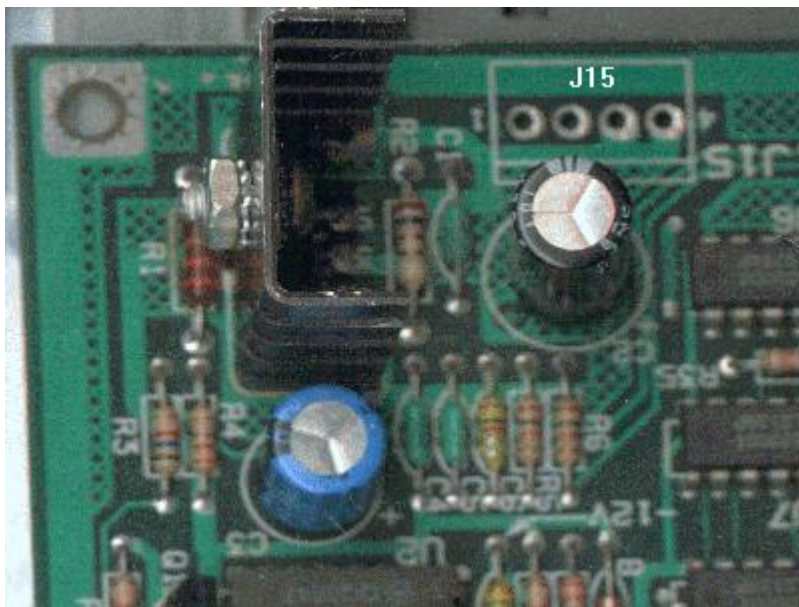
System11B CPU board are also very flexible, and can be used in any System11A or 11C game. The System11b CPU board is not quite as flexible as System11a or System11, because sys11b is missing the U1 sound amplifier and associated parts. The System11b CPU will work "as is" in a system11a or system11c game. But a sys11b CPU won't work in a System11 game (High Speed, Road Kings, Grand Lizard) unless the sound amplifier parts are added back into the sys11b CPU board. If using a system11b board in a system11c game, the CPU sound ROMs at U21 and U22 and 6802/6808 at U24 and 6821 at U9 are not used.

System11 CPU board Modifications:

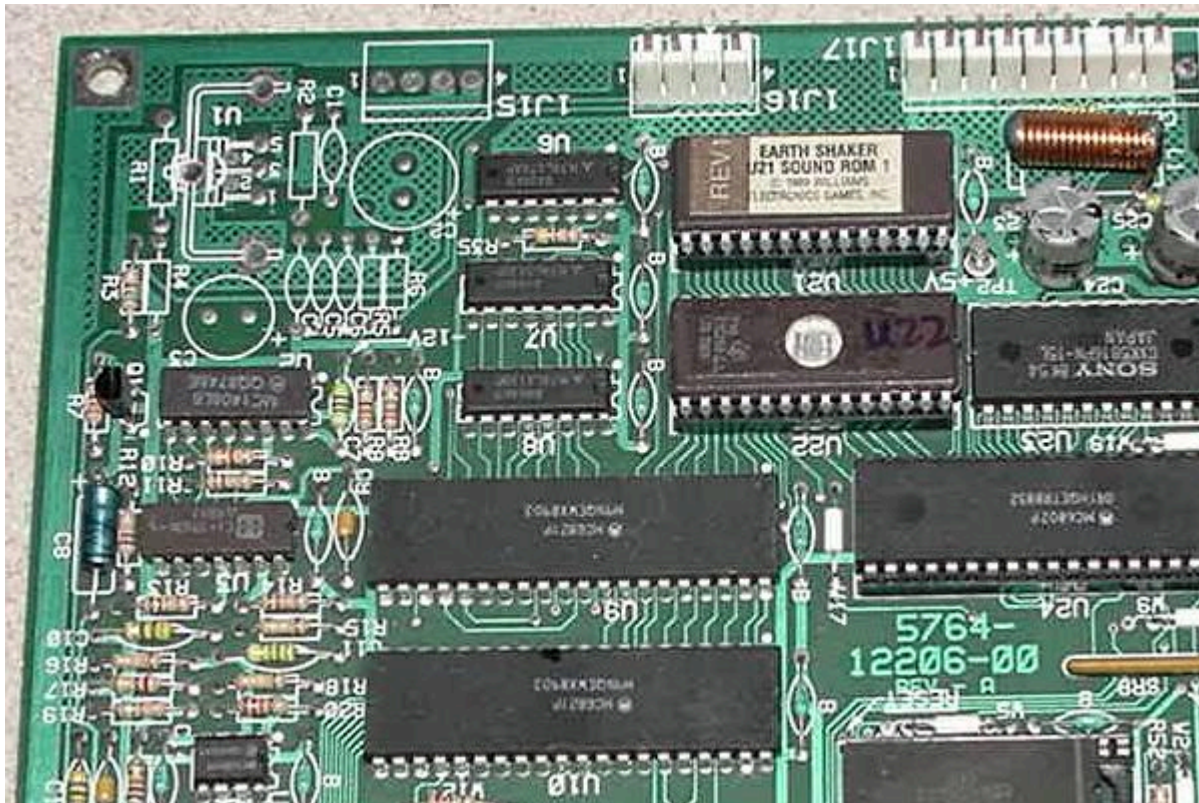
Modifying the original System11 CPU board with six 1N5234 or 1N4735 zener diodes. This is done for special solenoid switch input protection. Also J18 pins 6,7 should be jumped to ground, as shown here, so the system11 board's Special Solenoids will work properly in System11b and later games.



System11A sound section. Note missing 1J15 connector. The external sound board D-11581 has its own amplifier, so the CPU board's amplifier is not needed (hence the missing J15 connector). Picture by Frank.



System11B sound section. Note the missing U1 sound amp section which was removed since the external sound board has its own amp. This board from Earthshaker.



System11C missing sound section. It's all gone! The CPU board has no sound circuits. This board is from Rollergames.



Sound Board Generations.

For the first System11 games the external sound board was known as a "background sound" board, and the speech was handled by the CPU board's separate 6802/6821 at U24/U9 sound driven computer. Also the sound amplifier was on the CPU board. With System11a the external sound board D-11581 was a new board and more robust, with the sound amplifier used on this external sound board (though sys11a still had a CPU mounted amplifier, it was not used). The D-11581 sound board uses a 68B09E for the processor (and yes it must be a "B" and "E" version 68B09E chip to work, as the "B" gives a faster clock rate and the "E" gives an extended instruction set). With System11b, the D-11581 sound board had fewer parts (U6/U7/U18 not populated). With System11c all of the sound circuits on the CPU board was removed, and the external sound board did all the sound (and U6/U7/U18 were again populated). Note the sound board for system11a, b and c were the same board, but not necessarily with the same on-board components. The best version is the one with the most components (that is U6/U7/U18 populated).

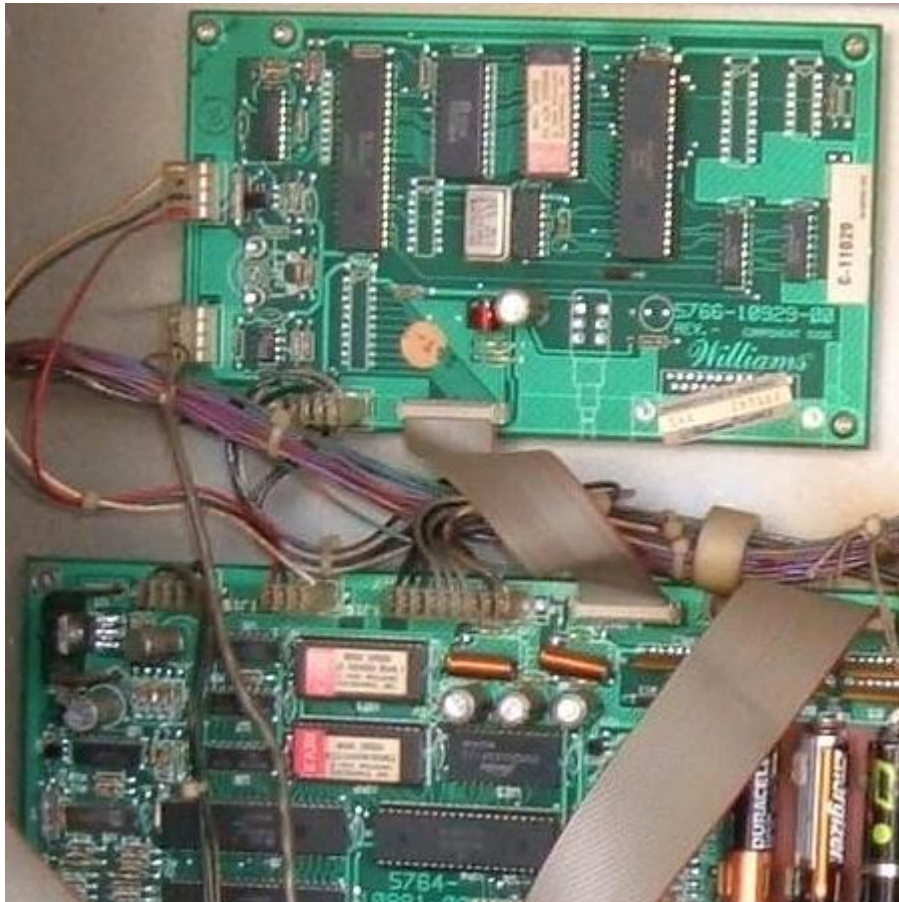
The D-11581 sound board has some jumpers, since the board can use 27128, 27256 or 27512 EPROMs. If W2 and W3 are removed (below chip U4), the board is set to use 27256 EPROMs. If W2 is installed and W3 removed, the board is set to use 27512 EPROMs*. With this information it may be possible to combine the Sys11a/Sys11b CPU board 27256 sound EPROMs into a single 27512, and the external sound board 27256 EPROMs U4/U19 into a single 27512. Then use a fully populated D-11581 sound board jumpered for 27512 EPROMs with a system11c CPU board in a system11a or system11b game (making the system11c CPU board just as versatile as an earlier system11 CPU board).

* Ok so maybe I lied about the sound board using 27512 EPROMs. It appears the system11c generation of the D-11581 sound board has two additional W10/W11 jumpers. For the board to be set to use 27512 EPROMs, W10 is installed and W11 is removed (in addition to W2 installed and W3 removed). The problem is on older D-11581 sound boards, there is no W10/W11 jumper. I believe this was the reason I could not get an older D-11581 to work with 27512 EPROMs. The schematics are of no help since this jumper is not shown. I believe the W10 jumper installed just ties all the EPROM's pin 1 together, and with W10 remove they are not tied together, and with W11 installed all the EPROM's pin 1 are tied to ground. But I may be wrong about that.

Another twist to the D-11581 is the U1 D/A (Digital/Analog) converter. This chips talks with the Yamaha sound chip at U3, using the sound EPROMs as data, and plays the digital music the game designer wants. But this U1 chip has a personality. As some may have noted, the Williams D-11581 sound board is also used in some Williams video games like Arch Rival. So it would be logical to think you could take an Arch Rival's D-11581 sound board, load it with system11 pinball sound EPROMs, and have a replacement sound board. Well you would think, but it appears that is not the case. The U1 D/A converter is different for Arch Rivals. I know because I tried to use an Arch Rivals D-11581 (27256 EPROMs) with Earthshaker EPROMs, and got some very interesting sounds. The sounds were very much different and more rap/drum oriented. It just did not work. Putting the Earthshaker U1 D/A converter in the Arch Rivals board fixed the problem, and the board then played the correct Earthshaker sound track.

If anyone has any solutions for the D-11581 27512 EPROM problem (correct jumpers) and the D/A converter problem, please contact me.

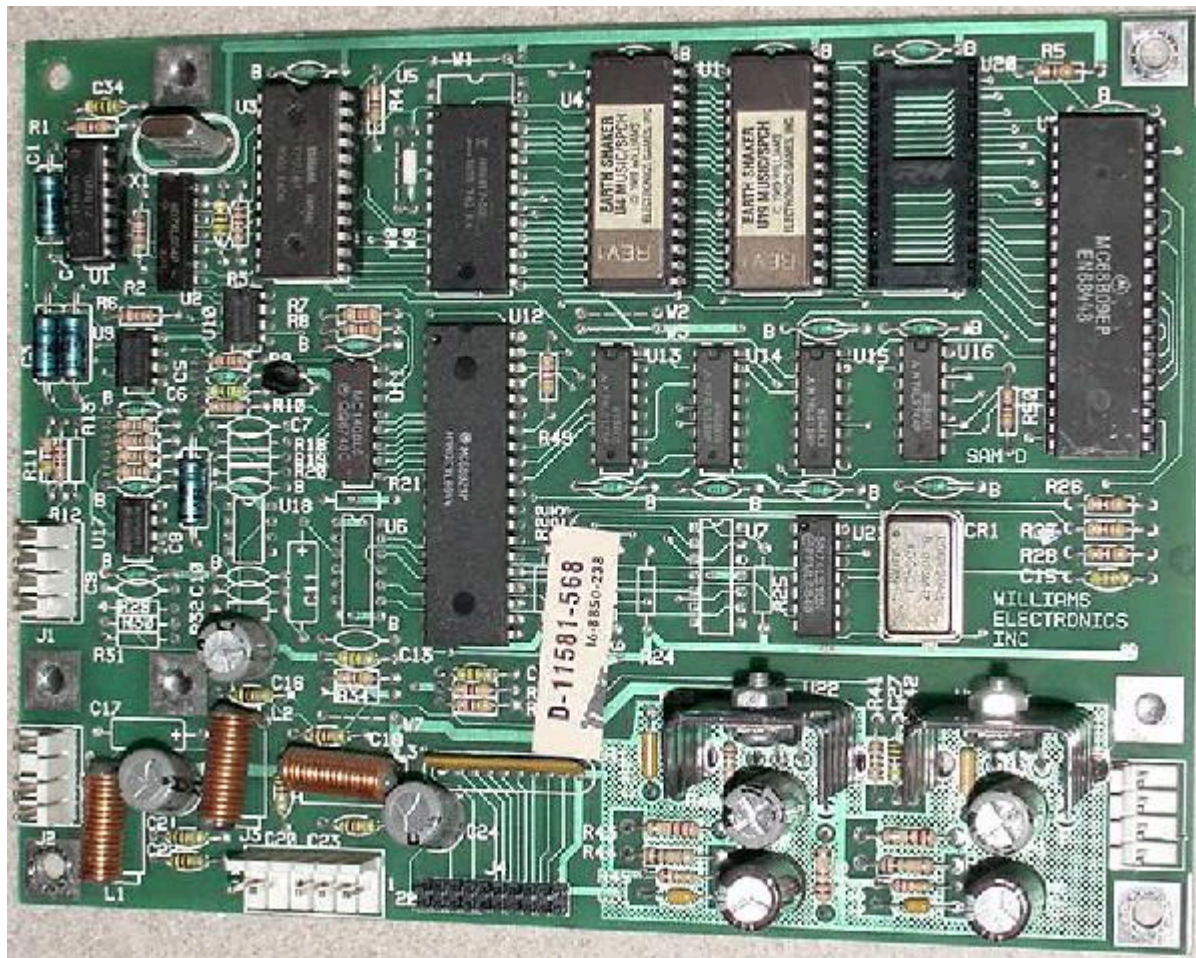
System11 background sound board. This board is from High Speed.



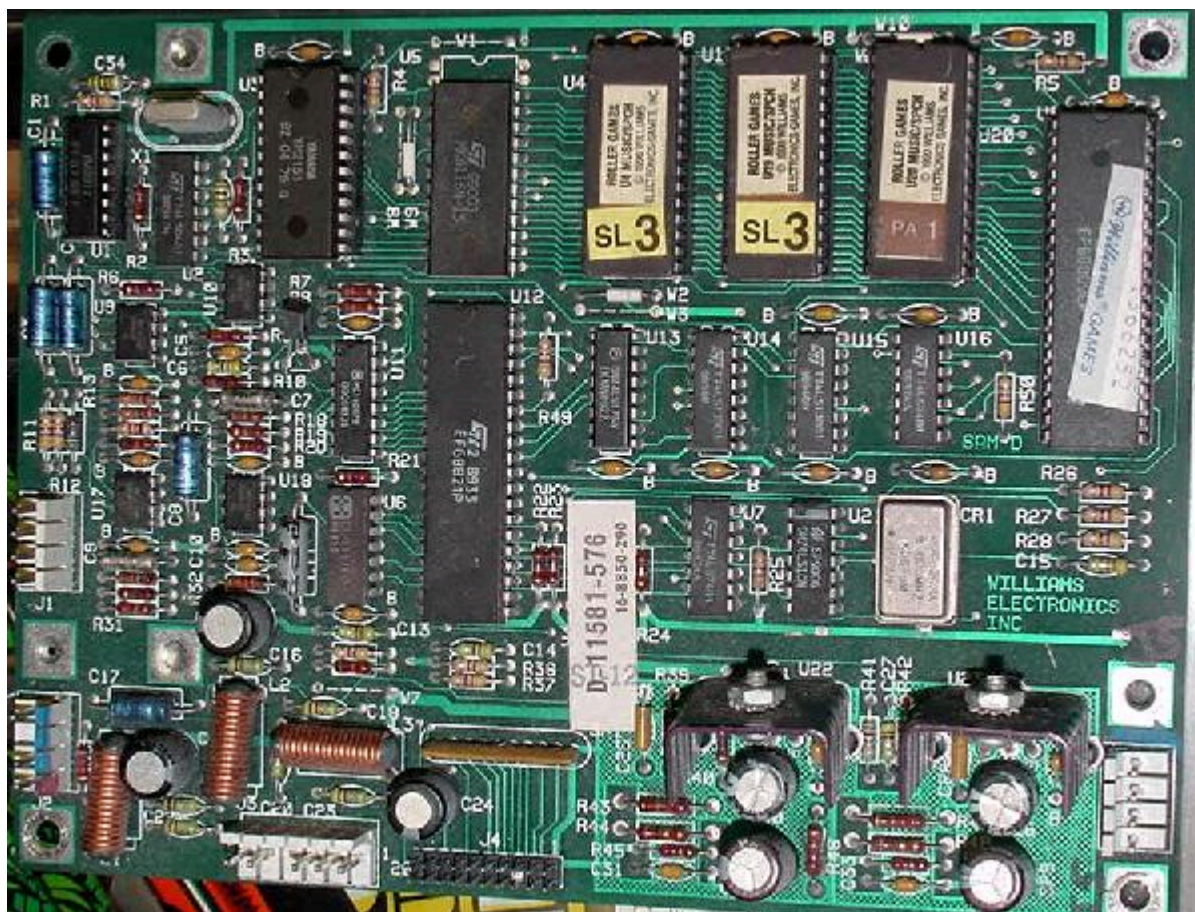
System11a sound board D-11581. This board is from F-14.



*System11b sound board D-11581 with U6/U7/U18 not populated.
This board has W2/W3 removed and is using 27256 EPROMs.
Board is from Earthshaker.*



*System11c sound board D-11581, but now it has more parts populated on the sound board. Note the installed W10 jumpers above chip U20, and the installation of jumper W2 so this board can use 27512 EPROMs.
This board is from Rollergames.*



English/German as the Default Language.

System 11 games used CPU jumper W7 (to the lower right of the battery holder) to specify the default language as English or German. If W7 is installed, English is the default. If W7 is removed, German is default. This was only used for early system11 games though. Later games (system11B/system11C) required both game ROMs U26 and U27 to be changed (language-specific versions), in addition to the jumper (but try the jumper first and if that doesn't work, then change the ROMs). Note the W7 jumper looks like there are two solder pads on the left side - really it's not that way, the W7 jumper goes to the outside left solder pad. Also if you install or remove the W7 jumper, you will need to remove the batteries to do a "hard reset" before the new W7 jumper takes effect.

Score Displays Used in System 11 Games.

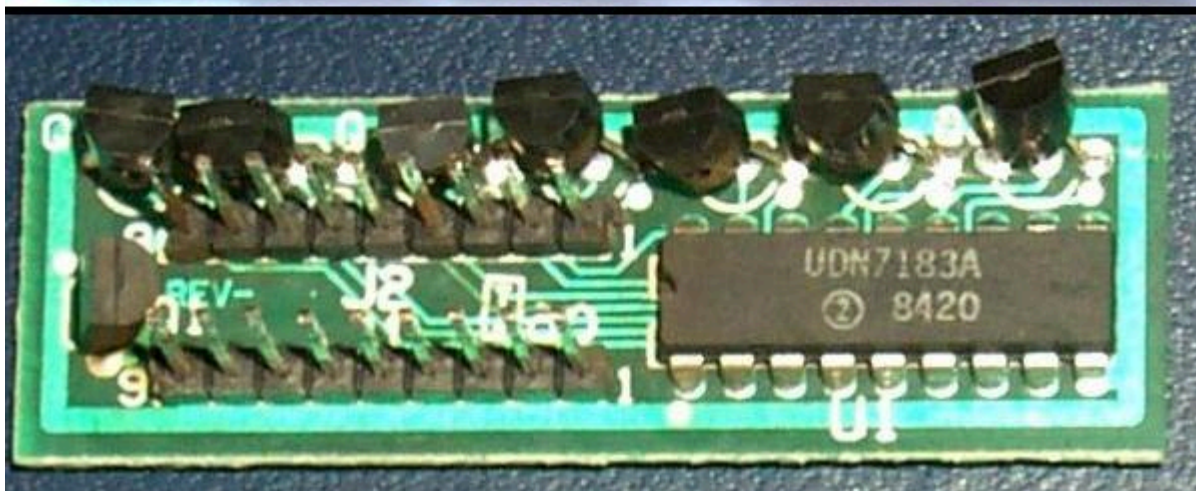
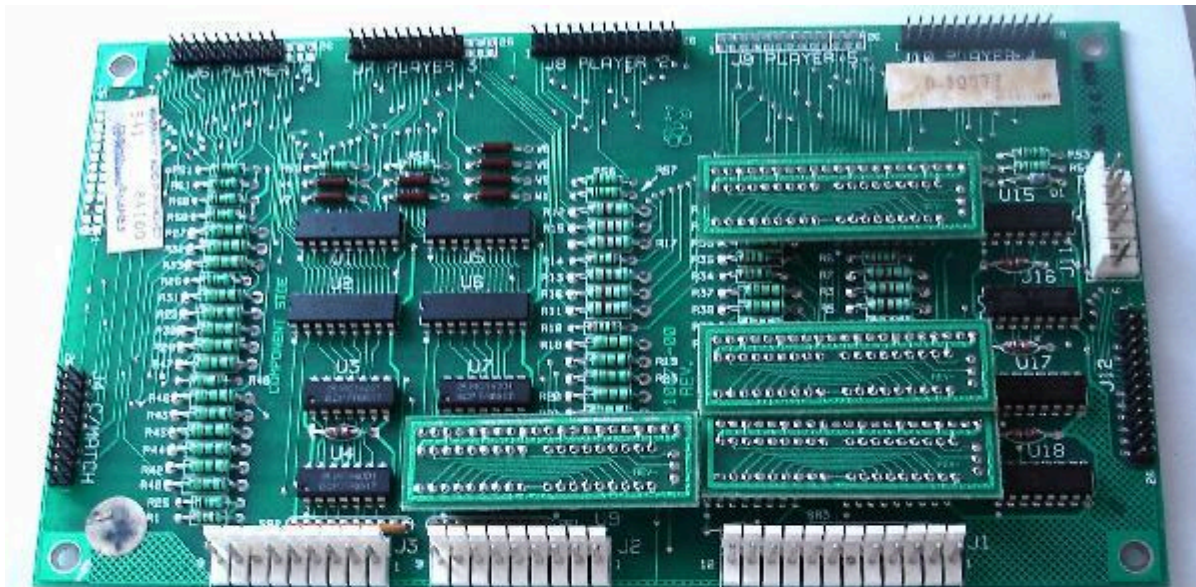
The score displays used in System 11 games varies. Initially (High Speed to Millionaire), Williams used five score displays: two 7 digit alpha-numeric displays, two 7 digit numeric displays, and one 4 digit numeric display (for the credits and ball in play). All five displays connect to a master display board via ribbon cables except on Pinbot and Millionaire (where all five display glasses were mounted on a master display panel board). Starting with F-14 Tomcat, Williams dropped the 4 character credit display, and all four display glasses were now mounted on a single master display board. The software now displayed credits and ball in play in the four displays (this saved some production costs). Then when Taxi came out, Williams switched to using two 16 character alpha-numeric displays.

If you are doing CPU board testing, the 4 display glass master control panel (D-11610) used from F-14 to Swords of Fury can be used on the earlier games High Speed to Millionaire (which used 5 display glasses). The only thing not seen will be the ball/credit display functionality. On newer games Taxi and later that uses two 16-character alpha-numeric glasses, the older display panels can not be used (the panels are plug compatible, but the resulting text on the 4 independent display glasses is not readable).

High Speed/Grand Liz/Road Kings score displays. The glass and the controller board are separate from each other and connected by ribbon cables. Note the 4-digit ball/credit display.



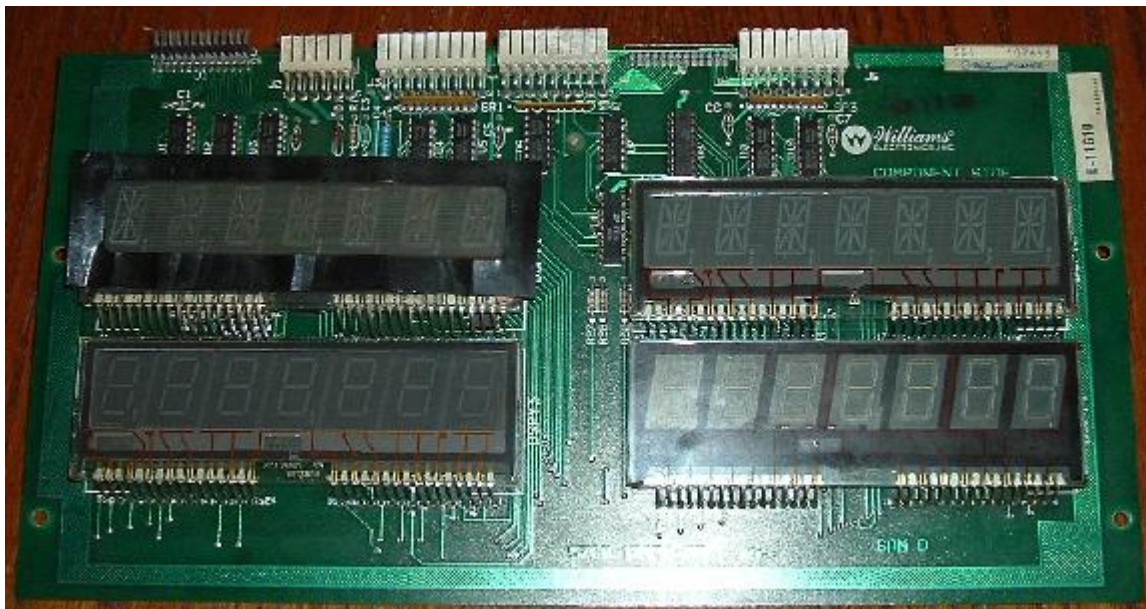
D-10977: This control board was sometimes used for High Speed/Grand Liz/Road Kings. Controls two 7-digit alphas, two 7-digit numerics, and one 4-digit ball/credit display. There are no 7180 chips on this board. Instead discrete MPS-A92 transistors and UND7183 chips were used. Notice the small rider boards.



D-15414: Millionaire/Pinbot style master display board with two 7-digit alpha-numerics, two 7-digit numerics, and a 4-digit ball/credit display (D-15414). Now controller and displays mounted on a single board. Picture by B.McCord.



D-11610: F-14 to Swords of Fury style master display board with two 7-digit alpha-numerics and two 7-digit numeric displays only (D-11610). Ball/credit display no longer used.



D-12232: Taxi and later style master display board with two 16-digit displays. Picture by B. McCord.



Power Supply.

High Speed to Swords of Fury (pre-Taxi) used the same D-8345 power supply as System9 games

(with a slight modification to the GI power supply connectors). This power supply output -12, +12, +5, -100, +100. In addition +18 (lamp matrix) and +25 (coils) power is turned around through the power supply to implement fuses (though it's not rectified by the power supply). A separate small "flipper" power supply board rectifies/outputs 50 volts for high power coils.

When the Auxiliary Power Supply board was introduced with Big Guns, the D-8345 power supply was not fully populated. For Big Guns the lamp matrix and 25 volt coil connectors were removed, as were these associated fuses, but the GI connectors were still present. By Banzai Run, the GI relay/connectors were gone, and the D-8345 power supply only had three fuses (one for high voltage, two for the 5/12 volt bridge rectifier). This under populated D-8345 power supply was used on Big Guns, Space Station, Cyclone, Banzai Run, Swords of Fury). The lower section with the GI connectors (on the lower right) was left blank (no GI relay or GI connectors). Also the 18 volt lamp matrix connector/fuse was removed, as was the 25 volt low-power solenoid connector/fuse.

With Taxi Williams changed to a smaller power supply board #D-12246 that outputted -12, +12, +5, -100 and +100 volts DC and had just three .156" Molex connectors (no longer was the rectangle 12 pin input connector used). This was done so the new power supply could not be used in pre-Taxi games. All other voltages (18 volts lamp matrix, 25 volt low-power coils, 50 volt high power coils) was generated and fused by the Auxiliary power supply board, so there was no need for this to occupy space on the new smaller power supply board.

Rottendog sells replacement system11 power supplies for around \$70, but they only work on pre-Taxi games as of 11/05 (Taxi and later games have different connectors and are not plug compatible with the Rottendog power supply).

Williams Schematics for System 11 Games.

Interestingly, Williams often didn't print their system 11 game manual schematics with the correct system 11 board! For example, many system 11c games had system 11b schematics printed in their manuals instead. For example, the Diner game manual has system 11b schematics, even though it's a system 11c game. There is usually a note at the bottom of the schematics about the sound section of the board that says, "removed in certain assemblies" to signify this. Keep this in mind when reading the schematics.

1e. Getting Started: Game List

Here are the list of games and their system generations. This is important to know before you begin repair.

Williams System 9

System9 games use five displays, four 7-digit numeric, one 4-digit numeric credit/ball display. Master display controller board separate from the displays.

- Star Light (some Star Light games used System7 boards and some used System9).
- [Space Shuttle](#), 12/84, #535
- Strike Zone shuffle alley, 1985. This is a system9 bowler. Uses SIX digit numeric displays (unlike all the other system 9 games which use Seven digit numeric displays). To do this, the Strike Zone uses the old system3-6 style master display board. So this shuffle alley is a combination of system9 CPU board and system3-6 displays/master display boards. The game does have speech.
- Sorcerer, 3/85, #532
- Comet, 6/85, #548

Williams System 11

Five displays, top two 7-digit alpha-numeric, bottom two 7-digit numeric, one 4-digit numeric credit/ball display. Master display controller board separate from the displays.

- [High Speed](#), 1/86, #541
Master display board D-10877. Does not use multiplexing, five score displays. Background music board D-11297.
- Alley Cats, 2/86
This is a Bowler, not a pinball game, but uses system11 boards.
- Grand Lizard, 3/86, #523
Does not use multiplexing. Background music board D-11297.
- Road Kings, 7/86, #542.
Sound board D-11298, hence the first game to use a Yamaha YM2151 sound chip on the sound board. Master display board D-10877.

Williams System 11a

- Pinbot, 10/86, #549.
Master display board D-15409 or D-15410 (still five score display glasses). Sound board

D-11298.

- Tic Tac Strike, 1986, bowling shuffle alley game.
- Millionaire, 1/87, #555.
Master display board D-15410. Last game with five score display glasses.
- [F-14 Tomcat](#), 3/87, #554.
First game with the new FL11630 parallel wound flipper coil, which used two flipper diodes (instead of one). The 4-digit ball/credit score display was dropped. Master display board D-11609 or D-11610 with just four display glasses, and the displays and controller board are combined onto one board.
- [Fire!](#), 8/87, #556
- Big Guns *, 10/87, #557.
First game to have Auxiliary power driver board and Special solenoids CPU controlled.

* System 11b started in the middle of "Big Guns" production. So "Big Guns" can have either System 11a or 11b boards.

System 11b

- Big Guns *, 10/87, #557, sound board D11581.
First game to have Auxiliary power driver board and Special solenoids CPU controlled.
- Space Station, 1/88, #552, sound board D11581 rev D or Sound board D-11298.
- Cyclone, 2/88, #564, sound board D11581 rev D.
- [Banzai Run](#), 7/88, #566, sound board D11581.
First game with an interconnect board, but the interconnect board used on this game is unique (D-12112). Master display board D-10877 with display board and display glass mounted remotely. This display board is unique to this game too.
- Swords of Fury, 8/88, #559, sound board D11581 rev D.
- [Taxi](#), 10/88, #553, sound board D11581.
First game with only two alpha-numeric 16 character displays and master display board D-12232 and displays mounted on board D-12308, new power supply D-12246. Also uses a special display board with another ribbon cable connector, which attaches to the third display.
- Jokerz, 1/89, #567.
This game only used a special stereo sound board D-12338 with different sound board power requirements and cabling. Also CPU board jumpers W1,W2,W4,W5,W7,W8,W11,W14,W16,W17,W19 must be installed.
- Earthshaker, 4/89, #568, sound board D11581.
- [Black Knight 2000](#), 6/89, #563, sound board D11581.
- Police Force, 9/89, #573. Uses the same special display board with another ribbon cable connector, which attaches to the third display, just like Taxi.
- [Transporter the Rescue \(Bally\)](#), 6/89, #2630
- [Elvira and the Party Monsters \(Bally\)](#), 9/89, #782, sound board D11581.
- Bad Cats, 12/89, #575
- Mousin' Around (Bally), 12/89, #1635, sound board D11581.
- Whirlwind, 4/90, #574, sound board D11581 rev D.

System 11c

- Game Show (Bally), 4/90, #985, sound board D11581.
- Roller Games, 5/90, #576, sound board D11581 rev D.
- Pool Sharks (Bally), 6/90, #1848, sound board D11581.
- [Diner](#), 8/90, #571
- Radical (Bally), 9/90, #1904
- Riverboat Gambler, 10/90, #1966, sound board D11581 rev D.
- Bugs Bunny Birthday Ball (Bally), 11/90, #396
- Dr.Dude (Bally), 11/90, #737
About 100 Dr.Dude games were made with the new WPC system boards.

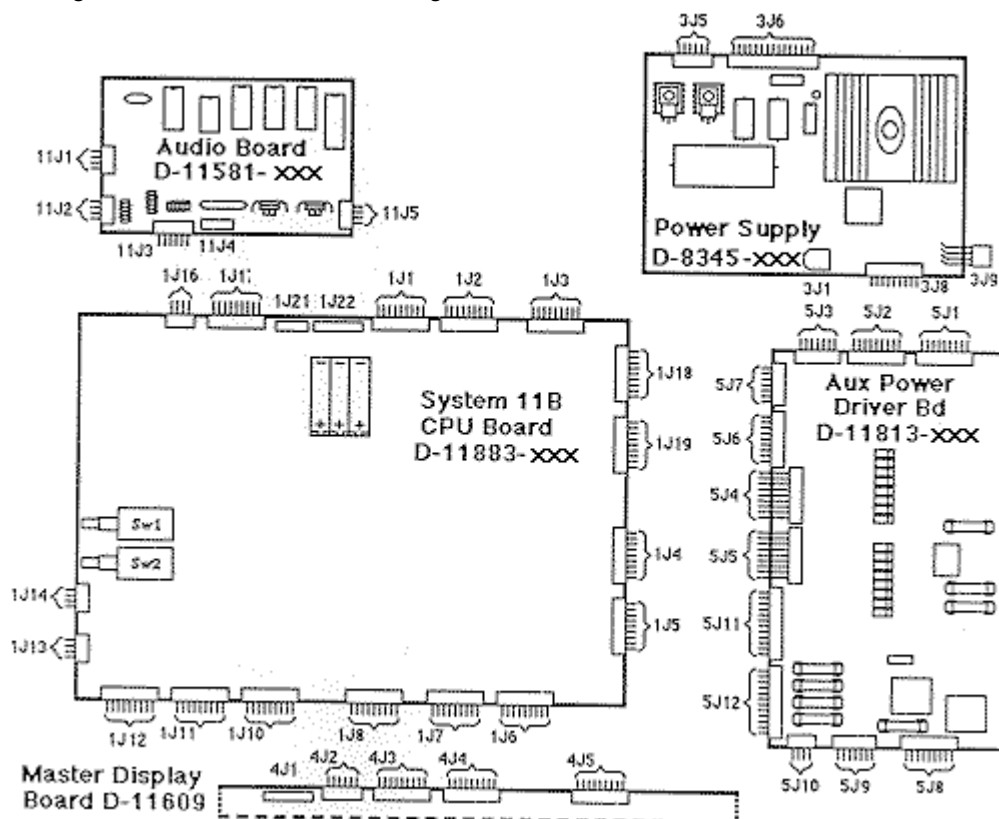
* Sound board D11581 rev D has resistor R18-21 and chips U6,U18 installed (where the prior versions do not).

1f. Getting Started: The Circuit Boards and How they Work

Both System9 and System11 uses a combination CPU and driver board (known as the "CPU" board). It contains all the logic and TIP122/102 driver transistors for the game. Starting with "Big Guns", an Auxiliary power driver board was added which houses another eight TIP36c transistors

for high voltage applications (but these TIP36c transistors are pre-driven by one of the existing TIP122/102 transistors on the CPU board).

The system 11 board layout, Big Guns and later. Games prior to this do not have an Auxiliary power driver board. Instead a smaller Flipper power supply board is in the same position. Note the board numbers ending in "xxx"; the three x's are the game number.



System 11 Board/Connector/Pin Numbering.

A prefix number precedes all connector "J" numbers. This prefix signifies which board the connector belongs to. The number immediately after the "J" is the connector number for that board. The number after a hyphen is the pin number for that connector. For example, connector 1J8-5 would be CPU board connector 8, pin 5. Here are a list of board number prefixes:

- 1 = CPU board.
- 2 = Interconnect board (Banzai Run and later).
- 3 = Power Supply board.
- 4 = Master Score Display board.
- 5 = Auxiliary power driver board (Big Guns and later), or Slave display board on early games.
- 6 = Backbox wiring
- 7 = Cabinet wiring
- 8 = Playfield wiring
- 9 = Insert board.
- 10 = Sound board.
- 11 = Audio board.
- 15 = 50v power supply board (Fire! and before).

Switched Solenoids (Multiplexing).

The basic concept of system 11 is the use of a switched solenoid ("multiplexing") design. Solenoids 1 to 8 (Q22-Q25 and Q30-Q33) have two banks: the "A" and "C" banks. This allows one TIP122/102 transistor to control two functions (usually multiplexed between a solenoid and a flasher). The eight bank selected TIP122/102 transistors can control 16 functions. A bank select relay controls which of the two functions any of the eight driving transistors control. The "A" bank consists primarily of coils. The "C" bank consists primarily of flasher lamps. Not all games use the multiplexing technique. High Speed and Grand Lizard (the first two system 11 games) do not multiplex. The games after Grand Lizard however required more driver transistors, and multiplexing was used to allow this.

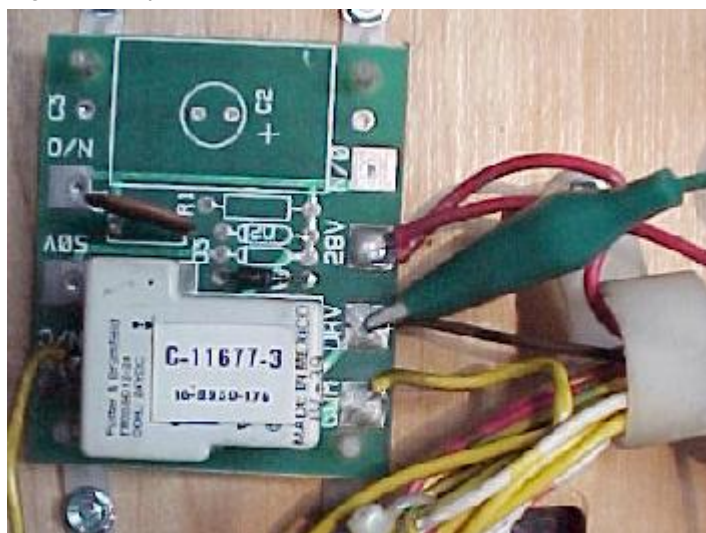
Multiplexing works like this; when the bank select relay is de-energized, solenoid power (25 volts) is connected to bank "A". Then only solenoids 1A to 8A can be driven by the driver transistors.

later 50 volts was used on pop bumpers and other random coils as needed like vertical up kickers. This was a sign of things to come, because the later WPC games went to 50 volts for all coils.)

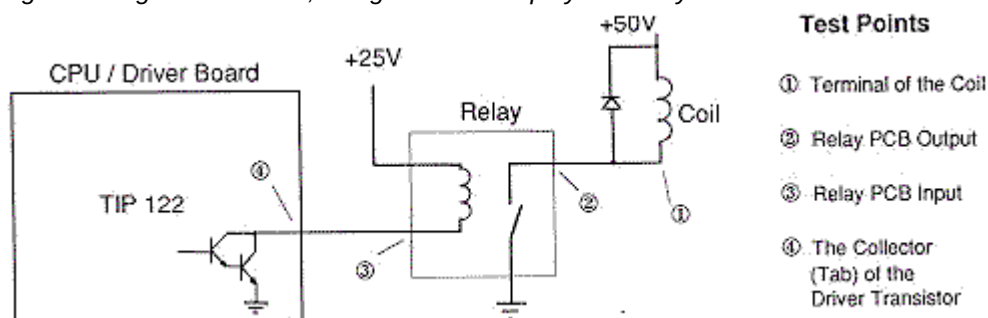
Unfortunately the CPU board was not designed to handle both 25 and 50 volts. On system11 and 11a games before Big Guns, a small relay board was often located under the playfield to drive the 50 volt coils. The CPU board's TIP122/102 transistor would activate the 25 volt relay on the relay board, completing the circuit for the 50 volt coil. The relay is really acting like a mechanical version of a TIP36 transistor (which was later used instead of the relay). On games Big Guns and before, there was a 50 volt fuse in the backbox for these coils too, and the 50 volts was fed from a small 50volt Power Supply board.

Starting with Big Guns, the Auxiliary Power Driver board was incorporated, which used eight TIP36c transistors. This meant that Williams no longer needed the under the playfield relays to drive the 50 volt coils. The CPU board's TIP120/122 was now used as a pre-driver for the Auxiliary Power Board's TIP36c transistors, which handled the 50 volts, and hence the relay boards were no longer used. Usually the pop bumpers and slingshots were 50 volts (five coils), leaving three more TIP36c transistors for other high powered coil functions. Also the Auxiliary Power Supply board has two bridge rectifiers (25 and 50 volts), and eight fuses just for the TIP36 controlled 50 volt coils.

The under the playfield relay board used on games before Big Guns to power the 50 volt coils.



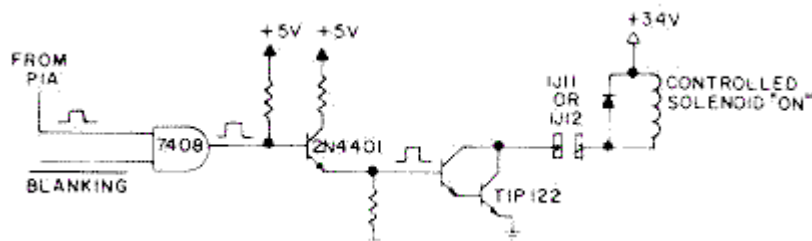
Robert Snicer's drawing of how early system 11 games handled the higher voltage 50 volt coils, using a under the playfield relay board.



Controlled Solenoids.

There are also eight "controlled solenoids" on system 11 games, in addition to the above "switched" solenoids. These are solenoids 9 to 16. These can control only one device (unlike the switched solenoids which are multiplexed). Note if your game doesn't use the above solenoid bank switching (like High Speed and Grand Lizard only), all solenoids are considered "controlled" solenoids.

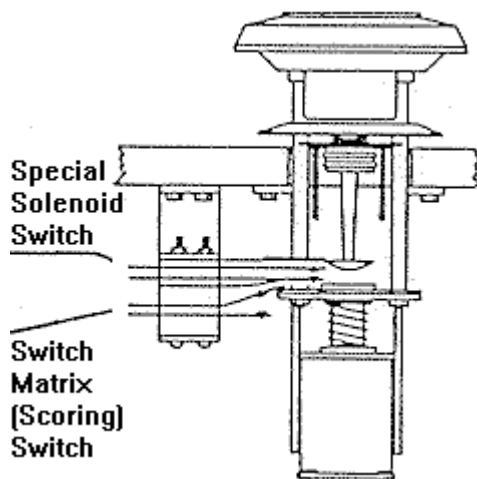
Williams' diagram of the controlled solenoids "On" state logic.



The Six Special Solenoids.

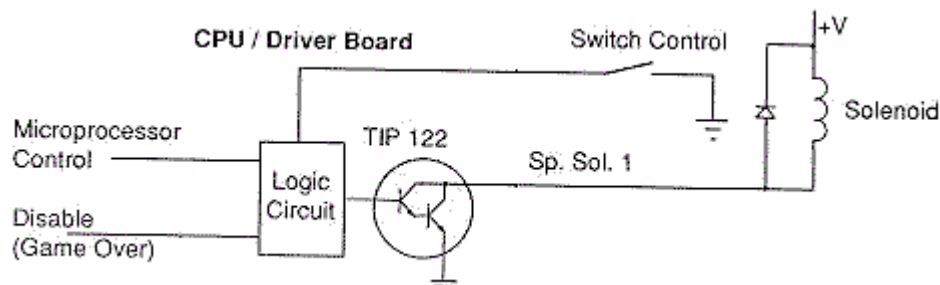
Williams' ROM software originally did not have CPU control of "special solenoids". These six solenoids includes the pop bumpers, kicking rubbers (slingshots), and kickback solenoids, and sometimes other coils. The original theory behind this was these items needed instant response. Having them controlled by the CPU would add enough of a delay to slow the solenoids down. After all, the CPU had to sense a switch closure of a pop bumper, then turn on a transistor which would fire the pop bumper coil. All this while the CPU was doing all the other chores it does (like scoring and controlling other devices). It was thought that this would not happen fast enough for good game play.

Norbert Snicer's diagram of a non-CPU controlled special solenoid pop bumper. Note the two switches; one to activate the pop bumper, and one connected to the switch matrix which controls the scoring and sound.



The down side to this was when a pop bumper switch was stuck on, the coil would "machine gun" (quickly pulsing on and off), and eventually either burn the coil or blow a fuse. On early system 11 games, the coils may not "machine gun", but instead may just lock on and burn the coil. This system also required a redundant switch, connected to the switch matrix for each coil, which controlled the scoring for this device. Also each special solenoid had to have its own fuse.

Norbert Snicer's diagram of a special solenoid. Note that either a direct playfield switch or the CPU can activate the logic circuit which turns on the TIP122/102, and activates the coil needed.



It should be noted that the system 11 special solenoids do have a TIP122 driver transistor (Q75, Q71, Q73, Q69, Q77, Q79). The only different between the special solenoids and other system 11 solenoids is how they are activated. "Normal" system 11 solenoids have a switch on the switch matrix which the CPU monitors. If the switch is closed, the CPU then activates the solenoid and scores the points. The special solenoids can also be used like this too. But they also have an option of having a switch on the playfield that directly controls the solenoid through chip logic, instead of the CPU. If the playfield switch is closed, this automatically activates the special solenoid's driver transistor through hardware logic, which fires the coil. If this switch gets stuck on,

the solenoid coil will lock on or "machine gun" (rapidly energize and de-energize). A CPU controlled solenoid switch will only fire the coil once when the switch is stuck (thus saving the coil/fuse).

Eventually Williams changed their mind on these "special solenoids", and made them CPU controlled. This means regardless of how long a solenoid's switch was closed, the solenoid would be energized only once by the CPU, and for a pre-determined time. So if a switch got stuck closed, the coil would not lock on or "machine gun" and burn (or blow a fuse). Williams started this with their last system 11a game "Big Guns".

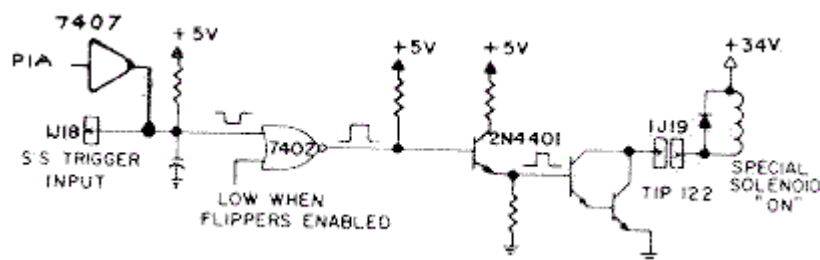
Special Solenoid Logic Flow.

For Special Solenoids (SSa to SSf), here is the logic flow. This is useful to know if you are having a problem with a special solenoid.

SSx:	6821 PIA	7407	7402	2N4401	TIP122
SSa:	U38 (pin 39) to U49	to U45	to Q74	to Q75	
SSb:	U41 (pin 39) to U49	to U45	to Q70	to Q71	
SSc:	U41 (pin 19) to U49	to U45	to Q72	to Q73	
SSd:	U38 (pin 19) to U49	to U45	to Q68	to Q69	
SSE:	U54 (pin 19) to U49	to U50	to Q76	to Q77	
SSf:	U54 (pin 39) to U49	to U50	to Q78	to Q79	

Since pre-Big Guns games use hardware logic to fire the solenoids, there are some other smaller (and easily damaged!) components that can fail too. Check capacitors C70 to C75 (.01 mfd), and resistor network SR20 (4.7k). If these become damaged, a special solenoid can "stick on". Even if your game is Big Guns or later (CPU controlled special solenoids), damage these components can cause problems.

Williams' diagram of the special solenoids "On" state logic.



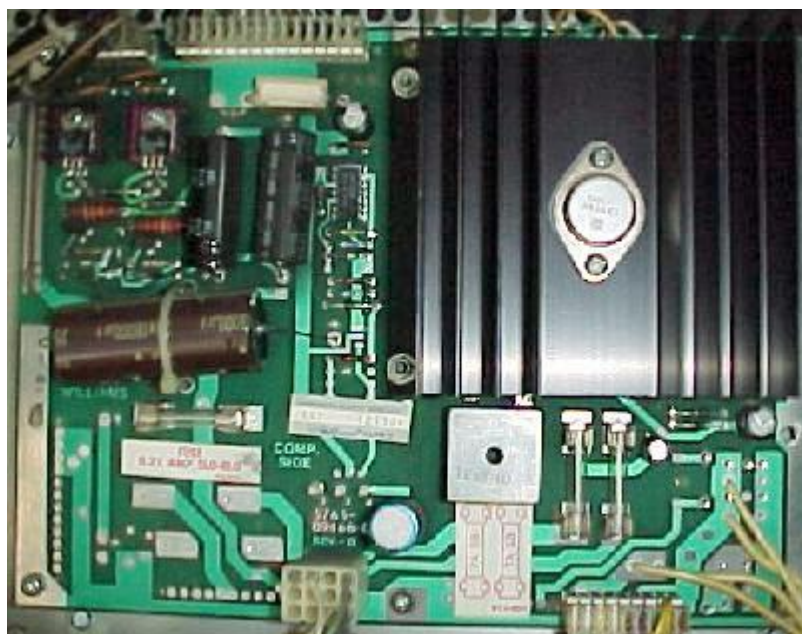
CPU Board Connector 1J18 (Special Solenoid Switches)

If your system 11 CPU board has no connector at plug 1J18, the special solenoids are CPU controlled. This connector is used for the "special switches" which control the special solenoids on games Fire! and before. This connector is no longer used on games Big Guns and later, because the special solenoids are now CPU controlled.

Power Supply board.

The power supply board provides +5 volts for all circuit boards. It also produces the +100 and -100 volts for the score displays. It also acts as a liaison for the +18 volts for the lamp matrix (no circuit, just a fuse) and the general illumination (GI) on games before Banzai Run. The power supply board does some mild processing of the solenoid voltage too.

The System 11 power supply D-8345-xxx (where xxx is the game number). This particular power supply version was used from Big Guns to Cyclone, because it lacks the relay in the lower right corner for the GI, but still has the GI connectors (that were moved to the Interconnect board on Banzai Run and later games).



There were basically four different system 11 versions of the power supply. The first version was D-8345-xxx (where xxx is the game number). This was used from High Speed to Swords of Fury. On these power supplies there are three different sub-versions of this board, all with the same prefix (D-8345) number:

- High Speed to Fire! (no Auxiliary power supply board, no Interconnect board): has GI connectors and GI relay on the power supply board.
- Big Guns, Space Station, Cyclone (Auxiliary power supply board, no Interconnect board): has GI connectors, but no GI relay (moved to the Auxiliary power supply board).
- Banzai Run, Swords of Fury (Auxiliary power supply board, Interconnect board): no GI connectors (moved to the Interconnect board), no GI relay (moved to the Auxiliary power supply board).

Starting with Taxi, a new power supply was used, number D-12246.

Low +5 volts from the Power Supply board.

A low +5 volts coming from the power supply can often be attributed to the 47 mfd 50 volt capacitor at C8. If this cap fails, a low +5 volts will result. If the large filter cap C10 (18,000 mfd 20 volts) wears out, this can cause low or bad +5 volts also.

The auxiliary power driver board started with the last System 11a game (Big Guns). Note the two bridge rectifiers at the bottom of the board (for the +50v and +25v solenoid voltages), the yellow bank select relay in the middle, the 16 diodes in the middle for the bank controlled solenoids, and eight TIP36 transistor at the top half to control eight +50 volt devices.



System 11's Auxiliary Power Driver board.

Starting with Big Guns, system 11 games had an additional "auxiliary power driver" (APD) board. This board was used to hold the:

- Bank select circuit and bank select A/C relay (NTE R14-11D10-24P).
- Two bridge rectifiers that supply the solenoid power (both 25 and 50 volts).
- Driver transistors (TIP36) for the 50 volt devices. This eliminated the small relay boards under the playfield. Up to eight TIP36c transistors can be used on this board. But if a game didn't use all eight, only the number used was actually installed!
- Eight fuses for the TIP36c controlled solenoids.
- Coil diodes. The coil diodes were no longer mounted on the coils themselves, but were instead installed on this board. This eliminated diodes breaking off the coils due to vibration. It also simplified coil replacement, as the operator no longer had to worry about which coil lead the diode's band was attached to on the coil.
- Solenoid fuses. The 50 volt coil fuses were mounted on this board.

With the addition of the auxiliary power driver board, the backbox became much neater.

Zero Ohm "Resistors" on the Auxiliary power driver board.

On many of the games with the new Auxiliary power driver board, Williams used zero ohm resistors as jumpers on this board. The reason they used these "resistors" was they could be automatically installed into the circuit boards with production machines before being wave soldered. Unfortunately, sometimes these zero ohm resistors fail and go open. For this reason, **system 11 games used actual wire jumpers instead of these zero ohm resistors. Check W1, W3, W4 and W6 for open zero ohm resistors on the APDB.**

The Interconnect board. In this game (Elvira), the interconnect board is mounted below the CPU board. Sometimes it's mounted on the side wall of the backbox.



The Interconnect Board.

Starting with Banzai Run, Williams started using an "interconnect board" inside the backbox (note

the Banzai Run interconnect board is unique compared to other interconnect boards). Often this long, skinny board was mounted beneath the CPU board, or on the sides of the backbox. This board was a "landing zone" for lamps (GI, controlled lamp power, flashers), the switch matrix, and coil power on the game. The lamp GI (general illumination) wiring from the playfield would come here, go through some fuses, and then continue on to the CPU board. This board also held the current limiting resistors for the flash lamps, and the bridge rectifiers for coil power and CPU lamp power. At this time Williams dropped the 'keep alive' power resistor R1 from the flash lamp circuit and only used R2 on the interconnect board. This greatly increased reliability because flash lamp resistor R1 (330 ohms 7 watts) failed quite often. The interconnect board also holds several MOC3010 opto-couplers for the flipper lane change circuit (games before the interconnect board did lane changes with a second switch on the flipper cabinet switch - these opto-couplers prevented sloppy operators from shorting the 50 volt flipper coil EOS switch voltage to the low-voltage lane change switches mounted next to them).

Though Interconnect boards are largely the same from game to game (except for Banzai Run), the value of the flasher power resistors do vary. For example, if another Interconnect board is transplanted from a different game title, the flashers may be too weak or too bright. Why does this happen? Because the 12 volt flasher bulbs are actually run off the 28 volt coil voltage. To limit the current so the 12 volt flasher bulbs don't implode immediately, Williams used power resistors (now mounted on the Interconnect board, formerly mounted under the playfield). The values of these resistors varies directly to the number of flasher bulbs in any particular circuit. This is also why multiple flasher bulbs are wired in series instead of parallel (meaning if one bulb dies, none of the other bulbs in that circuit will light). The more flasher bulbs wired in series, the more resistance, and the less ohm value needed for the corresponding power resistor on the interconnect board. For example if only two flasher bulbs are wired in a circuit, then the interconnect power resistor will be around 7 ohms. But if four flasher bulbs are wired in a circuit, the interconnect power resistor will probably be more like 3 ohms.

50 Volt Power Supply board.

All system 11 games *Fire!* and before have a small flipper power supply board to the right of the CPU board. This board was basically a fuse, a bridge rectifier, a resistor, and a few capacitors. It was used to provide power to the 50 volt flipper coils (since most of the other game coils were 25 volts). When the Auxiliary power driver board (APDB) was introduced in *Big Guns*, this board was combined into the APDB.

The 50volt power supply board on games Fire! and before.



Flipper Differences.

Flipper coils are actually two coils in one package. The "high power" side is a few turns of thick gauge wire. This provides low resistance, and therefore high power. The "low power", high resistance side is many turns of much thinner wire. This side of the coil is important if the player

holds the cabinet switch in, keeping the flipper coil energized. The high power low resistance side of the coil is only active when the flipper is first energized. But when the flipper is energized and at full extension, the low powered side of the flipper coil is used so the coil doesn't get hot and burn.

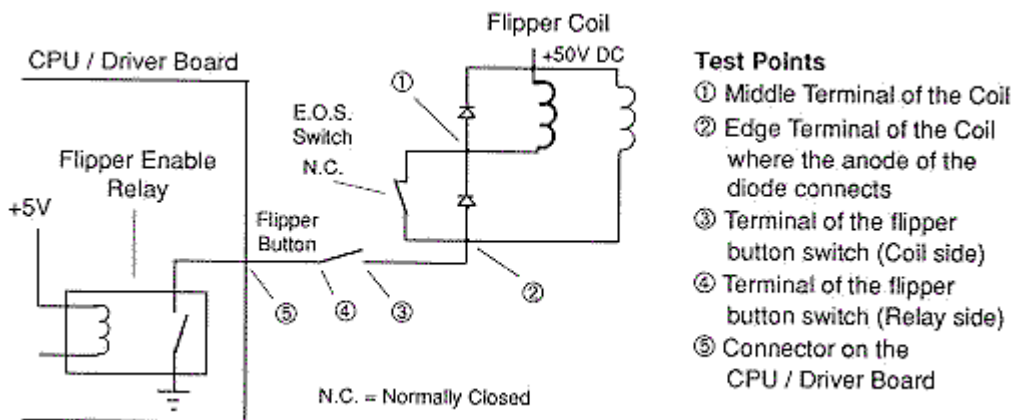
Flipper worked different on games High Speed to Millionaire. These games used a series wound FL23/600-30/2600 flipper coil. The common lug (where both the low and high powered coil wires were connected together) on these flipper coils was the middle of the three lugs. Also these coils used ONE diode across the two outside lugs. The EOS switch on these games, when opened, enabled BOTH the high power and low powered coils together. This style of flipper coil did NOT use a 2.2 mfd anti-spark EOS capacitor. The problem with this series wound coil was the "back spike" of current that occurred when the EOS switch was opened. This cause the EOS switch to excessively wear and pit.

With the introduction of F-14 Tomcat, Williams changed to the parallel wound FL11630 style flipper coil. This coil now used an outside lug as the common lug (where both the low and high powered coil wires were connected together). Also TWO diodes were used and required on these flipper coils. This parallel wound coil eliminated the "back spike" of current when the EOS switch opened. It also allowed the use of a 2.2 mfd 250 volt capacitor to further limit EOS switch sparking and pitting. Now when the EOS switch opens, this removed the high powered side of the coil from the circuit. The low powered side of the flipper coil is always in the circuit, but is essentially ignored when the high powered side is in the circuit. This happens because the current takes the easiest path to ground (the low resistance, high power side of the coil). The low power high resistance side of the flipper coil won't get hot if the player holds the flipper button in.

The CPU Board Flipper Relay K1.

The flippers are only enabled during game play and in diagnostic mode. The flipper enable relay is what turns the ground connected to the flippers off and on. This enable relay is located on the CPU board at K1, and is a 4P, 40 ohm, 6 volt relay. When you enter diagnostic mode, you should hear the flipper relay K1 click on (activating the flipper buttons).

*Norbert Snicer's drawing of the flipper circuit used from F-14 and later.
Notice the CPU board relay K1 that energizes the flippers during a game.*



2a. Before Turning the Game On: Check the Coil Resistance.

Any coil that has locked on (usually due to a shorted CPU board transistor) will heat up and could have a lower total resistance. This happens because the painted enamel insulation on the coil's wire burns easily, causing the windings to short against each other. This will lower the coil's resistance, causing the coil to get even hotter. Within a minute or so the coil becomes a dead short (less than 2 ohms), and usually blows a fuse (assuming the game has not been over-fused; in this case it can burn circuit board traces).

A very good idea for any unknown game just purchased is to check all the coils' resistance. If the game is new to you, and you have not powered it on, a quick check of coil resistance will tell you a lot about your new game. This takes about one minute and can save you hours of repair and diagnosing work.

If the CPU board driver transistor(s) are repaired, and the game is powered on with a dead-shortened coil, this will blow the same driver transistor(s) again when the coil is fired by the game for the first time! It can also cause the game to immediately reset if the coil is forced into activation (because this is shorting the coil voltage to ground with no load). There is no sense

making more work for yourself. So take 60 seconds and check all the coils' resistance BEFORE powering the game on for the first time.

In order to check coil resistance, put your DMM on its lowest resistance setting. Then put the DMM's red and black leads on each coil's lugs. A resistance of 2.5 ohms or greater should be seen. Anything less than 2.5 ohms, and the coil and/or driving transistor may be bad. Now remove the wire from one of the lugs of the coil, and test the coil again. If the resistance is still the same (low), the coil or diode is bad (and also perhaps the driving transistor). If the resistance is higher than 2.5 ohms, the coil is good but the solenoid driver board transistor is shorted and will need to be replaced. Lastly, the coil's 1N4004 diode could be shorted too, giving a false low coil resistance. Cut one diode leg from a coil lug and retest the coil's ohms.

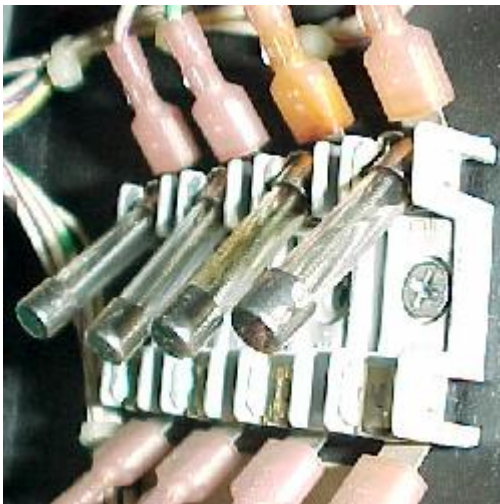
Remember when reconnecting the wires to the coil that the power wire (usually two wires or thicker wires) goes to the coil's lug with the BANDED side of the diode attached. The thinner wire is the coil's return path to ground via the driver transistor and attaches to the coil lug with the non-banded side of the diode attached.

If a low coil resistance is found, also suspect the associated driver transistor(s) as bad. A low resistance coil is a red flag, a warning, that there may be problems on the CPU board. Actually with System11 games, if a low resistance coil is found, I can pretty much guarantee that you will need to (should) replace of course the coil, but also all the silicon devices in its ground path (TIP driver transistor and probably the pre-driver transistor, and possibly the TTL chip that drives it). Always replace a TIP122 with the more robust TIP102.

2b. Before Turning the Game On: Check the Fuses

Check every fuse! Seems like such a simple thing, yet many of us forget to do it. Before you even turn the game on, check the fuses. Not only look for blown fuses, but especially over-fused circuits. For example, is there an 8 amp fuse where there should be a 5 amp? Is there a slow blow fuse where there should be a fast blow? Most of the fuses for these games are located in the backbox.

Removing one end of the fuse from the fuse block, for testing. These fuses are the four general illumination fuses as used on pre-Banzai Run games without an Interconnect board.



Testing Fuses: the Right Way.

Don't depend on your eyes or sense of smell to check fuses. First remove the fuse from its holder, (or at minimum remove just one end of the fuse from the holder). Don't try and test the fuse installed as it can give false readings, depending on the circuit. A perfectly good looking fuse could be blown, it happens all the time. Use a Digital Multi-Meter (DMM) to test the fuse. First remove the fuse from its holder, (or remove just one end of the fuse from the holder). Don't try and test the fuse installed as it can give false readings, depending on the circuit. Set your DMM to "continuity", put a lead on each end of the fuse, and buzz out those fuses. No buzz means fuse is bad.

(Side Note: a "buzz" on the DMM meter means zero resistance. If there is no "buzz", either the circuit is OPEN, or the resistance is 100 ohms or greater. If the meter doesn't have a continuity function, just use the lowest resistance setting. A good fuse will measure zero ohms.)

Another Reason to Pull a Fuse from its Holder to test it.

Always remove a fuse from its holder to test it. Do this because a particularly fatigued fuse will often fall apart as taken out of its holder. This may never be seen if the fuse is tested in its holder. This is especially true if the fuse tests 'good' then the fuse wire pulls away from an end-cap as it heats up. For this reason, regardless of the convenience, all fuses should always be pulled from its holder for testing.

Another Reason to Pull a Fuse from its Holder to test it.

Always remove a fuse from its holder to test it. Do this because a particularly fatigued fuse will often fall apart as you take it out of its holder. You may never see this if the fuse is tested in its holder. This is especially true if the fuse tests 'good' then the fuse wire pulls away from an end-cap as it heats up.

A blown fuse: where did it blow?

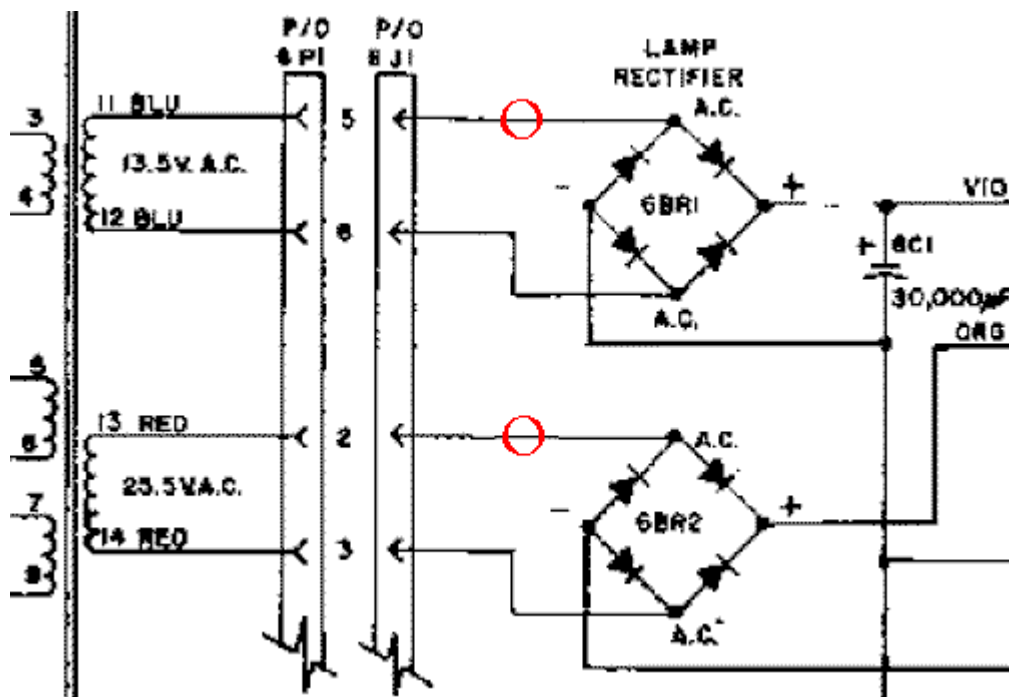
If you look at a blown fuse, most often it's blown in the middle of the fuse. This is where a fuse should blow, and indicates the fuse was doing its job (protecting an overloaded circuit). But if the fuse has blown at either end of the fuse, this could indicate another problem; a bad fuse holder. Keep this in mind, and examine the blown fuses you remove. If the fuse blew at either end, maybe you should replace its fuse holder too.

Adding fuses to the +25 volt solenoid bridge, and the +18 volt lamp matrix bridge. These two bridges are mounted inside the backbox on the right side, just above the large 30,000 mfd capacitor. Be sure to label the fuse holders with the correct amperage fuse (8 amp slo blo) and fuse type. (slow blow).

**Adding Fuses to System 11 games Fire! and before.**

Williams made a design error on games before Fire. They forgot to add a fuse going from the transformer to the bridge rectifiers powering the +25 volts for the solenoids, and the +18 volts for the lamp matrix. If either of these bridges shorts, or capacitors that smooth the voltages fails, you game could start on fire (no joke). This has happened, so it is recommended that you add two new fuses.

The red circles in this schematic show where the added fuses are installed.



It is very easy to modify your game (Fire! and before) and install these fuses. Just buy two fuse blocks at Radio shack. Then find the two bridges mounted on the inside of the backbox. There should be a lead on the bridge that is labeled "AC". Remove the wire going to this lead, and connect it to one lead of a fuse block. Connect the other lead of the fuse block back to the bridge. Do this for both the +25 volt solenoid voltage bridge, and for the +18 volt lamp matrix bridge. Install 8 amp slow blow fuses. Note some bridges will have both "AC" leads labeled. You can connect the fuse to either of these two "AC" leads.

Fuse Locations in System 11 games.

Though the following list will not apply to all system 11 games, here are the fuse types and locations of a typical system 11 game. All fuses are the slow-blow variety. Also there are probably other fuses in the backbox not located on circuit boards (for example on pre-Banzai Run games the general illumination fuses are located on fuse blocks inside the backbox).

Power Supply board.

- F1 = 1/10 amp. Earlier games score display voltage.
- F1 = 1/4 amp. Earlier games score display voltage.
- F1 = 3/8 amp. Score display voltage.
- F2 = 1/8 amp. Score display voltage.
- F2 = 2.5 amp. Earlier games +34 volt solenoid voltage.
- F3 = 1/8 amp. Score display voltage.
- F3 = 8 amp. Earlier games +18 volt lamp matrix power.
- F4 = 7 amp. -12 volts.
- F5 = 7 amp. +5, +12 volts.

Auxiliary Power Driver Board (Big Guns and later).

- F1 = 5 amp (slingshots)
- F2 = not used
- F3 = 2.5 amp (pop bumpers)
- F4 = 2.5 amp
- F5 = 2 amp (right flipper)
- F6 = 2 amp (left flipper)
- F7 = 4 amp (flipper bridge main power)
- F8 = 7 amp (special solenoids main power)

Interconnect Board (Swords of Fury and later).

- F1 = 5 amp (general illumination)
- F2 = 5 amp (general illumination)
- F3 = 5 amp (general illumination)
- F4 = 5 amp (general illumination)

50 volt Power Supply Board (Fire! and before).

- F1 = 5 amp (flipper bridge main power)

Diagnosing a Blown Solenoid Fuse.

If any of the game's solenoid fuse(s) blow immediately at power-on, here's some things to check:

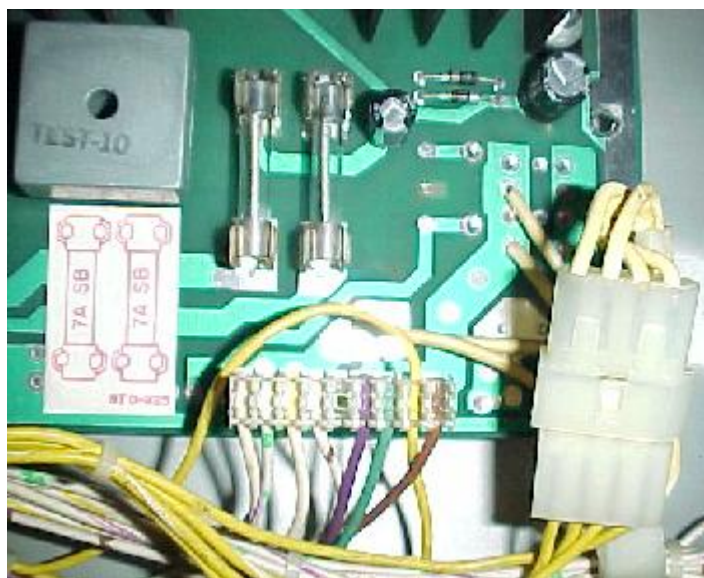
- First check the pop bumpers and slingshots to see if their "activation" switches are stuck closed (the switches that the ball triggers to make the bumper work). Prior to Big Guns and pre-System 11B, these "special solenoids" are unlike other coils in the game in that if their activation switch is stuck closed, it will keep the coil turned on (or will "machine gun"), and the fuse will blow.
- With the game off, use a DMM and check the ohm reading of each and every coil in the game. On two lug coils just put the meter leads on the coil's lugs (on three lug coils there is a "common" lug, measure the resistance between the common lug and the other two lugs). All coils should be 2.5 ohms or greater. If a coil is less than 2.5 ohms, the coil is bad and essentially a dead short, hence causing the solenoid fuse to blow.
- If the flipper solenoid fuse is blown, this is most often a bad or mis-adjusted EOS (End of Stroke) switch. See the flipper section for help with that.
- If a coil is less than 2.5 ohms, there is probably a reason this happened - the coil locked on, heated up, and melted the insulation off the coil's windings. This causes a short between the windings, and lowers the coil's overall resistance. Usually this most often due to a shorted driver transistor which keeps the coil energized while the game is powered on.
- For all coils found with less than 2.5 ohms of resistance, disconnect the wire(s) from one of the coil's lugs. For a two lug coil, there is a power wire which is usually a thicker wire or often two thicker wires. This brings power to the coil and is connected to the coil lug with the banded side of the diode attached. The other coil lug has the return wire which goes to the CPU board. I remove the thinner return wire because it is easier, but removing either will disconnect the coil from the circuit.
- Power the game on with a new solenoid fuse. Does the fuse still blow? If not, you have found the start of your problem (the bad coil - now you need to find out what made the coil bad!) If the fuse still blows, look for another bad coil or perhaps a short from the coil voltage to ground.

2c. Before Turning the Game On: Burnt GI Connectors & Non-Working GI

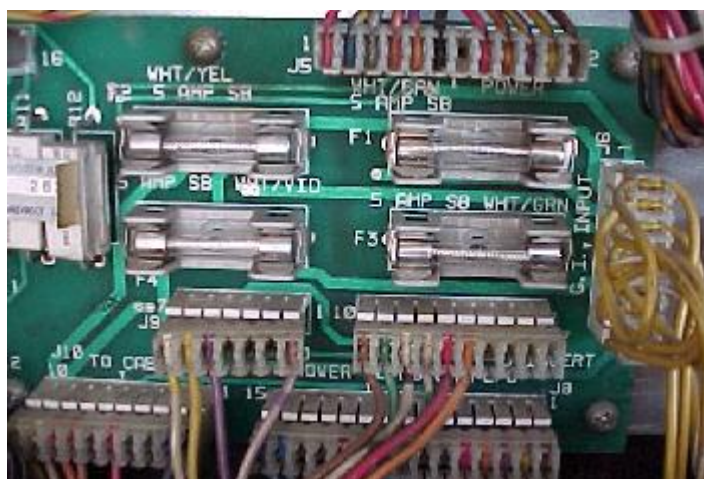
Often you'll get a new game, turn on the power and the General Illumination (GI) lights don't work. The whole game is somewhat dark. This can be caused from burned GI connectors, or a failed GI relay, or a fail fuse (see the section above for information on fuses and how to test them).

The GI connectors can get hot and fail. This happens because the Molex connectors don't always have enough surface area to handle the GI power requirements. The heat from the connector will cause the solder joints to fatigue which causes resistance (and more heat). The connector pins get so hot they soften the solder. All this causes more resistance, which causes more heat. It doesn't end till the board burns, the fuse heat fatigues and fails, or the connectors pins fall out (or burn!), and open the circuit.

The GI connector 3J8 on the lower right side of the power supply board. If any connector burns on a pre-Banzai Run system 11 game, it will probably be this one. Note the yellow wires to the left of this connector are the GI wires coming into the power supply board from the transformer. The connector for these yellow wires are on a "pigtail". This connector rarely burns. Also note since this is a Big Guns to Cyclone power supply board, there is no GI relay just above connector 3J8.



The GI connectors on the Interconnect board, for games *Banzai Run* and later. Note connector 2J6 on the far right with the yellow wires will be the one that is most likely to burn on this board.



The GI Relay(s).

Most system11 games have some sort of GI relay to turn all, or selected strings of general illumination lights off and on. For example, *High Speed* (which didn't use an A/C solenoid select relay), uses the power supply board's K1 relay to turn all the GI on and off. Other system11b and later games like *Elvira*, *Cyclone*, *BK2000*, etc, have two small relay boards, one mounted on the light box hinged panel, and the other under the playfield. These separate relays allow the playfield and backbox GI lights to be turned on and off independently.

Often the GI relays develop cold solder joints (regardless if it is mounted on a separate small circuit board, or the main power supply board). Resoldering these relay or connector solder joints will often fix many GI problems. But sometime the GI relay itself may have failed. The contact switches on the relay burn, and no longer conduct. The only solution to this problem is replacing the relay (usually a 24 volt DC, 10 amp, DPDT relay).

If either the playfield or backbox GI is not working, and there are no burnt connectors, the GI relay(s) are probably the problem. On games with multiple GI relays (like *Cyclone*, *Elvira*, etc), there's one relay for the backbox and one for the playfield. These two relays can be swapped; If the backbox and playfield GI relays are swapped and the problem moves, it is clearly a failed relay. If the problem does not move, it could be a CPU board driver transistor causing the problem.

Remember is there's a driver transistor on the CPU board that turns the GI relay(s) on and off. If this driver transistor is dead or shorted, this can cause a GI problem (either the GI being permanently on or permanently off). For example, on *Cyclone* Q9 is the driver transistor for the playfield GI relay, and there's a different driver transistor for the backbox GI relay. See the driver transistor section of this document for testing and diagnosing these transistors.

The Interconnect board and the GI.

Later system 11 games (Banzai Run and after), use an interconnect board which also "turns around" the GI power. These GI connectors 2J6, 2J7, 2J9, 2J10 on the interconnect board can also burn. The input GI connector at 2J6 (far right) will probably be the one that burns, but the output GI connectors at 2J10 (cabinet, coin door), 2J9 (playfield) and 2J7 (backbox) can also burn.

Fixing a Burnt Connector.

Fixing a burnt connector requires more than just replacing the connector! You also need to remove the board and **replace** the male pins. Molex connectors only have a life of 25 cycles. Add pinball vibration to the mix, and their life is even shorter (see pinrepair.com/connect for more details). Due to the tension of the pins and the potential heat in the GI circuit, the original plating is worn or burned off the connector pins. Also the terminal pins themselves weaken and provide less tension, which causes the gas-tight seal to fail. If only the connector terminal pins are replaced and the male circuit board header pins are not replaced, the resistance will still be there (from the cold or fatigued solder joints and tarnished pins). The new connector will burn in short order. Also only crimped-on trifurcon terminal pins should **always** be used during GI pin replacement.

When replacing the Male Header pins...

When you replace the circuit board male header pins for the GI, check the plated through holes for breaks. For any GI header pin hole in the circuit board, if there is a trace going to the pin on **both** sides of the board, a cracked plated through hole is very common! The plated through hole is how the trace on the bottom side of the circuit board connects to the trace on the top side of the board. If the plated through hole is cracked, the GI may not work even after a new header pin is installed!

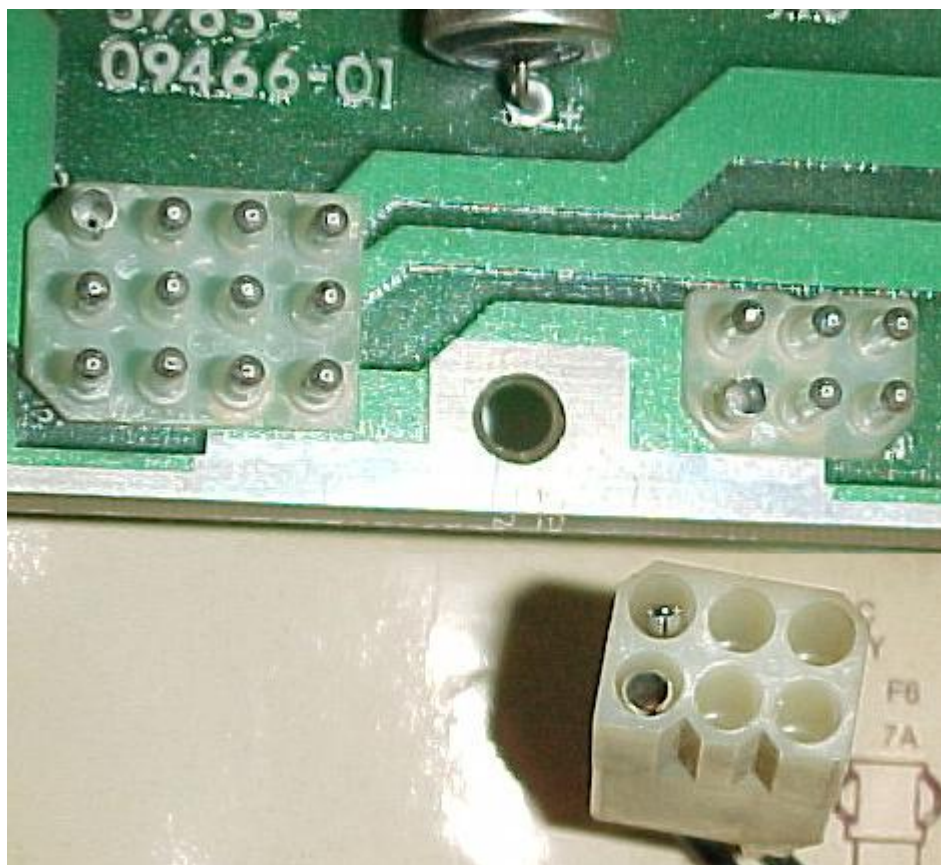
After the old GI header pins are removed, use a DMM set to continuity and "buzz out" each circuit board hole that has a trace connecting to it on **both** sides of the board. Put one lead of the DMM on the bottom trace, and the other DMM lead on the top trace. If a "buzz" is not heard, the plated through hole is cracked.

To fix a cracked plated through hole, stitch a bare wire through the hole. Then solder it on both sides of the circuit board on the connecting traces (some green solder mask will need to be removed to do this). Make sure there is enough room in the hole to insert the replacement header pins! Another way to fix a cracked plated through hole is to solder the new header pins on BOTH sides of the board. The traces on top of the board will be a bit tricky; suspend the header pins up a bit to get the soldering iron between the board and the pins.

A crimping tool (top), two different types of pins (left), and a new connector housing and male pins. Note the connector pins; the far left two pins are the crimp-on, single wiper type. The two pins on the right are insulation displacement pins, but with two wipers. It's ideal to use the crimp-on style pin, but with two wipers (not shown).



System 3 to System 11b power supply connectors. These mixed pin square wafer power supply connectors sometimes burn too.



Square Plug Power Supply Connectors 3J1/3J2, System 11b & Prior.

System 11b games and earlier used two square power supply plugs at 3J1 and 3J2 (newer system 11c games stopped using this style of connector). Sometimes these square plug power supply connectors get damaged (these connectors were used on Williams power supplies system 3 to system 11b, and on DataEast/Sega power supply until 1995). The six pin 3J2 is a ground connector, and usually does not get damaged. But the twelve pin 3J1 handles all the input voltages from the transformer to the power supply, so sometimes it gets burned. Finding the part numbers for these wafer style, mixed pin connectors was a real bear, as they were designed in 1971! So here are the part numbers for these wafer style, mixed pin connectors.

- 12 mixed pin PCB wafer connector, Molex part# 09-18-5121.
- 12 mixed pin wire connector, Molex part# 03-09-1122.
- 6 mixed pin PCB wafer connector, Molex part# 09-18-5061.
- 6 mixed pin wire connector, Molex part# 03-09-1062.
- Male .093" terminal pins for wire connector, Molex part# 16-06-0002.
- Female .093" terminal pins for wire connector, Molex part# 16-06-0001 (new Molex part# 43080-0001).

Click [here](#) for a Molex drawing of these parts.

Crimp-On Pin Connectors vs. Insulation Displacement Connectors (IDC) Plugs.

Insulation displacement connector (IDC) plugs are very convenient for an assembly line or automated procedure to install. No wire stripping is needed, the wire is just pushed onto the "V" in the pin, which cuts (displaces) the insulation to make contact with the wire. But they aren't very good in the long run. Many problems with games are attributed to these IDC plugs. A far better connector uses the crimp-on style of pin. You'll need a special tool to crimp them, but the reliability will be much higher. Only use crimp-on pin connectors when replacing burnt ones. Hand crimping tools include Molex WHT-1921 (part# 11-01-0015), Molex part# 63811-1000, Amp 725, or Radio Shack #64-410.

Trifurcon Connector Pins.

Molex makes a crimp-on .156" size female terminal pin called a "trifurcon" pin (not available in the .100" pin size). This style .156" pin differs than the "normal" pin; the metal material is more heat resistant, and it has three wiper contacts instead of just one. The more contact points means the female pin "hugs" the male header pin with greater surface area. I highly recommend these. They are available from Digikey (800-344-4539 or <http://www.digikey.com>). You can also view the specs for these pins at <http://www.molex.com/product/pcb/6838.html>. Compares these to the "normal" connector pin specs at <http://www.molex.com/product/pcb/2478.html>.

Note Molex sells these pins in "strips" or on a "reel". Do NOT buy connector pins this way! Always buy them in "bags" (separated). It's just too difficult to cut them when they are in strips. If

you don't do a good job cutting them, they won't insert into their plastic housing correctly. Also always get the tin plated version, NOT the gold plated pins.

- .156" Trifurcon terminal pins (three wipers), part# 08-52-0113 (tin plated phosphor bronze) for 18 to 20 guage wire. Tin plated phosphor bronze is the best pin material, as it has better spring, fatigue resistance and current capacity. But if this part number is not available, part# 08-50-0189 (tin plated brass) can be used instead. Great Plains Electronics, Mouser and Competitive Products (#06-2186) sells these.

Board Mounted Header Pins.

These are available in several styles. Get the most number of pins available, and cut the header to the size you need. They also come with a "lock" and without a lock. The lock variety is what you'll use the most.

- .156" header pins with lock (12 pins), part# 26-48-**1125**. This is the preferred variety. The 11 pin version part# is 26-48-1115.
- .156" header pins with no lock (12 pins), part# 26-48-**1121**. The 11 pin version part# is 26-48-1111.

* bold text denotes the number of pins, in this case, 12.

Connector Housings.

Sometimes you'll have to replace the plastic connector housing too if it is burnt, in addition to the pins within the housing. Get the most number of pins available, and cut the connector to the size needed. The high temperature black housings are nice, but are not required. Remember the PINS carry the current, not the plastic housings!

- .156" white housings (12 pins), part# 09-50-**3121**. The 11 pin version is part# 09-50-3111.
- .156" white housings (12 pins), part# 26-03-**4121**: Mouser. This particular housing is less expensive, and specially designed for Trifurcon terminal pins. The 11 pin version part# is 26-03-4111.

* bold text denotes the number of pins, in this case, 12.

Polarized Pegs.

A polarized peg is a small nylon plug that go into the connector housing so the housing is "keyed" (you can't plug it into the wrong board header pin connector). I highly recommend using these if you replace a connector housing.

- .156" polarized peg, part# 15-04-0219.

2d. Before Turning the Game On: Quick and Dirty Transistor Testing

Transistor Testing.

Whenever I get a new system 11 game, before I ever turn it on, I test all the TIP122/102 solenoid transistors. I do this because I'm already in the backbox (examining the fuses and the GI connectors), and a blown transistor can really confuse a system 11 game. This is the procedure I use, and it takes about 20 seconds to test all the TIP122/102 transistors:

- Make sure the game is off.
- Put your DMM (digital multi meter) on ohms (buzz tone).
- Put one lead on the ground strap in the backbox.
- Touch the other lead to the metal tab on the TIP122/102 transistors. These are the 16 transistors at the lower left side of the CPU board, the six transistor at the upper right, and the eight transistors at the far lower right of the CPU board in a vertical row.
- If you get zero ohms (buzz), the transistor is bad! (shorted on)

I replace the bad TIP122 transistor(s) with a more robust TIP102 immediately before I turn the game on. I also usually replace the associated pre-driver 2N4401 transistor too.

Testing the Power Supply.

Another good idea is to test the power supply before powering up the CPU board. This is pretty easy to do. On older power supplies, the two square plugs at the bottom of the power supply is the input power. The two plugs J5 and J6 at the top of the power supply is the output power. To test the power supply, on pre-Taxi games remove the two plugs J5,J6 from the top of the power supply, and turn the game on. On Taxi and later games the J5 connector is renamed J2 is at the bottom left corner. Check these voltages on these plugs with a DMM:

J5 connector (pre-Taxi) - high voltage for score displays:

- 3J5 pin 1 = ground (left most pin)
- 3J5 pin 3 = -90 to -105 vdc

- 3J5 pin 4 = +90 to +105 vdc
- 3J5 pin 6 = +4.9 to +5.2 vdc

J6 connector (pre-Taxi) - logic +5 and 12 volts for CPU board:

- 3J6 pin 2 = -12 to -15 vdc unregulated (left most pin)
- 3J6 pin 6 = +12 to +15 vdc unregulated
- 3J6 pins 7,8,9 = +4.9 to +5.2 vdc
- 3J6 pins 11,12,13 = ground

J2 connector (Taxi and later) - high voltage for score displays:

- 3J2 pin 1 = -90 to -105 vdc (right most pin)
- 3J2 pin 3 = +90 to +105 vdc
- 3J2 pin 5 = ground
- 3J2 pin 6 = +4.9 to +5.2 vdc

J1 connector (Taxi and later) - logic +5 and 12 volts for CPU board:

- 3J1 pins 1,2,3,4,5 = ground (bottom most pins)
- 3J1 pins 6,7,8,9 = +4.9 to +5.2 vdc
- 3J1 pins 11,12,13 = +12 to +15 vdc unregulated
- 3J1 pins 14,15 = -12 to -15 vdc unregulated

Fixing Bad Power Supply Voltages.

Of course check all power supply fuses first using a DMM and removing the fuses from their holder.

If the high voltage (+/- 100 volts) is missing or low or high, the score displays will not work. The HV section of the power supply will need to be rebuilt. The input 90 volts AC (3J1 pin 4 and pin 9 pre-Taxi) comes directly from the transformer, so there should be no issues with that (unless power supply HV fuse(s) are blown).

If the +5 volts is high, suspect a bad 2n6057/2n6059 voltage regulator on the power supply.

If the +5 volts (J6 pin 7,8,9 pre-taxi) is low, check J6 pin 6 (pre-taxi) for +12 to +15 volts DC. If this goes lower than 11 volts, there isn't enough "headroom" for the power supply to make +5 volts, and the game will not boot due to low +5 volts. You can also check the input AC voltage at 3J1 pin 10 and pin 11 (pre-taxi, should be about 12 volts AC). If the input 12 volts DC is low at J6 pin 6 (pre-taxi), put a DMM on low AC volts, and place the test leads on power supply capacitor C10. This should show less than .3 volts AC. If it is greater than this, replace cap C10 (15000 to 18000 mfd 20 volts). If J6 pin 6 (pre-taxi) is still below 12 volts, replace the BR1 bridge rectifier on the power supply. This should fix any low 12 volt DC input problems.

If J6 pin 6 (pre-taxi) is +12 to 15 vdc, this means the BR1 bridge rectifier on the power supply is good as is cap C10. Next this power goes to IC1 chip (723PC) and Q5 (2n6059/2n6059) to make regulated +5 volts. This should be 4.9 to 5.2 vdc (J6 pins 7,8,9 pre-taxi). If this voltage is low, first replace Q5 (2n6059/2n6059).

Lamp matrix voltage on pre-Taxi power supplies include 3J4 pins 5,6,7,8 (+18 vdc) and 3J4 pins 9,10,11,12 (ground). On Taxi and later power supplies, the lamp matrix power does not go to the power supply board at all (it is wired directly to the CPU board at 1J4 pins 1,2,8,9 and grounded at 1J5 pins 1,2,6,7). If the lamp matrix voltage is missing, the lamp matrix won't work. Power for this is supplied by a bridge rectifier mounted to the metal backpanel of the interior backbox, and the huge blue capacitor and fuse next to it.

On pre-BigGuns sys11 games, the solenoid voltage on the power supply include 3J3 pins 6,7,8,9 (+34 vdc) and 3J4 pins 1,2,3,4 (ground). This is the lower voltage solenoid power. Power for this is supplied by a bridge rectifier mounted to the metal backpanel of the interior backbox. The high voltage (50 to 70 volts DC) solenoid power is supplied by the smaller flipper power board beneath the power supply in the backbox. On sys11b games Big Guns and later, the Auxiliary Power Board has both the 35 and 70 volt solenoid voltages (there is no flipper power board and there is no backbox mounted low-power solenoid bridge rectifier). Also the power supply will *not* have a 3J4 connector for solenoid power.

The Auxiliary power board on sys11b games Big Guns and later handles the solenoid voltages completely: J10 pins 1-4 is solenoid ground, J11 pins 8-10 is +35 volts DC, and J11 pin 2-5 is +70 volts DC. The A/C relay is also mounted on the Auxiliary Power board (formerly mounted under the playfield on earlier games), and turns on and off +35 and +70 volts on the Auxiliary power board at pins J11 1-5 (35 volts) and J12 pins 9-12 (70 volts). This is done in conjunction with the A/C driver transistors which control either a flash lamp or a coil.

2e. Before Turning the Game On: Should I leave my Game Powered On?

This is a very common question. After all, arcades leave their games on almost continually with minimal damage (that you know of!). So why not do it with your game at home?

Although commercial pinball machines can handle being powered on continually, I would recommend you do not leave your games turned on when not in use. Here are some reasons:

- Electronic score displays on your game have a limited life, which is proportional to how much time they have been turned on.
- General illumination circuits will be stressed. Burnt pins and connectors are very common on games that are on for extended periods of time.
- Light bulbs don't last forever, and aren't all that easy to change on a playfield.
- The bulbs, displays, fans, and transformers only attract dirt when they are on. Leaving your game on means sucking dirt out of the air and depositing it into your machine.
- Heat generated by the general illumination lamps can warp playfield plastics or help delaminate backglass paint.
- Electricity is a precious resource. Conserve it! An electronic game from this era consumes about 4 amps in attract mode. So leaving your game on is like running a 240 watt light bulb. By comparison, an entire stereo system plus a television use about the same amount of power.

Leaving your pin on all the time can cost much more than any potential damage you could do turning it off and on as you need it.

3a. When things don't work: Replacing Components (Battery Corrosion, PIAs, etc.)

If you have found a bad component, now comes the hard part; replacing it! Transistors, bridge rectifiers, and most chips are not socketed. They are soldered directly into the driver board. Care must be taken when replacing a bad component.

Please see <http://pinrepair.com/begin> for details on the basic electronics skills and tools you will need when replacing circuit board components.

When replacing components, the object is to subject the board to the least amount of heat as possible. Too much heat can lift or crack the board's traces. Too little heat and you can rip out the plated-through holes when removing the part. New circuit boards are too expensive to replace. So you must be careful when doing this.

To remove a bad chip, just CUT it off of the board, leaving as much of its original lead(s) as possible. Then using needle nose pliers, grab the chip lead in the board while heating it with your soldering iron, and pull it out. You can clean up the solder left behind with a desoldering tool. This is the safest method of chip removal and should cause the least amount of damage to the board.

When replacing chips, always install a socket. Buy good quality sockets. Avoid "Scanbe" sockets at all costs! A good dual sided socket is preferred.

Battery Corrosion.

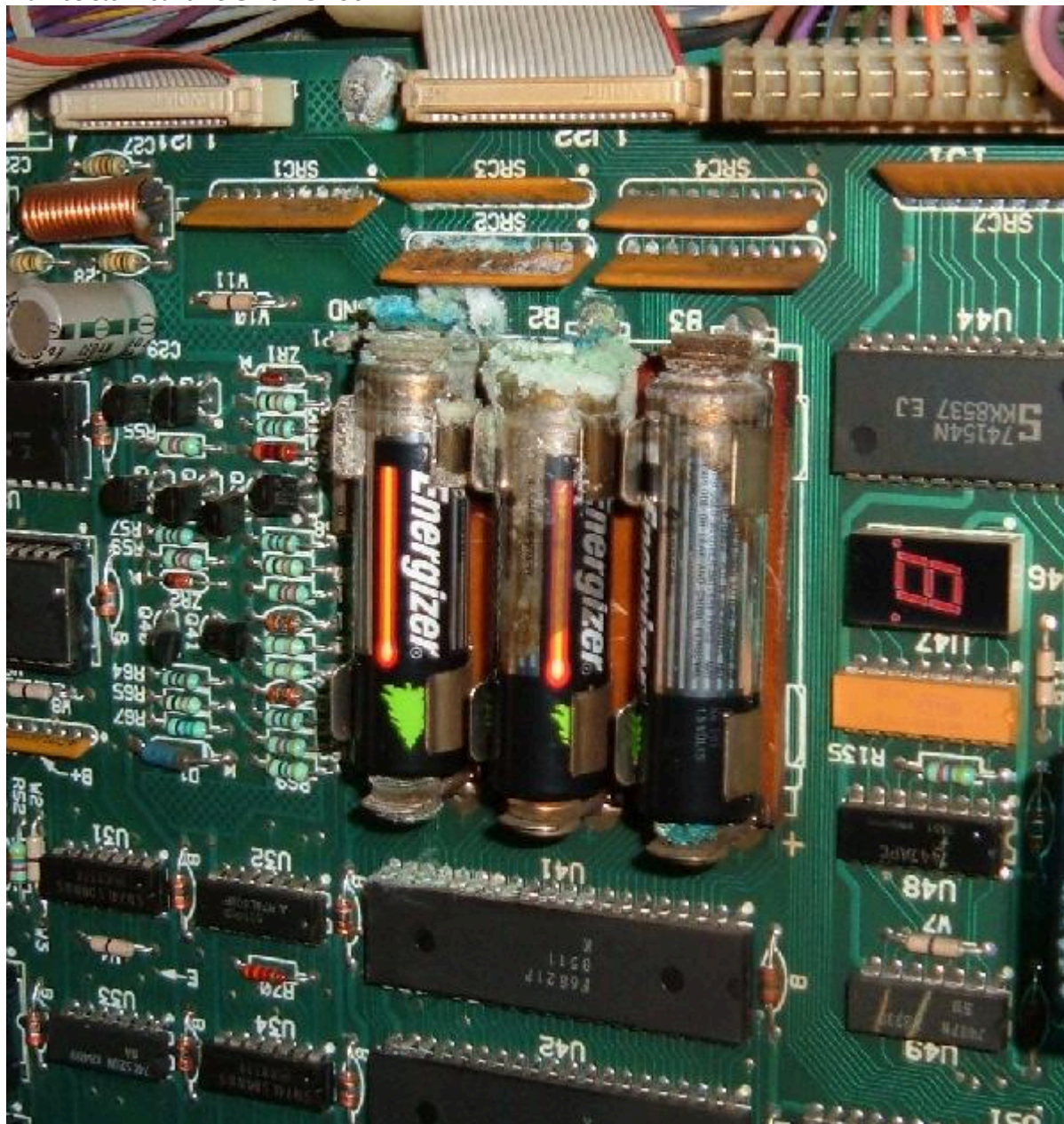
One BIG problem with system11 games are the batteries. The CPU board has mounted at the top center a 3-AA battery pack. This keeps audits, adjustments and high scores in memory when the game is powered off. But if the batteries are neglected and grow old, they can leak their corrosive fluids on the CPU board, ruining the board.

Batteries left to corrode on a High Speed. At first it doesn't look too bad...



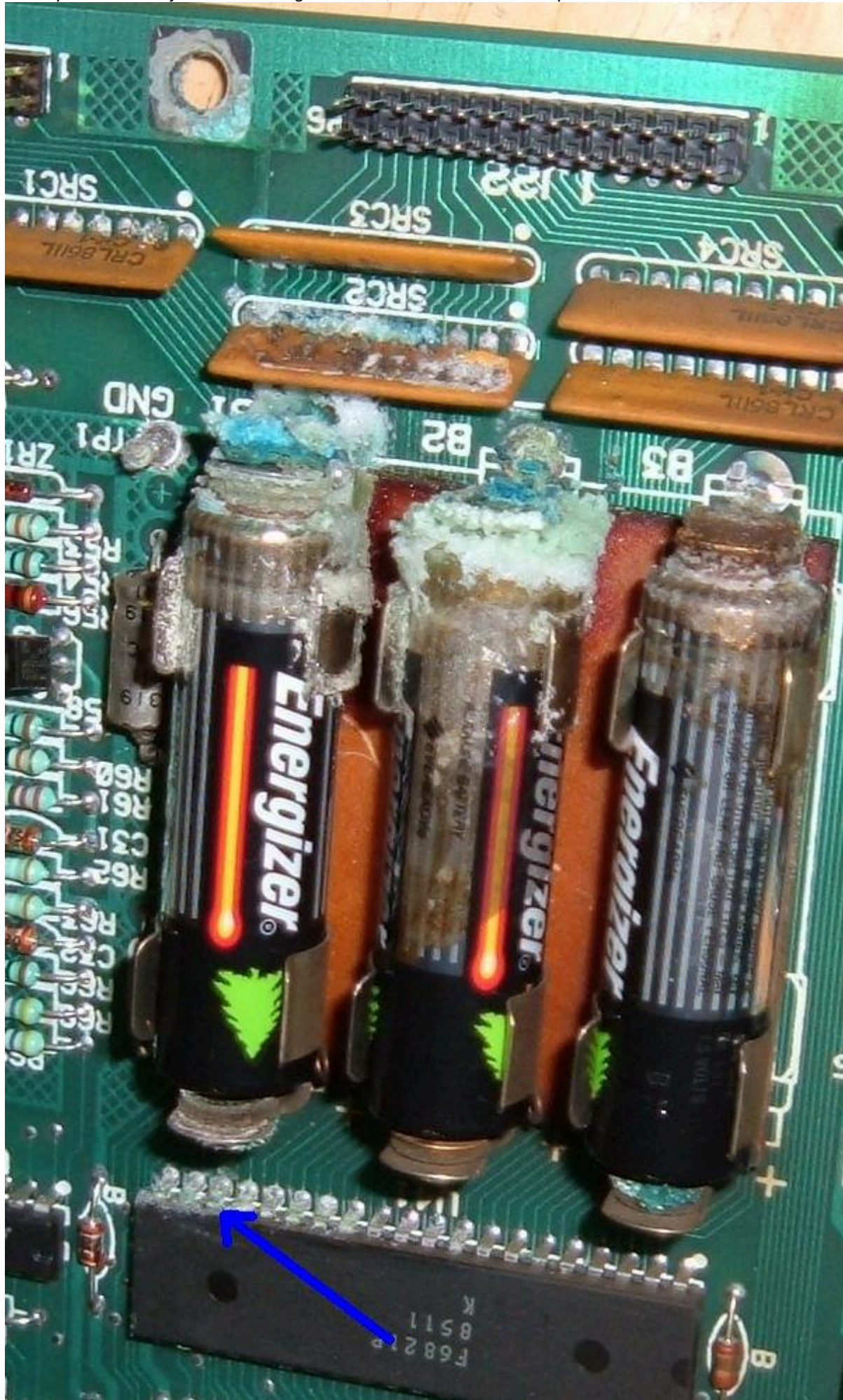
*...A closer look and you can see the corrosion has really taken its toll.
The original battery pack will be removed, the CPU board bead blasted in this area,
and the corroded traces/components replaced. Here PIA u41 will be replaced, along*

with resistor networks SRC1-SRC3.

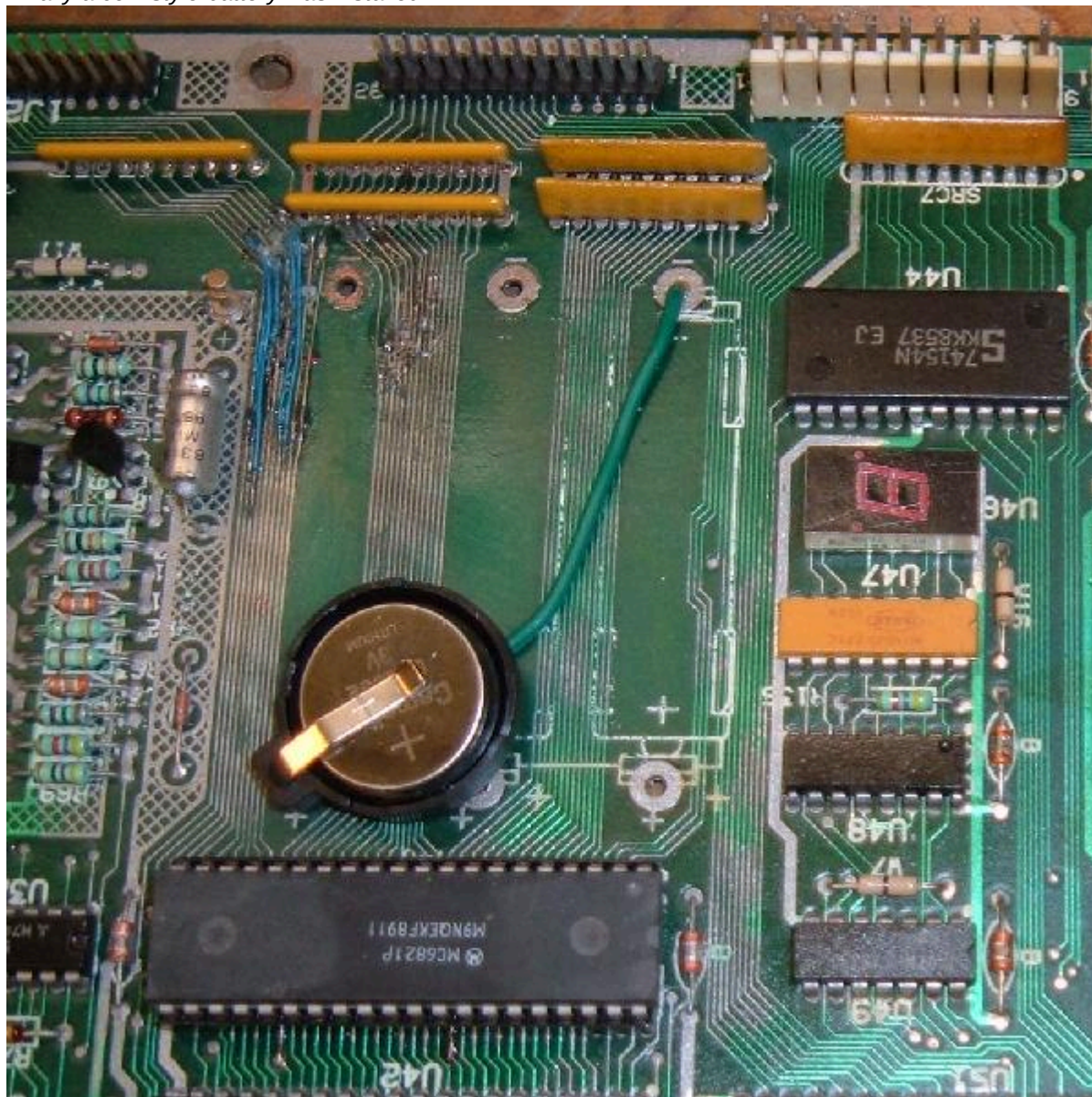


...Even closer. That's pretty nasty battery corrosion. The good news is this is "fresh" corrosion that hasn't dug in deep on the CPU board. Yes it

will require some major bead blasting to correct, but it should clean up OK.



Here's the board after bead blasting. Many of the traces underneath the battery holder were destroyed by corrosion, and had to be re-run with wire wrap style wire. PIA u41 was removed, the board under this chip bead blasted, and a socket installed. SRC1-SRC3 were replaced with SR resistor networks. Finally a coin style battery was installed.



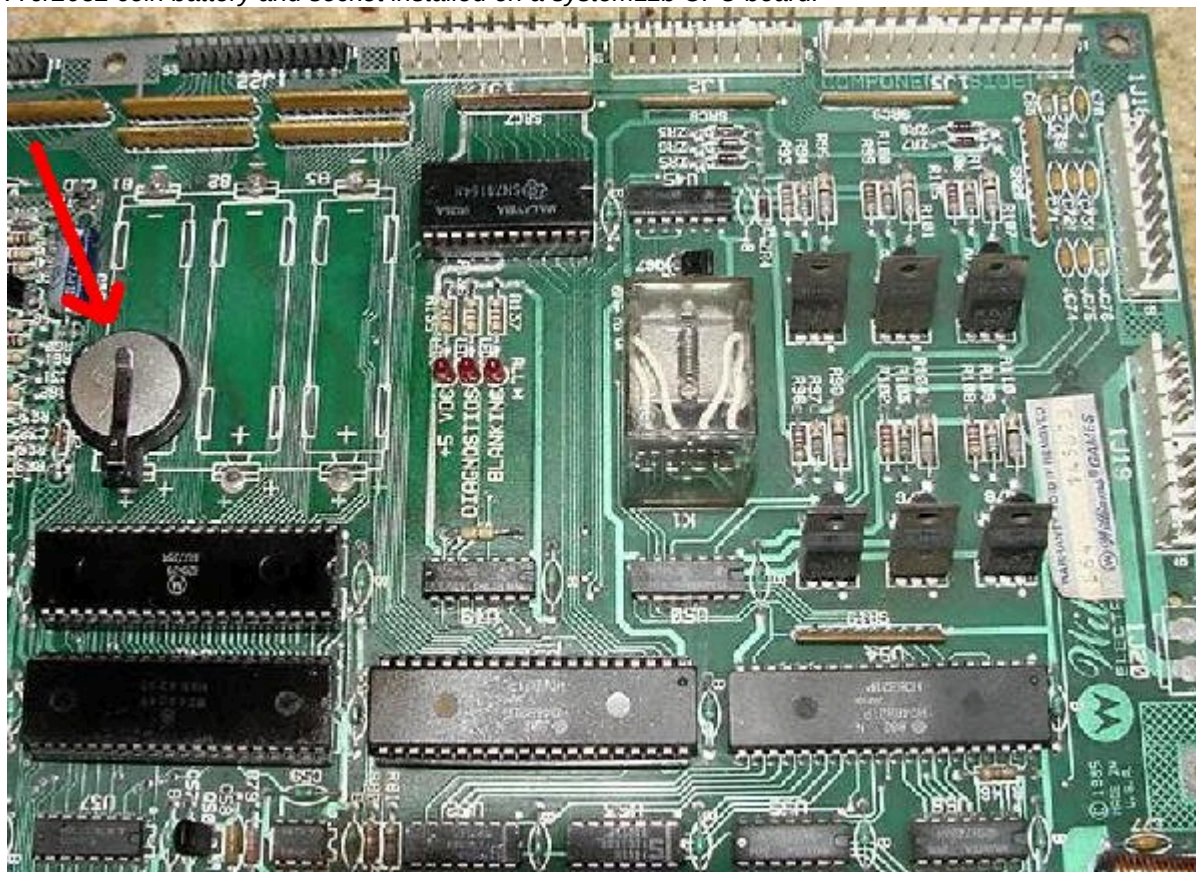
Battery corrosion can be a very serious problem. If a game is stored long enough with leaking batteries, the CPU board can be completely ruined. These are expensive board too (around \$300 or more.) Luckily Rottendog Amusements has reissued the sys11a-11c CPU board (but no sys11, as used on High Speed.)

An excellent approach to preventing battery corrosion is to mount a remote battery pack off the CPU board. This way if the batteries leak, you only ruin a \$5 battery holder instead of a \$250 CPU board. I highly suggest this approach. Another idea is to use a socketed coin style battery CR 2032. These work great on system11 games, last a very long time (up to 10 years), and don't leak. This is the approach I have been taking lately. The cost of the battery is about \$1, and the socket is also about \$1 (See [BG Micro](#) as an inexpensive place for these batteries and holders.) This is about the same cost as a remote battery pack and three AA batteries, and it's a cleaner installation.

A remote AA battery pack installed in a system11 game.



A cr2032 coin battery and socket installed on a system11b CPU board.

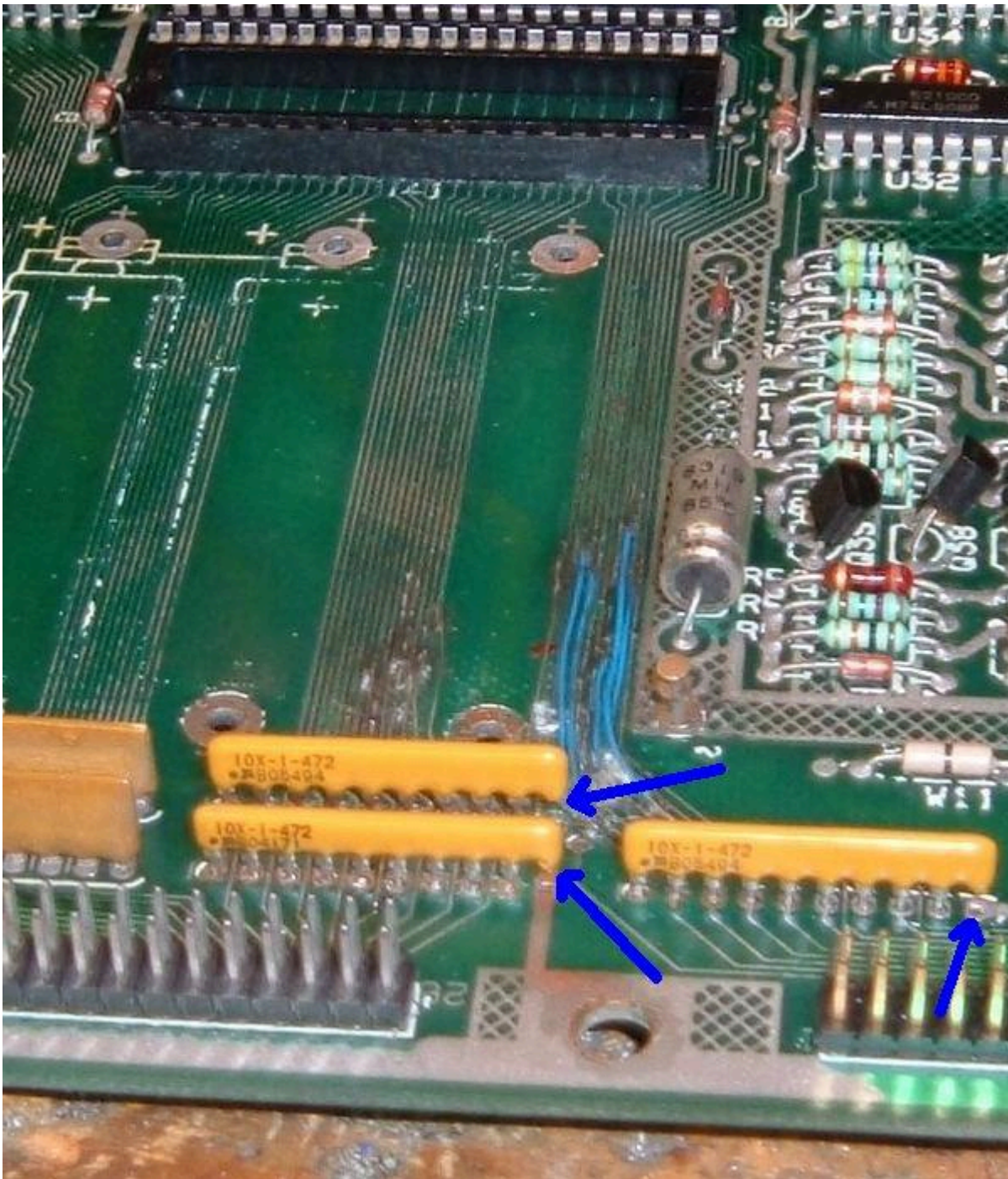


Battery Corrosion and Resistor/Capacitor SRC1-SRC5 Networks.

Just north of the batteries are SIP (single inline) 8r 8c 10 pin resistor/capacitor networks. These can often be compromised by battery corrosion. Unfortunately these are difficult to find (4.7k ohm with a 470 PFD capacitor network.) They can be replaced with a straight SR resistor network (with no capacitors). If this is done, a "4.7k ohm x 9R" bussed resistor network can be used, but it must be installed correctly! This bussed resistor network will have ten pins, with one "common" pin (the pin with the "dot"). The common pin 1 of the resistor network must go in the board's pin 1 (for +5 volts). BUT the last pin 10 of the resistor network leg should be cut and not installed in the CPU board, because this pin on the CPU board goes to ground (for the capacitor part of the SRC, which we are not using since we're installing a SR and not a SRC).

Replaced SRC networks with 4.7k 10 pin bussed SR networks at SRC1-SRC3.

Note pin #10 of the new SR network is cut and not installed.



Battery Corrosion Around the PIAs.

If battery corrosion has happened to a system11 CPU board, often the traces around the battery pack are compromised. If this has happened, there are a number of components directly below (and *underneath*) the battery holder than are often affected. For this reason, and to aid in the replacement of PIA chips for other reasons, I will list where these large 40 pin PIA chips connect below. This way you can "buzz out" the chip legs and make sure they still connect to their intended components (because often battery corrosion can eat these traces). If even a single trace to one of the PIA or support chips is broken, this can cause the entire CPU board to not "boot", or for game functions to not work. (For example, a score display segment to not light.)

PIA and Support Chip Connections.

Here's a list of all the PIAs connections (except for the sys11-sys11b sound PIA).

PIA U41 (display)

- pin 1 = gnd
- pin 2 (pa0) = src5 pin 2, j22 pin 11
- pin 3 (pa1) = src5 pin 3, j22 pin 12
- pin 4 (pa2) = src5 pin 4, j22 pin 13
- pin 5 (pa3) = src5 pin 5, j22 pin 14
- pin 6 (pa4) = src5 pin 6, j22 pin 15
- pin 7 (pa5) = src5 pin 7, j22 pin 16

- pin 8 (pa6) = src5 pin 8, j22 pin 17
- pin 9 (pa7) = src5 pin 9, j22 pin 18
- pin 10 (pb0) = src2 pin 2, j22 pin 19
- pin 11 (pb1) = src2 pin 3, j22 pin 20
- pin 12 (pb2) = src2 pin 4, j22 pin 21
- pin 13 (pb3) = src2 pin 5, j22 pin 22
- pin 14 (pb4) = src2 pin 6, j22 pin 23
- pin 15 (pb5) = src2 pin 7, j22 pin 24
- pin 16 (pb6) = src2 pin 8, j22 pin 25
- pin 17 (pb7) = src2 pin 9, j22 pin 26
- pin 18 (cb1) = src1 pin 2, j21 pin 11
- pin 19 (cb2) = u49 pin 5, sr19 pin 7
- pin 20 = +5
- pin 21 (r/w) = pin 21 on all PIAs, j21 pin 17, u36 pin 1, u13 pin 18, u16 pin 1
- pin 22 (cs0) = pin 22 on all PIAs
- pin 23 (cs2) = u37 pin 12
- pin 24 (cs1) = pin 24 on all PIAs (a13 address bus), u14 pin 6
- pin 25 (E) = pin 25 on all PIAs, u32 pin 10, u35 pin 1, u35 pin 3, u32 pin 13, u29 pin 10, u11 pin 18, j21 pin 19
- pin 26 (d7) = pin 26 on all PIAs, u16 pin 9, u28 pin 8, u25 pin 17, u26 pin 19, u27 pin 19
- pin 27 (d6) = pin 27 on all PIAs, u16 pin 8, u28 pin 13, u25 pin 16, u26 pin 18, u27 pin 18
- pin 28 (d5) = pin 28 on all PIAs, u16 pin 7, u28 pin 14, u25 pin 15, u26 pin 17, u27 pin 17
- pin 29 (d4) = pin 29 on all PIAs, u16 pin 6, u28 pin 7, u25 pin 14, u26 pin 16, u27 pin 16
- pin 30 (d3) = pin 30 on all PIAs, u16 pin 5, u28 pin 17, u25 pin 13, u26 pin 15, u27 pin 15
- pin 31 (d2) = pin 31 on all PIAs, u16 pin 4, u28 pin 4, u25 pin 11, u26 pin 13, u27 pin 13
- pin 32 (d1) = pin 32 on all PIAs, u16 pin 3, u28 pin 18, u25 pin 10, u26 pin 12, u27 pin 12
- pin 33 (d0) = pin 33 on all PIAs, u16 pin 2, u28 pin 3, u25 pin 9, u26 pin 11, u27 pin 11
- pin 34 (reset) = pin 34 on all PIAs, u15 pin 40, u43 pin 4, sr19 pin 3, q39, r69
- pin 35 (rs1) = pin 35 on all PIAs (a1), u25 pin 9, u26 pin 9, u27 pin 9, u11 pin 16
- pin 36 (rs0) = pin 36 on all PIAs (a0), u25 pin 10, u26 pin 10, u27 pin 10, u11 pin 14
- pin 37/38 (irq) = pins 37/38 on all PIAs, r70 (rt leg), u32 pin 1
- pin 39 (ca2) = u49 pin 3, sr19 pin 4
- pin 40 (ca1) = src1 pin 5, j21 pin 15

PIA U42 (display)

- pin 1 = gnd
- pin 2 (pa0) = src4 pin 2, j22 pin 3
- pin 3 (pa1) = src4 pin 3, j22 pin 4
- pin 4 (pa2) = src4 pin 4, j22 pin 5
- pin 5 (pa3) = src4 pin 5, j22 pin 6
- pin 6 (pa4) = src4 pin 6, j22 pin 7
- pin 7 (pa5) = src4 pin 7, j22 pin 8
- pin 8 (pa6) = src4 pin 8, j22 pin 9
- pin 9 (pa7) = src4 pin 9, j22 pin 10
- pin 10 (pb0) = src3 pin 2, j21 pin 3
- pin 11 (pb1) = src3 pin 3, j21 pin 4
- pin 12 (pb2) = src3 pin 4, j21 pin 5
- pin 13 (pb3) = src3 pin 5, j21 pin 6
- pin 14 (pb4) = src3 pin 6, j21 pin 7
- pin 15 (pb5) = src3 pin 7, j21 pin 8
- pin 16 (pb6) = src3 pin 8, j21 pin 9
- pin 17 (pb7) = src3 pin 9, j21 pin 10
- pin 18 (cb1) = src1 pin 3, j21 pin 12
- pin 19 (cb2) = src1 pin 4, j21 pin 13
- pin 20 = +5
- pin 21 (r/w) = pin 21 on all PIAs, j21 pin 17, u36 pin 1, u13 pin 18, u16 pin 1
- pin 22 (cs0) = pin 22 on all PIAs
- pin 23 (cs2) = u37 pin 10
- pin 24 (cs1) = pin 24 on all PIAs (a13 address bus), u14 pin 6
- pin 25 (E) = pin 25 on all PIAs, u32 pin 10, u35 pin 1, u35 pin 3, u32 pin 13, u29 pin 10, u11 pin 18, j21 pin 19
- pin 26 (d7) = pin 26 on all PIAs, u16 pin 9, u28 pin 8, u25 pin 17, u26 pin 19, u27 pin 19
- pin 27 (d6) = pin 27 on all PIAs, u16 pin 8, u28 pin 13, u25 pin 16, u26 pin 18, u27 pin 18
- pin 28 (d5) = pin 28 on all PIAs, u16 pin 7, u28 pin 14, u25 pin 15, u26 pin 17, u27 pin 17
- pin 29 (d4) = pin 29 on all PIAs, u16 pin 6, u28 pin 7, u25 pin 14, u26 pin 16, u27 pin 16
- pin 30 (d3) = pin 30 on all PIAs, u16 pin 5, u28 pin 17, u25 pin 13, u26 pin 15, u27 pin 15
- pin 31 (d2) = pin 31 on all PIAs, u16 pin 4, u28 pin 4, u25 pin 11, u26 pin 13, u27 pin 13

- pin 32 (d1) = pin 32 on all PIAs, u16 pin 3, u28 pin 18, u25 pin 10, u26 pin 12, u27 pin 12
- pin 33 (d0) = pin 33 on all PIAs, u16 pin 2, u28 pin 3, u25 pin 9, u26 pin 11, u27 pin 11
- pin 34 (reset) = pin 34 on all PIAs, u15 pin 40, u43 pin 4, sr19 pin 3, q39 (top leg), r69 (rt leg)
- pin 35 (rs1) = pin 35 on all PIAs (a1), u25 pin 9, u26 pin 9, u27 pin 9, u11 pin 16
- pin 36 (rs0) = pin 36 on all PIAs (a0), u25 pin 10, u26 pin 10, u27 pin 10, u11 pin 14
- pin 37/38 (irq) = pins 37/38 on all PIAs, r70 (rt leg), u32 pin 1
- pin 39 (ca2) = src1 pin 8, j21 pin 18
- pin 40 (ca1) = src1 pin 6, j21 pin 16

PIA U38 (switch matrix)

- pin 1 = gnd
- pin 2 (pa0) = u30 pin 4
- pin 3 (pa1) = u30 pin 3
- pin 4 (pa2) = u30 pin 10
- pin 5 (pa3) = u30 pin 11
- pin 6 (pa4) = u39 pin 4
- pin 7 (pa5) = u39 pin 3
- pin 8 (pa6) = u39 pin 10
- pin 9 (pa7) = u39 pin 11
- pin 10 (pb0) = u40 pin 2
- pin 11 (pb1) = u40 pin 17
- pin 12 (pb2) = u40 pin 4
- pin 13 (pb3) = u40 pin 15
- pin 14 (pb4) = u40 pin 6
- pin 15 (pb5) = u40 pin 13
- pin 16 (pb6) = u40 pin 8
- pin 17 (pb7) = u40 pin 11
- pin 18 (cb1) = gnd
- pin 19 (cb2) = u49 pin 11, sr19 pin 8
- pin 20 = +5
- pin 21 (r/w) = pin 21 on all PIAs, j21 pin 17, u36 pin 1, u13 pin 18, u16 pin 1
- pin 22 (cs0) = pin 22 on all PIAs
- pin 23 (cs2) = u37 pin 11
- pin 24 (cs1) = pin 24 on all PIAs (a13 address bus), u14 pin 6
- pin 25 (E) = pin 25 on all PIAs, u32 pin 10, u35 pin 1, u35 pin 3, u32 pin 13, u29 pin 10, u11 pin 18, j21 pin 19
- pin 26 (d7) = pin 26 on all PIAs, u16 pin 9, u28 pin 8, u25 pin 17, u26 pin 19, u27 pin 19
- pin 27 (d6) = pin 27 on all PIAs, u16 pin 8, u28 pin 13, u25 pin 16, u26 pin 18, u27 pin 18
- pin 28 (d5) = pin 28 on all PIAs, u16 pin 7, u28 pin 14, u25 pin 15, u26 pin 17, u27 pin 17
- pin 29 (d4) = pin 29 on all PIAs, u16 pin 6, u28 pin 7, u25 pin 14, u26 pin 16, u27 pin 16
- pin 30 (d3) = pin 30 on all PIAs, u16 pin 5, u28 pin 17, u25 pin 13, u26 pin 15, u27 pin 15
- pin 31 (d2) = pin 31 on all PIAs, u16 pin 4, u28 pin 4, u25 pin 11, u26 pin 13, u27 pin 13
- pin 32 (d1) = pin 32 on all PIAs, u16 pin 3, u28 pin 18, u25 pin 10, u26 pin 12, u27 pin 12
- pin 33 (d0) = pin 33 on all PIAs, u16 pin 2, u28 pin 3, u25 pin 9, u26 pin 11, u27 pin 11
- pin 34 (reset) = pin 34 on all PIAs, u15 pin 40, u43 pin 4, sr19 pin 3, q39 (top leg), r69 (rt leg)
- pin 35 (rs1) = pin 35 on all PIAs (a1), u25 pin 9, u26 pin 9, u27 pin 9, u11 pin 16
- pin 36 (rs0) = pin 36 on all PIAs (a0), u25 pin 10, u26 pin 10, u27 pin 10, u11 pin 14
- pin 37/38 (irq) = pins 37/38 on all PIAs, r70 (rt leg), u32 pin 1
- pin 39 (ca2) = u49 pin 9, sr17 pin 10
- pin 40 (ca1) = gnd

U37 Support Chip.

This chip is very important in regards to address lines connecting to the PIAs.

- pin 1 (a10) = u13 pin 7, u33 pin 10
- pin 2 (a11) = u13 pin 9, u26 pin 23, u27 pin 23
- pin 3 (a12) = u13 pin 12, u26 pin 2, u27 pin 2
- pin 4 (a15) = u15 pin 25
- pin 5 (a14) = u15 pin 24
- pin 6 (vma) = u32 pin 9
- pin 7 = n/c
- pin 8 = gnd
- pin 9 = n/c
- pin 10 = u42 pin 23
- pin 11 = u38 pin 23

- pin 12 = u41 pin 23
- pin 13 = u51 pin 23
- pin 14 = u34 pin 9, u37 pin 14
- pin 15 = u12 pin 1, u34 pin 10
- pin 16 = +5

(following not verified.)

PIA U54 (lamp matrix)

- pin 1 = gnd
- pin 2 (pa0) = u55 pin 13
- pin 3 (pa1) = u55 pin 11
- pin 4 (pa2) = u55 pin 9
- pin 5 (pa3) = u56 pin 3
- pin 6 (pa4) = u56 pin 1
- pin 7 (pa5) = u56 pin 13
- pin 8 (pa6) = u56 pin 11
- pin 9 (pa7) = u56 pin 9
- pin 10 (pb0) = u53 pin 12
- pin 11 (pb1) = u53 pin 10
- pin 12 (pb2) = u53 pin 2
- pin 13 (pb3) = u53 pin 4
- pin 14 (pb4) = u52 pin 1
- pin 15 (pb5) = u52 pin 4
- pin 16 (pb6) = u52 pin 12
- pin 17 (pb7) = u52 pin 9
- pin 18 (cb1) = gnd
- pin 19 (cb2) = u49 pin 1, sr19 pin 10
- pin 20 = +5
- pin 21 (r/w) = pin 21 on all PIAs, j21 pin 17, u36 pin 1, u13 pin 18, u16 pin 1
- pin 22 (cs0) = pin 22 on all PIAs
- pin 23 (cs2) = u37 pin 14, u34 pin 9
- pin 24 (cs1) = pin 24 on all PIAs (a13 address bus), u14 pin 6
- pin 25 (E) = pin 25 on all PIAs, u32 pin 10, u35 pin 1, u35 pin 3, u32 pin 13, u29 pin 10, u11 pin 18, j21 pin 19
- pin 26 (d7) = pin 26 on all PIAs, u16 pin 9, u28 pin 8, u25 pin 17, u26 pin 19, u27 pin 19
- pin 27 (d6) = pin 27 on all PIAs, u16 pin 8, u28 pin 13, u25 pin 16, u26 pin 18, u27 pin 18
- pin 28 (d5) = pin 28 on all PIAs, u16 pin 7, u28 pin 14, u25 pin 15, u26 pin 17, u27 pin 17
- pin 29 (d4) = pin 29 on all PIAs, u16 pin 6, u28 pin 7, u25 pin 14, u26 pin 16, u27 pin 16
- pin 30 (d3) = pin 30 on all PIAs, u16 pin 5, u28 pin 17, u25 pin 13, u26 pin 15, u27 pin 15
- pin 31 (d2) = pin 31 on all PIAs, u16 pin 4, u28 pin 4, u25 pin 11, u26 pin 13, u27 pin 13
- pin 32 (d1) = pin 32 on all PIAs, u16 pin 3, u28 pin 18, u25 pin 10, u26 pin 12, u27 pin 12
- pin 33 (d0) = pin 33 on all PIAs, u16 pin 2, u28 pin 3, u25 pin 9, u26 pin 11, u27 pin 11
- pin 34 (reset) = pin 34 on all PIAs, u15 pin 40, u43 pin 4, sr19 pin 3, q39 (top leg), r69 (rt leg)
- pin 35 (rs1) = pin 35 on all PIAs (a1), u25 pin 9, u26 pin 9, u27 pin 9, u11 pin 16
- pin 36 (rs0) = pin 36 on all PIAs (a0), u25 pin 10, u26 pin 10, u27 pin 10, u11 pin 14
- pin 37/38 (irq) = pins 37/38 on all PIAs, r70 (rt leg), u32 pin 1
- pin 39 (ca2) = u49 pin 13, sr19 pin 2
- pin 40 (ca1) = gnd

PIA U10 (sound/solenoids)

- pin 1 = gnd
- pin 2 (pa0) = u9 pin 2
- pin 3 (pa1) = u9 pin 3
- pin 4 (pa2) = u9 pin 4
- pin 5 (pa3) = u9 pin 5
- pin 6 (pa4) = u9 pin 6
- pin 7 (pa5) = u9 pin 7
- pin 8 (pa6) = u9 pin 8
- pin 9 (pa7) = u9 pin 9
- pin 10 (pb0) = u18 pin 1
- pin 11 (pb1) = u18 pin 12
- pin 12 (pb2) = u18 pin 9
- pin 13 (pb3) = u18 pin 5
- pin 14 (pb4) = u17 pin 1
- pin 15 (pb5) = u17 pin 12
- pin 16 (pb6) = u17 pin 9

- pin 17 (pb7) = u17 pin 5
- pin 18 (cb1) = gnd
- pin 19 (cb2) = u50 pin 11
- pin 20 = +5
- pin 21 (r/w) = pin 21 on all PIAs, j21 pin 17, u36 pin 1, u13 pin 18, u16 pin 1
- pin 22 (cs0) = pin 22 on all PIAs
- pin 23 (cs2) = u12 pin 5
- pin 24 (cs1) = pin 24 on all PIAs (a13 address bus), u14 pin 6
- pin 25 (E) = pin 25 on all PIAs, u32 pin 10, u35 pin 1, u35 pin 3, u32 pin 13, u29 pin 10, u11 pin 18, j21 pin 19
- pin 26 (d7) = pin 26 on all PIAs, u16 pin 9, u28 pin 8, u25 pin 17, u26 pin 19, u27 pin 19
- pin 27 (d6) = pin 27 on all PIAs, u16 pin 8, u28 pin 13, u25 pin 16, u26 pin 18, u27 pin 18
- pin 28 (d5) = pin 28 on all PIAs, u16 pin 7, u28 pin 14, u25 pin 15, u26 pin 17, u27 pin 17
- pin 29 (d4) = pin 29 on all PIAs, u16 pin 6, u28 pin 7, u25 pin 14, u26 pin 16, u27 pin 16
- pin 30 (d3) = pin 30 on all PIAs, u16 pin 5, u28 pin 17, u25 pin 13, u26 pin 15, u27 pin 15
- pin 31 (d2) = pin 31 on all PIAs, u16 pin 4, u28 pin 4, u25 pin 11, u26 pin 13, u27 pin 13
- pin 32 (d1) = pin 32 on all PIAs, u16 pin 3, u28 pin 18, u25 pin 10, u26 pin 12, u27 pin 12
- pin 33 (d0) = pin 33 on all PIAs, u16 pin 2, u28 pin 3, u25 pin 9, u26 pin 11, u27 pin 11
- pin 34 (reset) = pin 34 on all PIAs, u15 pin 40, u43 pin 4, sr19 pin 3, q39 (top leg), r69 (rt leg)
- pin 35 (rs1) = pin 35 on all PIAs (a1), u25 pin 9, u26 pin 9, u27 pin 9, u11 pin 16
- pin 36 (rs0) = pin 36 on all PIAs (a0), u25 pin 10, u26 pin 10, u27 pin 10, u11 pin 14
- pin 37/38 (irq) = pins 37/38 on all PIAs, r70 (rt leg), u32 pin 1
- pin 39 (ca2) = u9 pin 40
- pin 40 (ca1) = gnd

PIA U51 (solenoids)

- pin 1 = gnd
- pin 2 (pa0) = u44 pin 23
- pin 3 (pa1) = u44 pin 22
- pin 4 (pa2) = u44 pin 21
- pin 5 (pa3) = u44 pin 20
- pin 6 (pa4) = diagnostic LED
- pin 7 (pa5) = ?
- pin 8 (pa6) = ?
- pin 9 (pa7) = sr19 pin 9, jumper w7
- pin 10 (pb0) = src9 pin 2, j3 pin 9
- pin 11 (pb1) = src9 pin 3, j3 pin 8
- pin 12 (pb2) = src9 pin 4, j3 pin 7
- pin 13 (pb3) = src9 pin 5, j3 pin 5
- pin 14 (pb4) = src9 pin 6, j3 pin 4
- pin 15 (pb5) = src9 pin 7, j3 pin 3
- pin 16 (pb6) = src9 pin 8, j3 pin 2
- pin 17 (pb7) = src9 pin 9, j3 pin 1
- pin 18 (cb1) = u14 pin 10
- pin 19 (cb2) = sr20 pin 2, c68, j3 pin 10
- pin 20 = +5
- pin 21 (r/w) = pin 21 on all PIAs, j21 pin 17, u36 pin 1, u13 pin 18, u16 pin 1
- pin 22 (cs0) = pin 22 on all PIAs
- pin 23 (cs2) = u37 pin 13
- pin 24 (cs1) = pin 24 on all PIAs (a13 address bus), u14 pin 6
- pin 25 (E) = pin 25 on all PIAs, u32 pin 10, u35 pin 1, u35 pin 3, u32 pin 13, u29 pin 10, u11 pin 18, j21 pin 19
- pin 26 (d7) = pin 26 on all PIAs, u16 pin 9, u28 pin 8, u25 pin 17, u26 pin 19, u27 pin 19
- pin 27 (d6) = pin 27 on all PIAs, u16 pin 8, u28 pin 13, u25 pin 16, u26 pin 18, u27 pin 18
- pin 28 (d5) = pin 28 on all PIAs, u16 pin 7, u28 pin 14, u25 pin 15, u26 pin 17, u27 pin 17
- pin 29 (d4) = pin 29 on all PIAs, u16 pin 6, u28 pin 7, u25 pin 14, u26 pin 16, u27 pin 16
- pin 30 (d3) = pin 30 on all PIAs, u16 pin 5, u28 pin 17, u25 pin 13, u26 pin 15, u27 pin 15
- pin 31 (d2) = pin 31 on all PIAs, u16 pin 4, u28 pin 4, u25 pin 11, u26 pin 13, u27 pin 13
- pin 32 (d1) = pin 32 on all PIAs, u16 pin 3, u28 pin 18, u25 pin 10, u26 pin 12, u27 pin 12
- pin 33 (d0) = pin 33 on all PIAs, u16 pin 2, u28 pin 3, u25 pin 9, u26 pin 11, u27 pin 11
- pin 34 (reset) = pin 34 on all PIAs, u15 pin 40, u43 pin 4, sr19 pin 3, q39 (top leg), r69 (rt leg)
- pin 35 (rs1) = pin 35 on all PIAs (a1), u25 pin 9, u26 pin 9, u27 pin 9, u11 pin 16
- pin 36 (rs0) = pin 36 on all PIAs (a0), u25 pin 10, u26 pin 10, u27 pin 10, u11 pin 14
- pin 37/38 (irq) = pins 37/38 on all PIAs, r70 (rt leg), u32 pin 1
- pin 39 (ca2) = sr20 pin 3, c69, j3 pin 11 (comma)

- pin 40 (ca1) = u14 pin 13

3b. When thing don't work: Locked-On & Non-Working Coils/Flashlamps (Checking Transistors/Coils)

If a coil is "stuck on" when the game is turned on, a shorted driver transistor is often the cause. If a coil does not work (and the fuses are good!), an open driver transistor could be the cause. This section will help diagnose this, and other related faults.

Introduction.

In a working game, the first thing to remember on all coils and flashlamps is the power is **always** present at all coils/flashlamps. All these devices are waiting for is the backbox driver board to complete the their power circuit to ground, causing the coil or flashlamp to energize.

Essentially the driver board is a big computer controlled grounding plane. Through the game ROM program, the CPU, and the PIAs (Peripheral Interface Adaptors), the game can control which driver board transistor can "sink" a ground, and hence complete a particular coil's power path (causing the coil/flashlamp to energize for a short period of time).

The way the driving logic works is as so: the CPU, which is running the game ROM program, wants to energize a coil. It tells the a PIA (Peripheral Interface Adaptor) to turn on the appropriate coil. This in turn drives a 7408/7402 chip, which then turns on a small "pre-driver" 2N4401 transistor. So far this is all done with "logic level" 5 volts. Then the pre-driver transistor turns on a much bigger TIP120/TIP102 transistor. This final link in the chain is what ultimately completes the coil's path to ground, causing the 28 volt coil to energize momentarily.

A potential problem with this system is if ANY part in the chain shorts, everything else down the chain turns on, and a coil locks-on. Typically this is last link in the chain, the TIP120/TIP102/TIP36 driver transistor, becoming "shorted" internally (because this device is in direct line with the 28 or 50 volt solenoid voltage, where the other devices are "buffered" from this voltage). But it could be any of the other parts too! (the 2N4401 pre-driver transistor, the 7408/7402 chip, or the PIA chip!) It could even be ALL these devices short on!

So instead of the CPU controlling the driver transistor (and hence its associated coil/flashlamp), the coil/flashlamp becomes lock-on (permanently energized), because the path to ground is shorted inside one or many of the controlling devices. So if a coil (or several coils) or flashlamps are locked-on, the TIP120/TIP102/TIP36 is at minimum is usually the cause. But the big problem is if the TIP120/TIP102/TIP36 driver transistor shorts, sometimes the "backlash" can ruin the parts behind it (2N4401, 7408/7402, PIA) that control the transistor.

Solenoid Power Circuit.

On sys9 and pre-sys11b games, the 28 volt solenoid circuit consists of a bridge rectifier mounted on the backbox. Like the lamp rectifier, its a 35amp, 400 volt bridge rectifier. After that, the power goes to the power supply board, and thru a 47volt varistor used to protect the coils from a voltage spike (if the voltage goes above 47 volts, the MOV varistor shorts, which will blow the main solenoid fuse).

On sys11b and later games with auxiliary power boards, the 28 volt bridge is now located there. In addition there is a 50 volt bridge for high power coils on this same board. The under-PF mounted 50 volt relay boards are discontinued on these games, as the Auxiliary power board has TIP36 transistors (pre-driven by the CPU board's TIP102/TIP122 transistor) to drive the 50 volt coils. The driver board and auxiliary power board driver transistors are the most probable source of solenoid problems.

What do the Driver Transistors Do?

Basically, a driver transistors completes each coil's path to ground. There is power at each coil, all the time. The driving transistor is "turned on" by the game's software, through a TTL (Transistor to Transistor Logic) chip. When the transistor is turned on, this completes the coil's power path to ground, energizing the coil. Driver transistors also work the CPU controlled lamps and flash lamps, causing a lamp to "lock on".

Sometimes these driver transistors short "on" internally. This completes a coil or flash lamp's power path to ground permanently, making it "stuck on", as soon as the game is turned on. Also a shorted pre-driver transistor, or shorted TTL chip (which controls the transistors) could be the problem (though a shorted driver transistor is the most common cause). To fix this, the defective component (and perhaps some other not defective, but over stressed componets) will need to be replaced.

Driver Transistor Operation.

As described above, the main driver transistor (a TIP122/TIP102 or TIP36) completes the coil or

flash lamp's power path the ground, energizing it. But there are other components involved too!

Each driver transistor has a "pre-driver" transistor. In the case of a TIP122/TIP102 (the most common driver transistor), this is a smaller 2N4401 transistor.

Starting with System11b, under-playfield relay boards were no longer used for high-power (50 volt) kickers and other devices. In System11 and System11a, a TIP122 would turn on a small relay board mounted under the playfield. This in turn would momentarily turn on a 50 volt device (like a ball kicker). But with System11b and the Auxiliary Power board, this approach was abandoned. Instead of the under-playfield relay boards, TIP36 transistors mounted on the Aux Power board were used. These are pre-driven by the CPU board TIP122 transistors. If the main driver transistor is a TIP36c, this is pre-driven by both a TIP122/TIP102 and a smaller 2N4401 transistor. The bigger TIP36c transistor is an even more robust than the TIP122/TIP102, and controls very high powered 50 volt coils (like the up kickers, etc).

Then before even the smaller 2N4401 pre-driver transistor, there is a TTL (Transistor to Transistor Logic) 7408 or 7402 chip. But the first link in the chain is the 6821 PIA chip. This is what in affect turns on the TTL and smaller 2N4401 pre-driver transistor, which then turns on the TIP122/TIP102 (which then turns on the TIP36c, if used for the coil/flash lamp in question), and energized the device.

This series of smaller to bigger transistors is done to isolate the hi-powered coil voltage, from the low-power logic (5 volts) on the driver board. Also the 7408/7402 and PIA chips (all operating at +5 volts), which really controls the transistors, can not directly drive a high power TIP122/TIP102 or TIP36c transistor (which is controlling the coil's high voltage by sinking the ground).

If ANY of these components in the chain have failed, a coil/flashlamp can be stuck on, and will energize as soon as the game is powered on!

I have a Stuck-on Coil (or Flashlamp), What should I Replace?

A short summary (before reading all the info below). The following procedures will test the driver and pre-driver transistors in question. If either is bad, it will need to be replaced. When replacing either a driver or pre-driver transistor, replace them both (or in the case of a TIP36, replace the TIP122/TIP102 and smaller 2N4401 transistor)! A shorted transistor will cause the other transistors in the link to be stressed, and they should all be replaced.

Inside the front cover of the game manual is a list of each coil used in the game. Also listed are the driving transistor(s) for each coil. Use this chart to determine which transistors could potentially be bad. Also use the schematics.

If after replacing the driver transistors the coil/flashlamp is still stuck on, then replace the TTL 7408 or 7402 logic chip. This TTL chip can also go bad. If there is still a problem, the 6821 PIA in front of the TTL chip can also die.

Also remember to test the resistance of a coil BEFORE replacing the driver transistors. If a coil gets hot, it can burn the painted enamel insulation off the coil windings. This lowers the overall resistance of the coil because adjacent windings short together. If resistance gets much below 3 ohms, the coil becomes a "short", and will fry its associated driver transistors immediately!

A Coil just Does Not Work - What is Wrong?

Driver transistors can go "open" too. This means all the logic prior to the open transistor could be working fine, but the coil will not energize. If there is power at the coil, this is something to consider (but first see the test procedures below to make sure the coil itself is actually OK). Also if in the past a TIP transistor on the CPU board shorted and burned, sometimes board traces can break. This will also prevent a coil from firing.

Another consideration are the CPU board connectors. On the special solenoids in particular, the male header pins on the CPU board that connect the switches to the CPU, and which connect the CPU back to the coil, can have cracked solder joints. Often resoldering these male connector pins on the CPU board can fix non-working coil problems.

Do the Transistor Test Procedures work 100%?

In short, no. But they do work about 98% of the time, and are an excellent starting point. But yes, a transistor can test as "good", but still be bad. (But if a transistor tests as bad it's pretty much guaranteed that it is bad.) The DMM test procedures test the transistors with no load. Under load, a transistor may not work.

Types of Transistors Used on a System11 Driver Board.

There are basically four types of coil driver transistors used on a system 11 CPU board:

- **TIP122/TIP102:** used for transistor Q69, Q71, Q73, Q75, Q77, Q79 (special solenoids), Q81-Q87 (lamp return rows), Q22-Q25 and Q30-Q33 (multiplexed solenoid drivers), Q6 (coin lockout relay), Q7 or Q8 (bank select relay), Q9 and Q14-Q17 (miscellaneous devices). These transistors are used to control all solenoids and flashers. When replacing a TIP122 on a system 11 game, always replace a TIP122 with a TIP102 instead. The TIP102 is a more robust version of the TIP122. Equivalent transistors include TIP122 = NTE261, TIP102 = NTE2343.
- **TIP36c:** used for transistor Q1-Q8 on the Auxiliary power driver board System11b and later. (if your game is earlier than Big Guns, you do not have this board, and hence do not have any TIP36c transistors). These transistors control the high voltage 50 volt coils that were previously controlled by under-the-playfield relay boards. NTE393 is an equivalent transistor.
- **2N4401:** Q2-Q5, Q10-Q13, Q18-Q21, Q26-Q29, Q34-Q38, Q41, Q67-Q68, Q70, Q72, Q74, Q76, Q78. Used as a pre-driver for all the TIP122 transistors. NTE123AP is an equivalent transistors.
- **TIP42:** Q52, Q54, Q56, Q58, Q60, Q62, Q64, Q66. Used to control the lamp matrix columns. The TIP42 switches the +18 volts on for any particular lamp column. NTE197 is an equivalent transistor.

Special Solenoids.

On system9 and system11, there are six coils on the CPU board known as "special solenoids". Special solenoids work differently than the other CPU controlled coils. Special solenoids are used in pop bumpers and slingshot kickers, and since they must act quickly, the CPU does not control them. Closing of a special solenoid's playfield trigger switch enables solenoid power directly through TTL (Transistor to Transistor) chip logic and two transistor, without any processing by the CPU chip. A second switch matrix switch is closed when a special solenoid pulls in, which tells the CPU to score the solenoid points (CPU controlled solenoids do not need this second switch). Hence the special solenoid trigger switches are not part of the switch matrix, where the scoring switch is. Note there are six special solenoids on the system9 and sys11 CPU board. And the Special solenoids always run at 28 volts (they are not designed for the more powerful 50 volt coils).

At the time, it was felt that the clock speed of the CPU was not fast enough to give quick acting pop bumpers and slingshot kickers, as the CPU was doing other things like monitoring the switch matrix and running the lamp matrix and score displays. This opinion was all changed with Big Guns and system11b. Now Williams felt that the Special Solenoids should be CPU controlled. This was an excellent idea because now the Special Solenoids are "one shot" coils. That means if a pop bumper or slingshot switch gets stuck closed, the coil fires ONCE and only for a set duration of time. On System9 and System11/11a games, if a slingshot or pop bumper activation switch is stuck closed, the associated coil stays locked on! This burned up a lot of coils and driver transistors in these sys9 and sys11/11a games, which would blow the solenoid fuse and put the game out of operation.

But the story doesn't end there with Special solenoids. The control of special solenoids on all system9 and sys11/sys11a games is directly through playfield control via the playfield trigger switches. But interestingly, special solenoid can also be controlled by the CPU too. This can be seen when running the internal game diagnostics, and the game turns the special solenoids on and off in the coil test. Because of these "dual trigger" (two ways to turn on) functionality of the special solenoids, these can be more problematic than the other 16 "CPU controlled" coils on the game.

Special solenoids use a 7408 chip, a 7402 chip, and two transistors (2N4401 pre-driver and a TIP120/TIP102 driver). This is one more TTL circuit than the CPU controlled coils use. A special solenoid operates if the playfield switch pulls one 7408 input low. The other 7408 input can be pulled low by the CPU via a PIA (and this is what is done in the diagnostic solenoid test). So a special solenoid could work in diagnostic test but not work in game mode (or vice versa). This confuses a lot of people because the diagnostics show a coil a "working", yet when playing the game the same coil does not respond.

Also the the special solenoid playfield switch trigger has a 100 ohm 1/2 watt resistor and a 22 mfd 100 volt electrolytic capacitor (the positive lead connected to the resistor) in parallel to the switch. This is different than CPU controlled coils that use a switch matrix switch to turn them on (switch matrix switches only have a 1N4004 diode on the switch).

Again the thing about special solenoids that is really freaky is this: the diagnostics can show the special solenoids as working, but in game play they may not work! The opposite is also true; a special solenoid could work in the game, but not in diagnostics. This happens because there are two different and distinct triggers for the special solenoids. That is, playfield trigger for the special solenoids uses different hardware logic then the diagnostic trigger for special solenoids. This can be very confusing.

The logic flow for the special solenoids works like this: the PIA and the playfield trigger switch feed to the same 7408 chip. (Note the playfield trigger switch goes first thru a pullup 4.7k resistor which sometimes go open or out of spec causing problems.) The 7408 is an 'OR' TTL chip, meaning if either of the switch input are triggered (playfield or PIA), the TTL output turns-on the special solenoid circuit energizing the coil. The OR'ed 7408 trigger signal then goes to a 7402 chip, which goes to a 2N4401 pre-driver transistor, and finally to a TIP120 or TIP102 driver transistor (which ultimately sinks the ground and fires the coil). So if a special solenoid only works in game mode and not diagnostics, the problem has to be the 7408 chip or the PIA chip. If the special solenoid only works in diagnostic mode and not game mode, the problem has to be the pullup 4.7k resistor, the 7408 chip, or the playfield switch (and associated cap/resistor on the switch) or connector for the playfield switch (common to have cracked solder joints on this connector). If a special solenoid works with one trigger but not the other, the 7402 and everything connecting after it (pre-driver, driver transistors, coil, etc.) are fine.

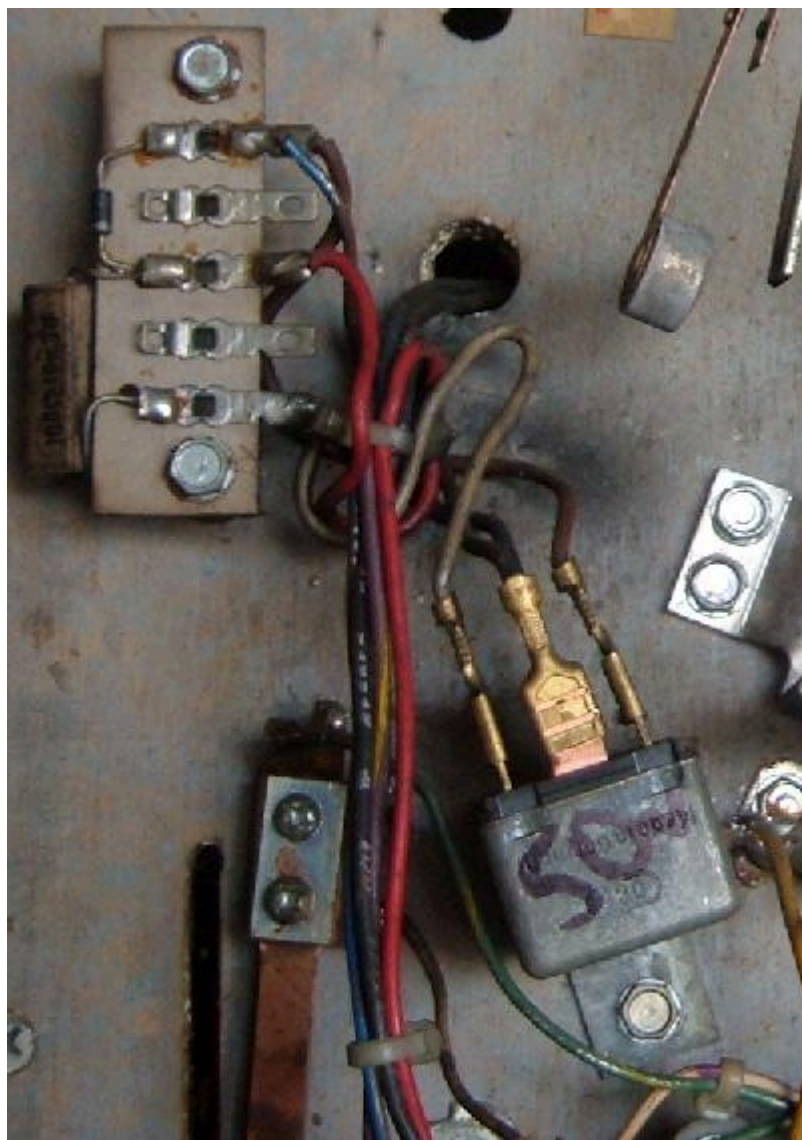
To confuse things even more, the Special solenoids have yet another switch involved. This is the scoring switch, which is part of the switch matrix (unlike the special solenoid trigger switch). So each pop bumper and slingshot have a second physical switch mounted on the playfield device. This switch closes as the coil energizes. This switch matrix switch in turns tells the CPU to score the device (but does *not* tell the CPU to fire the coil). So if there's a pop bumper or slingshot that works fine (energizes), but does not score, often it's because this secondary switch matrix switch is mis-adjusted or broken.

The 50 volt Coils.

On all system11 games, some coils such as the flippers and the kick backs are high voltage 50 volt coils. On system11 and system11A games (Fire! and before), to power these coils a "50 volt power supply board" is used (the power supply board itself does not supply this voltage). The 50v power supply board is a 6"x4" skinny board on the far right side of the backbox. This board has an output voltage fuse for the flippers, and an input voltage fuse mounted to the inside of the backbox wood. Sometimes the 50 volt's bridge rectifier on this board dies. If there is voltage at the AC lugs of the bridge, but no DC voltage on the positive and negative leads, the bridge has failed and gone open (remember, the "offset" lug of the bridge is the positive lug, and the negative lug is directly opposite). Bridges also can short, which will cause the associated fuse to blow immediately.

On games before Big Guns (system11 and system11a), Williams used under-playfield mounted 50 volt relays for the 50 volt coils. The TIP122 CPU board transistor would momentarily energize the relay, which would in turn momentarily complete ground to the 50 volt coil. This required a relay board to be mounted under the playfield for each 50 volt coil. Williams stopped this with System11b (Big Guns and later). Instead they used the Auxiliary Power Board, which had large eight TIP36 transistors to drive the 50 volt coils. The CPU board's TIP122 transistors would pre-drive the Auxiliary Power Board's TIP36 transistors, which would sink 50 volts and energize the 50 volt playfield coils (this means the TIP36 transistors became essentially solidstate 50 volt relays). The Auxiliary Power Board (APB) also had eight fuses, one for each of the TIP36 transistors. Hence the 50v power supply board is no longer used on system11B and later games. The Auxiliary Power Board also had a bridge rectifier for the 25 and 50 volts circuit. The APB board is mounted directly below the power supply board on the right side of the backbox.

A 50 volt relay mounted under the playfield on High Speed for the left side kickback. The diode for this relay is mounted on the circuit board at the upper left.



The Bank Selected Solenoids and the A/C Relay.

On all System 11 games, transistor Q7 or Q8 controls the "solenoid A/C select relay". This is the relay that controls which bank (either "A" or "C") that TIP122 transistors Q22-Q25 and Q30-Q33 will control. This means one TIP122 transistor can control two different devices (usually a flasher on bank C, and a solenoid on bank A). This was called "multiplexing". Note on System11 and System11A games before Big Guns (no Auxiliary power driver board), transistor Q7 controls the solenoid A/C select relay. On System11B and System11C games Big Guns and later (with an Auxiliary power driver board), then Q8 controls the A/C relay mounted on the Auxiliary power driver board. The solenoid A/C select relay is a 24 volt DC, 10 amp, DPDT relay (NTE R14-11D10-24P).

The location of the A/C relay is somewhat confusing. On High Speed and Grand Lizard there is no A/C relay, as these two games do not use system11's multiplexing feature. On F-14 Tomcat the A/C relay is mounted under the playfield. Finding the A/C relay on system11A games can be challenging because under the playfield can be a lot of relays! This happens because prior to Big Guns all playfield 50 volt coils are activated via a relay board. The 50 volt and A/C relay board are identical looking. Just remember on system11 and system11A the relay on the power supply is for the General Illumination, and the relay on the CPU board is for the flippers. The remaining playfield mounted relays are either a 50 volt coil relay or the A/C relay.

Also remember some System 11 games do not utilize the solenoid A/C select relay. The first two system 11 games (High Speed and Grand Lizard) do not. There are enough transistors on the CPU board so there is no need for any transistors to be shared between two devices. In this case transistor Q7 is used for a 50 volt kickback lane relay (on High Speed), and the A/C select relay not used.

The A/C relay on a F-14 Tomcat (system11A), mounted on the bottom edge of the playfield (center relay; the other two relays are 50 volt coil relays).



Test the Solenoid A/C Select Relay.

If the solenoid A/C select relay is not working correctly, there will be strange game operation issues. A messed up A/C relay, if stuck on bank "C", won't give any power to coils 1 to 8. If the relay is constantly energized (stuck on bank "C"), it's probably because its driver transistor (either Q7 or Q8) is shorted. If it's stuck on bank "A" (where the relay sits at rest) or won't energize, the flashlamps in the game won't work. If the A/C relay won't energize, the game will fire coils when its supposed to energize flash lamps! Or vice versa. This can give some very strange game behavior.

If you are wondering what a particular relay does, any individual relay can be activated quite easily. All relays should have a 1N4004 diode associated with the relay. On playfield mounted relays, this diode is mounted right next to the relay on the relay board. On the power supply, CPU board, or Auxiliary Power Board the 1N4004 diode is right above the relay. With the game in diagnostic mode, connect an alligator test clip to ground. Touch the other end of the test clip to the NON-BANDED side of the relay's diode. This will energize the relay. If the game is in solenoid test and is testing one of the eight multiplexed solenoids, manually energizing the A/C relay should switch the test from the solenoid to a flasher (assuming the game is using a flasher for that multiplexed transistor). If it is a GI relay, when the relay is energized the GI lights it controls should turn off. If it is the flipper relay, energizing it should activate the flippers via the cabinet flipper buttons (note the flipper relay will automatically be energized when entering dianostic mode, so try this in attract mode).

The solenoid A/C select relay can also be tested via its driving transistor. To do this, take your aligator test wire and connect it to the metal tab on transistor Q7 (Fire! or before) or Q8 (Big Guns or later). Then with the game on and in attract or diagnostic mode, touch the other end of the aligator clip to the ground strap in the backbox. You should hear the A/C select relay click on and off. This will make finding the relay a bit easier on system11 and system11A games.

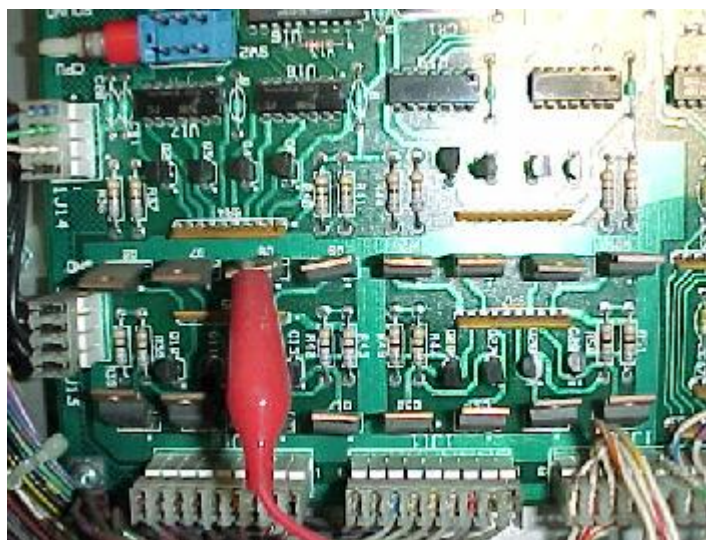
If you don't hear the A/C relay "click", you should now test transistor Q7/Q8. The quick and easy way to do this is:

- Turn the game off.
- Put your DMM on ohms (buzz tone).
- Put one lead on the ground strap in the backbox.
- Touch the other lead to the metal tab on transistor Q7/Q8.
- If you get zero ohms (buzz), the transistor is bad! (shorted on)

Is the Solenoid A/C Select Relay Bad?

Be aware that relays can go bad too. This can especially happen if transistor Q7/Q8 locks on for some time, and leaves power to the relay turned on. The relay can actually get so hot, it burns the relay contacts together. Also, the solder joints on the A/C select relay can go "cold" or fatigue too. This often will make an A/C select relay not work (but reflowing the relay solder joints can often fix this). If you need to replace the A/C select relay, it's a 24 volt DC, 10 amp, DPDT relay.

Turning on Relay A/C to test both coils/flashers that driver transistors Q22-Q25 and Q30-Q33 control. Here transistor Q8's metal tab is grounded with an alligator clip.



Special Solenoid Logic Flow.

For Special Solenoids (SSa to SSf), here is the logic flow. This is useful to know if you are having a problem with a special solenoid. I've had the Special Solenoid TIP/2n4401 transistor(s) lock on, and back flow to kill its driving 7402 (this is quite common). But it can go back one step further and kill the 7407 at U49 too causing multiple other Special Solenoids to lock-on. Luckily having a Special Solenoid kill its PIA is rare.

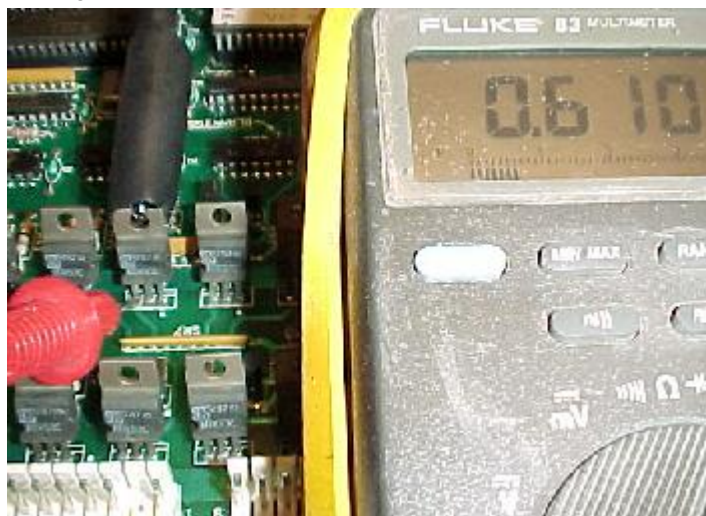
SSx:	6821 PIA	7407	7402	2N4401	TIP122
SSa:	U38 (pin 39)	to U49	to U45	to Q74	to Q75
SSb:	U41 (pin 39)	to U49	to U45	to Q70	to Q71
SSc:	U41 (pin 19)	to U49	to U45	to Q72	to Q73
SSd:	U38 (pin 19)	to U49	to U45	to Q68	to Q69
SSe:	U54 (pin 19)	to U49	to U50	to Q76	to Q77
SSf:	U54 (pin 39)	to U49	to U50	to Q78	to Q79

Since pre-Big Guns games use hardware logic to fire the solenoids (they are not CPU controlled), there are some other smaller (and easily damaged!) components that can fail too. Check capacitors C70 to C75 (.01 mfd), and resistor network SR20 (4.7k). If these become damaged, a special solenoid can "stick on". Even if your game is Big Guns or later (CPU controlled special solenoids), damage these components can cause problems.

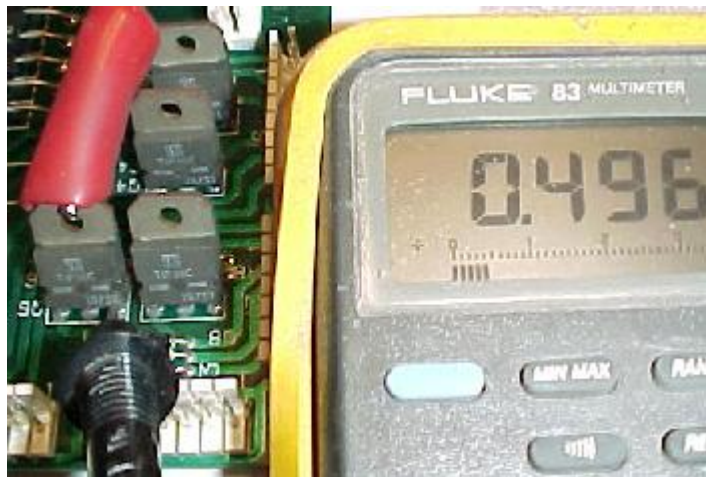
Transistor Testing procedures, circuit board out of the game.

If you have a circuit board out of the game for some reason, I would suggest testing all the solenoid/flasher driver transistors. It only takes a moment, and will ultimately save you time. To test a transistor, you'll need your digital multi-meter (DMM) set to the "diode" position.

Testing a TIP122/102 transistor on the CPU board.



- **TIP122/102:** Put the **black** lead of your DMM on the center lead or on the metal tab of the transistor. Put the red lead of your DMM on each of the two outside legs of the transistor one at a time. You should get a reading of .4 to .6 volts. Any other value, and the transistor is bad and will need to be replaced.

Testing a TIP36c on the Auxiliary power driver board.

- **TIP36c:** Put the **red** lead of your DMM on the center lead or on the metal tab of the transistor. Put the black lead of your DMM on each of the two outside legs of the transistor one at a time. You should get a reading of .4 to .6 volts. Any other value, and the transistor is bad and will need to be replaced.

Testing a 2N4401 pre-driver on the CPU board.

- **2N4401** (pre-drivers): Put the **red** lead of your DMM on the center lead of the transistor (note this transistor doesn't have a metal tab). Put the black lead of your DMM on each of the two outside legs of the transistor one at a time. You should get a reading of .4 to .6 volts. Any other value, and the transistor is bad and will need to be replaced.

Testing a TIP42 lamp matrix column driver transistor on the CPU board.

- **TIP42:** Put the **black** lead of the DMM on the base lead (as facing the transistor, the left leg) of the transistor. Put the red lead of the DMM on each of the two transistor legs (center and right), one at a time. The DMM should show a reading of .4 to .6 volts. Any other value, and the transistor is bad and will need to be replaced.

Most often transistors short when they go bad. This will usually give a reading of zero or near zero, instead of .4 or .6 volts.

To enter diagnostics, the red center button must be in the "down" position (as shown here). If the center button is "up", you will enter the audits menu instead.



Testing Transistors/Coils, circuit boards installed in a (near) WORKING game, using the Diagnostics Test.

If your game powers on, you can use the diagnostics to test most devices. From the attract mode:

- Press the center red button inside the coin door to the "down" position.
- Press the black button closest to the coin door once.
- Press the center red button inside the coin door to the "up" position.
- Press the black button closest to the coin door to move from test to test.

Solenoid Doesn't Work during Diagnostic Tests.

If a solenoid doesn't work from the diagnostic tests, here's what to check. Turn the game off before doing this.

- Check all the fuses. A non-working solenoid could be as easy to fix as just replacing a fuse.
- Find the solenoid in question under the playfield. Make sure the wire hasn't fallen off or become cut from the coil (a very common problem).
- If the above is correct, make sure the windings of the coil haven't broken off from the solder lugs. If one has broken, you can re-solder it. Make sure you sand the painted enamel insulation from the wire before re-soldering.
- Check the coil diode. For games before Big Guns, the coil diode will be right on the coil, with the banded side of the diode connecting to the power side of the coil. For games with an Auxiliary power driver board (Big Guns and later), the coil diodes are mounted on the auxiliary power driver board (flipper coils should always have diode(s) though, for all system 11 and WPC games). Moving the diodes away from the coil increases reliability as the diode is not subject to the jarring and heat a coil can produce. It also eliminates the need for the operator to know which coil wire goes to the banded side of the diode when replacing a coil!

A Coil doesn't Work, What To Do.

The following procedures will start at the coil, and work back to the CPU board, testing components. This will eliminate things and make finding the problem easier.

Testing for Power at the Coil.

Most pinball games (including system 11) have power at each and every coil at all times. To activate a coil, GROUND is turned on momentarily by the driving transistor to complete the power path. Since only ground (and not power) is turned on and off, the driving transistors have less stress on them. With this in mind, if we artificially attach a coil to ground, it will fire (assuming the game is turned on and in attract mode).

- Turn the game on and leave it in "attract" mode.
- Lift the playfield.
- Put your DMM on DC voltage (100 volts or greater).
- Attach the black lead of your DMM to the metal side rail.
- Touch the red lead of your DMM on either/both/all lugs of the coil in question.
- You should get a reading of 25 to 80 volts DC. Switch your red test lead to the other lug of the coil, and you should get the same voltage again. On flipper coils, test all the lugs of the coil for power. If you don't get any voltage reading, no power is getting to the coil. If you don't get power at all lugs, then you have a broken winding on the coil itself. Replace the coil or fix it (if the winding is an outside winding, you can remove the paper label and unwrap a turn of wire, sand the insulation off, and resolder the coil winding to the lug).
- If no power is getting to the coil at any lug, it may be a coil that is A/C relay selected. Push the center red coin door button down, and press the black button closest to the coin door.

This will put the game in diagnostics mode. This should de-energize the solenoid A/C relay, and turn the power to the coil in question on. If there is still no power, put an alligator clip on the metal tab of transistor Q7 (Fire! and before) or Q8 (Big Guns or later) to activate the solenoid A/C relay, and retest for power at the coil again.

- A broken solenoid A/C relay can cause power to not get to a coil. But this will affect more than one coil. Cold solder joints on the A/C relay-to-board solder pads can do this too.
- If no power is getting to the coil, a wire is may be broken somewhere. Trace the power wire. Remember, the power wires are "daisy-chained" together. So if there is a break in the wire at a previous coil, the coils down stream will not get power.

Coil Test to Make sure Coil is Good.

Here's another method of testing coils, which is more "low-level". This will test if the coil itself is good, and that there is power at the coil.

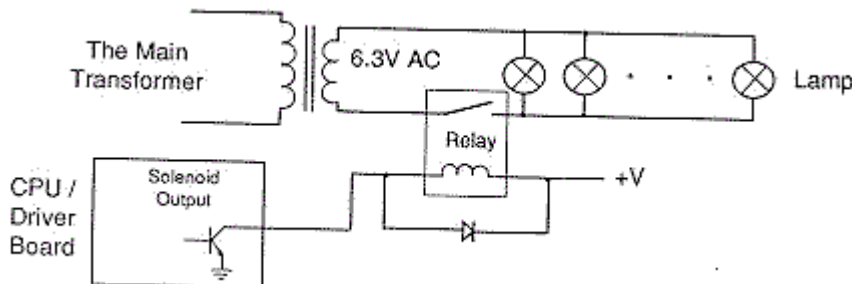
- Game is on and in "attract" mode, and the playfield lifted.
- Connect an alligator clip to the metal side rail of the game.
- Momentarily touch the other end of the alligator clip to the GROUND lead of the coil in question. This will be the coil lug with the single (thinner) wire attached. On coils with a diode, the ground coil lug is the one with the non-banded side of the diode connected. On FL11630 flipper coils F-14 Tomcat and later, touch the middle lug to ground.
- The coil should fire (if you accidentally touch the alligator clip to the power side of the coil, the game will reset and/or blow a fuse, as you are shorting solenoid high voltage directly to ground).
- If the coil does not fire, it may be a coil that is A/C relay selected. Push the center red coin door button down, and press the black button closest to the coin door. This will put the game in diagnostics mode. This should de-energize the solenoid A/C relay, and turn the power to the coil in question on.
- If the coil still does not fire, either the coil itself is bad, or the coil's fuse is blown and power to the coil is not present.

Testing the under-the-playfield Relay Board.

On most system 11 games, there are a mix of some 50 volt and 25 volt coils. For games without Auxiliary power driver boards, most games use relay boards to power the 50 volt coils. A TIP122/102 transistor on the CPU board energizes the under the playfield relay board. This relay then turns on the ground to the 50 volt coil, motor or other device. This was done because the original TIP122 on the CPU board can't handle the current draw of a 50 volt coils or motors.

The under the playfield relay boards were no longer required on 50 volt coils with games that had an Auxiliary power driver board (APDB). That's because the APDB had TIP36c transistors to control the 50 volt coils, replacing the need for the small relay boards. But some games even with the APDB still used under the playfield relay boards for other uses (like turning off specific strings of GI lights, like on Big Guns).

Under-the-playfield relay boards were still used after the introduction of the Auxiliary power driver board. Their function was to turn sections of the playfield GI lamps off.



It is easy to test the under the playfield relay boards. Connect an alligator clip wire to the "DRV" lead on the small relay board. Connect the other end of the alligator clip to ground (the game's metal side rail). This will energize the relay (you should hear a "click"), and the device it powers should also operate (or possibly turn off a section of GI lights).

Cracked Solder Joints on the 50 Volt Coil Relay Boards.

On system11 games before Big Guns (those games without the Auxiliary Power Driver board), Williams used small relay boards under the playfield to drive the 50 volt coils. Since the games were really set up for 28 volts coils, a 28 volt relay was driven by the driver board TIP122/102 transistor, and hence the relay would turn on the 50 volt power to the desired coil. This was only on games before Big Guns, as the Auxiliary Power Driver board utilized eight TIP36c transistors which could handle the 50 volt coils without using a relay board.

On the games using relay boards for the 50 volt coils, often the solder joints on the under the playfield relay boards are cracked or "cold". If you are having a problem with any device that is controlled by a relay board, re-solder all the header pin solder points on the relay board. Also the 150 ohm resistor used on the relay board can burn and go open.

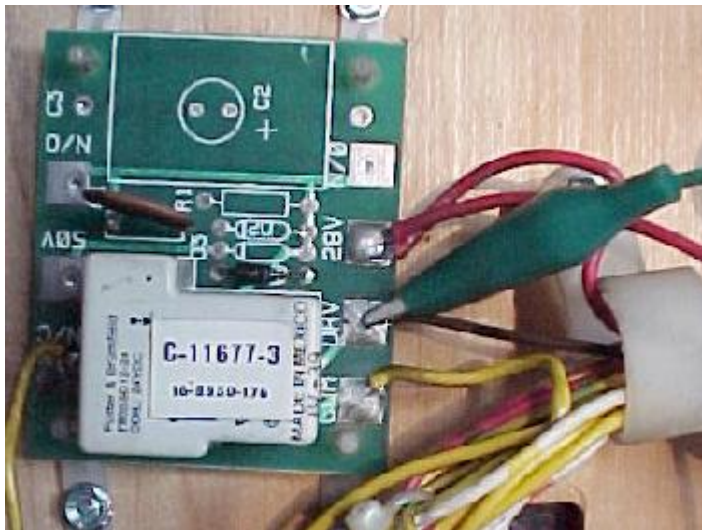
Locked On 50 Volt Coil, Sys11/Sys11A.

If a system11 and system11a game (Fire! and before) have a 50 volt coil is locked on at game power-on, this is usually a under-playfield mounted relay board problem. The CPU board TIP122 turns the 28 volt relay board's relay on. This in turns completes the 50 volt power path to the coil through the relay switches. If the 1N4004 diode on the 50 volt coil fails, the relay switches for this relay will take extreme abuse (big huge blue spark). Eventually this will burn the switch contacts to the point where they weld together. And this will cause the 50 volt coil to lock-on at game power on.

To fix this problem first make sure the coil's 1N4004 diode is not broken or shorted. Then look at the associated relay on the relay board. You will have to remove the plastic cover from the relay, which should pry off. Examine the switches on the relay, and adjust/clean as needed. To clean these relay switches, use a Flexstone file (this unlike cleaning the gold contact playfield switches, which should **never** be filed). Also make sure the 150 ohm resistor on the relay board is not burnt. When done, push the plastic cover back on the relay (it should snap in place).

To test the relay and the 50 volt coil it powers, use the game's diagnostics. Or ground the non-banded side of the 1N4004 diode on the relay board (which complete 28 volts power to the relay board). This will energize the relay, which in turns energizes the 50 volt coil. Note there are two 1N4004 diodes on the relay board. The one you want to ground is the non-banded side of the 1N4004 diode that is by itself (**not** next to the 150 ohm resistor).

Grounding the "DRV" lead with an alligator clip on the under the playfield relay board to energized the device this board controls.



Testing TIP122/102 Transistor and Down-Stream Wiring/Coil.

If the coil fires in the above test, you may have a transistor problem. You can also test the everything from the TIP122/102 downstream to the coil itself (but note this does not test the transistor itself). **Only do this for the TIP122 or TIP102 transistors! Damage can occur if this test is done on other transistors (like TIP42 or TIP36).** This test will test everything from the CPU board's TIP122/102 down to the coil itself. If this test passes, and your coil still doesn't work in game play, you have a problem more "up stream". All that is left is the TIP122/102, the 2N4401 pre-driver transistor, the logic TTL chip that ultimately controls the whole process (a 7402 for the special solenoids or 7408 for the standard solenoids), and possibly the driving PIA 6821 chip.

- Game is on, and in diagnostic mode (push the center red coin door button down, and press the black button closest to the coin door; this will put the game in diagnostics mode).
- Remove the backglass.
- Find the transistor that controls the coil and/or flasher in question (refer to the manual).
- Attach an alligator clip to the grounding strap in the bottom of the backbox.
- Momentarily touch the other lead of the alligator clip to the metal tab on the TIP122/102 transistor (only works on these).
- The coil or flasher should fire.
- If the coil or flasher does not fire, it may be a transistor that is multiplexed through the solenoid A/C relay.
- To energized the solenoid A/C relay (which will fire the other coil/flasher that is multiplexed), attach an alligator clip to the grounding strap in the backbox. For games prior to Big Guns

(no Auxiliary power driver board), connect the other end of the alligator clip to the metal tab of transistor Q7. For games Big Guns and later (with an Auxiliary power driver board), connect the other end of the alligator clip to the metal tab of transistor Q8. This will energize the solenoid A/C relay on the power supply or the Auxiliary power driver board.

- If the coil or flasher does not fire, and the coil or flasher did fire in the previous test, you probably have a wiring problem. A broken wire or bad connection at the connector would be most common. It is also possible you have a bad driver or pre-driver transistor. Use your meter and test the transistors on the board (see [Transistors Testing Procedures](#) for details).

I've Done the Above Tests & they Work, but the Coil still doesn't work in Game mode.

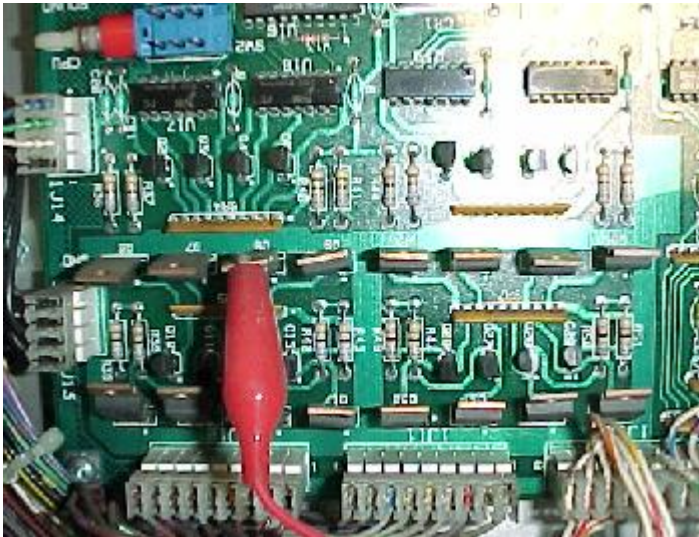
You have preformed all the above tests and replaced/tested the coil, TIP36c, TIP122/102 and/or the 2N4401 transistors. But the coil still doesn't work in game mode!

If the coil in question is a special solenoid (pop bumpers, slingshots), you need to look at the driving components. There are some other smaller (and easily damaged!) components that can fail too for the special solenoids. Check capacitors C70 to C75 (.01 mfd), and resistor network SR20 (4.7k). If these become damaged, a special solenoid can "stick on". Even if your game is Big Guns or later (CPU controlled special solenoids), damage to these components can cause special solenoid problems.

There are more components that needs to be tested or replaced too, if the transistors themselves are good. This is the hardware logic chips that drive the pair of TIP102/2N4401 transistors:

- Q2/Q6, Q3/Q7, Q10/Q14, Q11/Q15: 7408 at U17, 6810 PIA at Uxx.
- Q4/Q8, Q5/Q9, Q12/Q16, Q13/Q17: 7408 at U18, 6810 PIA at Uxx.
- Q18/Q22, Q19/Q23, Q26/Q30, Q27/Q31: 7408 at U19, 74LS374 at U28, 6810 PIA at U54 or U38.
- Q20/Q24, Q21/Q25, Q28/Q32, Q29/Q33: 7408 at U20, 74LS374 at U28, 6810 PIA at U54 or U38.
- Q68/Q69, Q70/Q71, Q72/Q73, Q74/Q75: 7402 at U45 (special solenoids).
- Q76/77, Q78/Q79: 7402 at U50 (special solenoids).
- Q80, Q81, Q82 TIP122/102 lamp row drivers: 7406 at U55, 6810 PIA at U54.
- Q83, Q84, Q85, Q86, Q87 TIP122/102 lamp row drivers: 7406 at U56, 6810 PIA at U54.

Turning on Relay A/C to test both coils/flashers that driver transistors Q22-Q25 and Q30-Q33 control. On Big Guns and later, transistor Q8's metal tab is grounded with an alligator clip. On pre-Big Guns (games with no Auxiliary power driver board), transistor Q7's metal tab is grounded instead.



Installing a New Transistor.

If you have determined a coil's transistor is bad, there are a few things to keep in mind. Most TIP122/102 transistors also have a "pre-driver" transistor (2N4401 or NTE123AP).

If you replace a coil's TIP122/102 transistor, it's a good idea to also replace its corresponding pre-driver. It will be located near the TIP transistor. See the schematics to determine the specific pre-driver transistor(s).

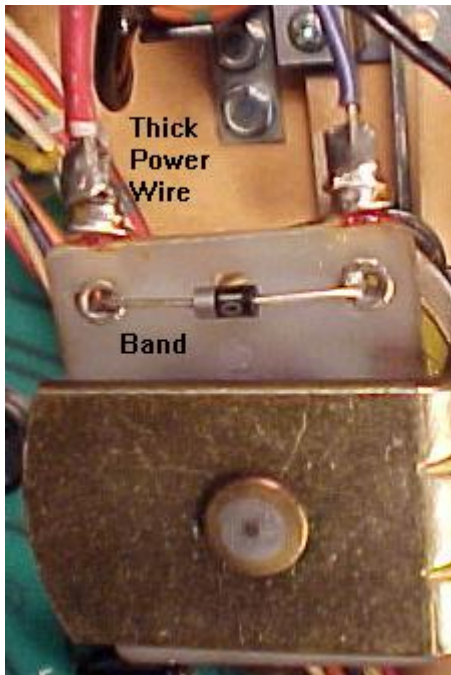
Game with an Auxiliary power driver board (Big Guns and later), use a bigger TIP36c driver transistor for high voltage devices. These TIP36c transistors have TWO pre-drivers: a TIP122/102 and a 2N4401 transistor. Again, if the TIP36c has failed, it's a good idea to replace both corresponding pre-driver transistors.

Replacing the pre-driver transistors is optional (if they test Ok). You can always test these pre-drivers instead of just replacing them. But if the driver transistor has failed, the pre-driver was probably over-stressed too. It is a good idea to replace the pre-driver transistor(s) too.

Replace TIP122 transistors with TIP102?

This is a very common question. The TIP102 is a more "hardy" transistor than the TIP122, but works exactly the same. So why not replace a bad TIP122 with a TIP102? Well, actually I recommend the TIP102 over the TIP122. Some people may argue with this, claiming the TIP102 will not go bad as quickly, and therefore can cause more heat and damage the circuit board before or while it fails. But if the transistor is already shorted, it really doesn't matter if it's a TIP122 or TIP102. It's still shorted, and will still cause the same heat and damage. So my recommendation is to replace all bad TIP122 transistors with the more robust TIP102. After all, this is what Williams started using on their next generation of pinball boards (WPC) after system 11. The TIP102 is far more robust than the TIP122 (which frankly was barely robust enough for the given job).

The diode mounted right on the coil, games Fire! and before. Note the thicker red power wire on the left goes to the banded side of the diode. The thinner wire on the right lug goes to ground.



Coil Diodes.

On all electronic pinball games, each and every CPU controlled coil must have a coil diode. This diode is VERY important. When a coil is energized, it produces a magnetic field. As the coil's magnetic field collapses (when the power shuts off to the coil), a surge of power as much as twice the energizing voltage spikes backwards through the coil. The coil diode prevents this surge from going back to the circuit board and damaging components, or causing the CPU to get confused (which often results in a game reboot).

If the coil diode is bad or missing, it can cause several problems. If the diode is shorted on, coil fuse(s) will blow. If the diode is open or missing, strange game play will result (because the CPU board is trying to absorb the return voltage from the coil's magnetic field collapsing). At worst a missing or open diode can cause the driver transistor or other components to fail.

When Replacing a Coil Diode...

Remember to always install a coil diode with the banded end of the diode to the power wire coil lug! The power lug is the one with the thicker red or purple wire connected to it. This is usually the lug with TWO wires connected to it (because the power wires "daisy chain" from coil to coil). If you install a diode in reverse, it will instantly short and be ruined when power is applied.

Diodes Mounted on the Coil.

Sometimes a diode lead breaks on the coil from vibration. When replacing a coil, the repair person can install the coil wires incorrectly (the power wire should always be attached to the coil's lug with the banded side of the diode). To prevent this, **Williams moved the coil diodes off the coils and onto the Auxiliary power driver board starting with Big Guns**. This isolates the coil diode from vibration and eliminates the possibility of installing the coil's wires in reverse. This was done on most coils **except** the flipper coils.

The coil diodes on a FL11630 flipper coil, F-14 Tomcat and later. Note the solo center blue wire and the blue wire on the right lug goes to the EOS switch. The left lug (or gray/yellow) is the "hot" wire. The second blue/violet wire on the right lug continues to the cabinet switch and ultimately ground. Note the orientation of the diode bands.



Testing Diodes on the Auxiliary power driver board.

If you suspect a problem with a coil diode (game resets during multi-ball when lots of coils are firing), you can test the coils on the Auxiliary power driver board. Just set your DMM to "diode" setting, and put the black lead on the banded side of the diode, and the red lead on the non-banded side. You should get a reading of .4 to .6 volts. If you reverse the leads, you should get a null (no) reading.

Testing a coil diode on the Auxiliary power driver board.



Test a Diode on a Coil?

You can test coil diodes. They do fail and they do break. This is true mostly for just the coil diodes that are actually mounted on the coil itself. Testing coil diodes is somewhat a waste of time. If you suspect a coil diode is bad, just cut the old one off and solder on a new one. They are so inexpensive, it's not really worth trying to test them. Most bad coil diodes are physically broken, and you can usually see the damage.

But if you want to test a coil diode, you can. If the coil diode is mounted on the coil, you will need to clip one end of the diode off the coil lug to test it (that's why just replacing the diode is a good idea if you suspect a problem). If your game has an Auxiliary power driver board (Big Guns or later), the coil diodes are mounted on the Auxiliary power driver board. These rarely go bad.

Use your DMM set to "diode" setting, and test the board mounted coil diode. With the black lead on the banded side of the diode and the red lead on the non-banded side, you should get between .4 and .6 volts. Reverse the leads (red lead to banded side of diode), and you should get a null reading. If you don't get this reading, cut one lead of the diode from the circuit board, and repeat the test. If you still don't get these results, replace the diode with a new 1N4004 diode.

Test the Coil Resistance with a DMM.

After replacing the driver transistor, ALWAYS measure the resistance of the associated coil. This is important. If a coil gets hot (because its driver transistor was shorted), it can burn the painted enamel insulation off the coil windings. This lowers the overall resistance of the coil because adjacent windings short together. If resistance gets much below about 2.8 ohms, the coil becomes a "short", and will fry its associated driver transistors very quickly!

To test the coil's resistance, it is best to remove the attached wire from one (either one) of the coil's lugs. Then set the DMM to low resistance, and put the DMM leads on the lugs of the coil. Most coils should be in the 5 to 15 ohm range, but could go as high as 150 ohms, or as low as 2.5 ohms. If the coil is below that, it should be replaced with a new coil of the same type. Coils with resistance below 2.5 ohms are basically a dead short, and this will fry its associated driver transistor.

Installing a New Coil.

Many replacement coils will come with a diode soldered across its solder lugs. On System 11 games Big Guns and after, all coils except the flipper coils have the diode mounted on the Auxiliary power driver board. For these coils you can cut the diode off the coil before installing. You can then solder the coil wires to either coil lug. You can leave the diode in place, but you must install the coil wires correctly. Though you still have the circuit board mounted diode as protection, you could damage the coil's driver transistor.

For games Fire! and before (with no Auxiliary power driver board), the coil's ground wire (usually the smaller wire) MUST go to the lug of the coil with the non-banded side of the diode. The power wire connects to the lug with the banded side of the diode. If you have the wires reversed, this essentially causes a shorted diode, which destroys the diode.

Coil Doesn't Work Check List.

If a coil doesn't work in a game, here's a check list to help determine the problem.

Before you start, is the coil stuck on? (Hint: is there heat, smoke and a bad smell?). If so, the coil's driving transistor has probably failed. Turn the game off and check the driving transistor, and replace if needed. See [Transistors Testing Procedures](#) for more info.

If the coil just doesn't work, here's a list of things to check:

- Have the power wires fallen off the coil's solder lugs?
- Is the coil damaged? Has the internal winding broken off the coil's solder lug?
- Is there power at the coil? See [Testing for Power at the Coil](#) for more details.
- If there is no power at the coil, check its fuse. Always remove the fuse from the fuse holder, and check the fuse with a DMM set to ohms.
- Check the other coils that share one of the same wire colors. Are they working too? If not, suspect the fuse that handles these coils.
- Power to coils are often ganged together. If the power wire for this coil has fallen off a previous coil in the link, power may not get to this coil.
- Using your DMM and its continuity test, make sure the coil connects to the correct connector/pins on the CPU board.
- With the game on, ground the non-banded diode side of the coil, to see if the coil works.
- Check the driving transistor. Usually this transistor will short on when it fails, but not always.
- With the game on, momentarily ground any of the 28 volt coil's associated TIP122/102 driver transistor's metal tab. The coil should energize. This checks the wiring from the driver board to the playfield coil.

End of System 11 Repair document Part One.

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- * Go to the [Pin Fix-It Index](#) at <http://pinrepair.com>